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Preface

I started thinking seriously about robots the way many people my age do: by building software and wondering what happens when the software gets a body.

Over a few weeks in 2026, I tried to write the laws I wished existed. The first was a bill for my own country, the *Rancangan Undang-Undang tentang Robotika dan Sistem Kecerdasan Artifisial Terwujud*, written with its Naskah Akademik in the format Indonesian law requires. The second was a discussion draft for the United States Congress, the Safe and Accountable Robotics Act of 2026. Neither has been introduced. Both taught me the same lesson. Every clause about torque limits or incident reports was secretly a clause about people: who carries the risk, who keeps the gains, who gets to say no, and who is remembered when something goes wrong.

Law could not hold all of that. A bill can say a robot must stop when a child steps in front of it. It cannot easily say what a society owes the warehouse worker whose shift was automated, the driver in Jakarta who has no employer to send him a layoff notice, or the grandmother whose evenings are now spent talking to a machine because her family works abroad. Those are questions for sociology, and this book is my attempt to ask them carefully.

I set myself one rule. Everything in these pages had to be true as of October 2026. There are no imagined robot uprisings and no invented 2040 headlines. When a forecast appears, it is labeled as a forecast and attributed to whoever made it. The real situation is dramatic enough. Humanoid robots are walking out of laboratories, governments are building walls around supply chains, and the oldest questions about work and dignity are coming back in metal.

I am seventeen. I do not pretend to have lived through the changes I describe. But my generation will spend its whole working life inside the arrangements being designed right now, and I would rather help design them than inherit them.

M. Bagus Prabowo Semarang, October 2026

Introduction: The Social Life of Machines

> "We shape our tools and thereafter our tools shape us." — a line usually attributed to Marshall McLuhan, and most likely written by his colleague John Culkin (1967)

A machine is never only a machine

Picture a delivery robot waiting at a crosswalk. Engineers see sensors, a battery, a path-planning algorithm. Look again and you see a set of relationships. Someone bought the robot and expects a return. Someone, possibly in another country, watches its camera feed and takes over when it gets confused. Someone trained the model that tells it a curb from a shadow, often by labeling images for a few dollars an hour. A city decided, or failed to decide, whether it could use the sidewalk. A wheelchair user now shares that sidewalk with it. A courier who used to carry the same package is doing something else, or nothing.

None of these relationships is visible in the machine. All of them are produced by it. That is what I mean when I say robots are *social arrangements*. They do not just perform tasks. They rearrange who does what, who owns what, who is watched, who is responsible, and who matters.

Sociology has a long tradition of looking at technology this way. Langdon Winner asked in 1980 whether artifacts have politics. Science and technology studies taught us that "technical" choices embed social ones. Feminist scholars showed how household technology reshaped, rather than abolished, domestic labor. This book applies that tradition to the newest and perhaps most physically intimate technology of all: machines that can move through our streets, enter our homes, touch our bodies, and increasingly decide for themselves how to do so.

Why now

Three things changed in the 2020s.

First, robots left the cage. For half a century the typical industrial robot worked behind a safety fence, and the fence was a social contract: the machine stays here, the humans stay there. Collaborative robots,

warehouse fleets, sidewalk couriers, and now humanoids broke that contract. Risk moved onto people who never agreed to carry it.

Second, robots gained general-purpose brains. Large models trained on text, images, and video began to be connected to arms and legs. A robot no longer needs to be programmed for each motion. It can be told what to do in words and learn from demonstration. That makes robots more useful and less predictable at the same time.

Third, robots became geopolitical. Supply chains for actuators, magnets, batteries, and chips run through a small number of countries. In 2025 and 2026 the United States opened a national security investigation into robotics imports and placed foreign-produced advanced robotics devices on the Federal Communications Commission's Covered List. China staged international humanoid robot games. Robots joined chips and electric vehicles as instruments of national power.

The questions this book asks

The book is organized around five clusters of questions.

Part I, The Machine Arrives, asks what robots are. Chapter 1 traces how they moved from the factory fence to the sidewalk. Chapter 2 asks what kind of thing a robot is: a tool, an agent, a pet, a person? I argue that robots should remain property, never persons, but that their autonomy creates real gaps in responsibility that law must close.

Part II, Work and Capital, asks who wins and who loses. Chapter 3 examines the evidence on robots and jobs. Chapter 4 asks the question I think matters most and is discussed least: who owns the gains of automation? Chapter 5 uncovers the humans hidden inside supposedly autonomous machines, the teleoperators and data workers spread across the Global South.

Part III, Living With Machines, moves from the economy into everyday life. Chapter 6 looks at care robots and companionship. Chapter 7 examines the home as a site of data extraction. Chapter 8 follows robots into public space, onto sidewalks, into police work, and into the line of fire of people who resist them.

Part IV, Power and Order, turns to institutions. Chapter 9 asks what it means to own a machine that its maker can switch off. Chapter 10 maps the geopolitics of embodied AI. Chapter 11 treats law itself as a social technology and argues for anticipatory regulation.

Part V, Futures Without Fiction, comes home. Chapter 12 looks at robots from Indonesia and the Global South, where most workers are informal and the standard tools of labor protection do not reach. Chapter 13 asks the oldest question in a new form: what is a human being worth when machines can do more and more of what humans do?

A note on method and honesty

I wrote this book with the help of AI research and drafting tools. It would be strange to write about intelligent machines and pretend otherwise. Every factual claim has been checked against sources listed in each chapter's notes. Where the evidence is contested, I say so. Where I give an opinion, I mark it as mine.

The goal is not to predict the future. The goal is to notice that the future of robots is being decided now, in procurement contracts, standards committees, trade investigations, terms of service, and draft bills, and that most of the people it will affect are not in the room. A sociology of robots is, in the end, an argument for getting them in the room.

Part I: The Machine Arrives

Chapter 1: From Fence to Sidewalk: How Robots Left the Factory

> "He works with his hands like a robot. He builds houses. He can work." — Karel Capek, *R.U.R.* (1920), trans. Paul Selver and Nigel Playfair

A word for forced labour

The word came before the machine. In 1920 the Czech writer Karel Capek published a play called *R.U.R.*, short for *Rossumovi Univerzální Roboti*, or Rossum's Universal Robots. The first performance took place on 2 January 1921, and not in Prague, as planned. The National Theatre had postponed its premiere, and nobody told an amateur company in Hradec Králové, so the amateurs opened first. By 1923 the play had been translated into thirty languages.[1] Capek gave the credit for its most famous word to his brother Josef. In Czech, *robota* means the forced labour that serfs owed on their lords' land, and it comes from *rab*, meaning slave.[1]

People who know the word but not the play are often surprised by what Capek's robots are. They are not made of metal. They are artificial organisms, built from a synthetic substance that behaves like living matter, and they look enough like people that the heroine, Helena Glory, takes the general manager's secretary for a woman. The secretary corrects her: she was not born at the factory, she was made there.[1] Skin is kneaded in troughs and nerves are spun on bobbins, and the bodies are assembled like cars. One critic has called Capek's robots a picture of "the traumatic transformation of modern society by the First World War and the Fordist assembly line."[1]

The plot is also more interesting than its reputation. Helena comes to the island factory as the representative of a League of Humanity, which wants to free the robots and pay them wages. The managers find this absurd, because, they explain, robots do not *like* anything. Ten years later, robots run the world economy and human births are falling. Then the robots revolt and kill almost everyone. They spare one man, Alquist, the clerk of works, and the reason they give is the line at the top of this chapter: he works with his hands like a robot. He builds houses. He can work.[1]

I start here because the play, written a full generation before any working industrial robot existed, already contains the three questions this book is about. The first is about labour: who does the work, and what happens to the people whose work is taken over? The second is about status: is a robot a thing, a worker, or something that might one day ask for wages? The third is about worth. The robots of *R.U.R.* decide that a human being deserves to live because he is *useful*, which is exactly the standard the factory owners had applied to the robots. Capek saw that a society that measures beings by their output will, sooner or later, measure its own members the same way.

Asimov called *R.U.R.* "a terribly bad" play that is "immortal for that one word."[1] I think he was wrong about the first half. The play is clumsy in places, but it understood something that much later writing about robots forgot. It never treats the robot as a gadget. It treats the robot as a social arrangement: a way of organising work, property, and recognition among people. The robots are a labour policy before they are a technology.

This chapter follows the robot's real history, which turned out quite unlike Capek's. For most of that history, real robots were kept apart from people by an actual fence. Then, in stages, the fence came down. First the robot went into the home, then onto the shared warehouse floor, then out to the sidewalk, and now, in a few factories, it walks on two legs. My argument is that each step moved risk. Behind the fence, the danger of the machine was carried by a small number of trained workers, under rules their employer was bound by. On the sidewalk, it is carried by anyone who happens to be passing, and most of them never agreed to anything.

Behind the fence

The first real industrial robot did not look like a person and did not want anything. Unimate was a hydraulic arm with five degrees of freedom. Its program was stored on a magnetic drum, and it weighed about 4,000

pounds. George Devol filed the patent behind it, titled “Programmed Article Transfer,” in December 1954. It was granted in June 1961, the year the first Unimate went to work on a General Motors line at the Inland Fisher Guide Plant in Ewing Township, New Jersey.[2] Its job was to take hot die-cast parts from a casting machine and handle them, a task that exposed workers to toxic fumes and to the risk of losing a hand. A local newspaper account quoted on the Unimate’s Wikipedia page describes it dropping red-hot door handles into pools of cooling liquid, so that “a man” no longer had to touch car parts “just made from molten steel.”[2]

Devol’s partner, Joseph Engelberger, was frank about the goal: mechanical workers that would replace humans in factories.[2] For the next four decades the industry did roughly that, with one important rule. The robot and the human were kept apart. Industrial arms are fast, strong, and blind. They move along programmed paths whether or not a person is in the way. The engineering answer was the cage: a fence of steel mesh, a gate fitted with a switch that stops the machine when it opens, and a procedure called lockout–tagout, under which a worker going inside must first cut the power and put a personal lock on the switch.

The fence works only if people respect it, and the first death we know of shows what happens when the work itself pushes them past it. On 25 January 1979, at Ford’s casting plant in Flat Rock, Michigan, a twenty-five-year-old named Robert Williams was one of three operators of a five-story automated storage system built by a division of Litton Industries. The system’s one-ton transfer vehicles moved castings to and from high shelves. When it gave wrong inventory readings (or, by another account, when it was running too slowly), Williams was asked to climb into the racks and get the parts himself. On the third level he was struck from behind by a transfer vehicle and killed instantly. His body lay there for about thirty minutes before co-workers, worried by his absence, found him.[3]

The case that followed is the first chapter in the legal history of robot harm, and its structure tells us something. Williams’s family sued the manufacturer, and in 1983 a jury awarded his estate $10 million, then the largest personal injury award in Michigan history. It found that there were not enough safety measures to prevent such an accident. Litton then tried to recover from Ford, arguing that Ford had not sent Williams to the training Litton offered and had let him into the racks without engaging the lockout system. The Michigan courts refused, because Litton had already settled with the estate.[3] Even in this first case, responsibility was spread across a chain: the maker, the employer, the procedure, the worker. Nobody blamed the machine. Nobody thought to.

There is a second lesson. Williams did not wander into danger by chance. He was *asked* to go in, because the automated system was not doing its job and production had to continue. That pattern will come back again and again in this book. The fence is a promise of separation, and production pressure keeps breaking the promise. Still, inside the factory the promise was made to a specific, limited group of people, employees, who were covered by labour law, by workplace safety rules, and in many countries by unions. The risk was real, but it was concentrated, and someone was clearly responsible for managing it.

Behind the fence, robots multiplied. Today the standard way to measure them is “robot density,” the number of operational industrial robots per 10,000 employees in manufacturing. The figures the International Federation of Robotics (IFR) published in April 2026, which cover 2024, are revealing. South Korea has the highest density in the world, with 1,220 robots per 10,000 manufacturing workers. Singapore follows with 818, then Germany with 449 and Japan with 446. The IFR calls Japan the “world’s predominant robot manufacturing country.” The United States ranks eighth with 307, and the global average is 132.[4] China’s density, on revised labour statistics, is a more modest 166, which ranks twenty-second. Its absolute numbers dwarf everyone else’s: about two million operational units, roughly four and a half times Japan’s stock, and 54 percent of all robots installed worldwide in 2024, some 295,000 units.[4] By the IFR’s September 2026 count, the global operational stock reached a record five million industrial robots in 2025, with more than 600,000 new installations that year, 59 percent of them in China.[5]

Why Korea and Japan? Part of the answer is industrial structure. The IFR points to Korea’s electronics and automotive industries, the two largest customers for industrial robots.[4] Part of it is demography: ageing societies with shrinking labour forces have strong reasons to automate. Seen from Semarang, these numbers are a reminder that “the robot” is not a single global event. It arrives unevenly, shaped by what a country makes, how old its workers are, and who can pay. Chapter 12 returns to what this means for Indonesia. The

point for now is narrower. For most of the period these statistics describe, nearly all of those millions of machines were inside fences, in places members of the public never go.

Through the front door

The first robot that ordinary people let into their lives was not humanoid either. It was a disc about thirty-four centimetres across and about nine centimetres high, designed to slide under the bed.

The idea for the Roomba began in 1989 with Joe Jones, an engineer at MIT's artificial intelligence laboratory, who built early versions out of LEGO. Two companies turned the prototypes down in the 1990s. In 1999 S.C. Johnson agreed to fund the project at iRobot, then pulled out after spending one or two million dollars, and iRobot went ahead alone. The first Roomba went on sale in September 2002.[6] The Smithsonian's National Museum of American History, which holds an original unit, describes it as the first successful domestic robot.[6]

The early Roomba was, by design, not very clever. It followed the philosophy of Rodney Brooks, iRobot's chief technology officer, who argued that robots should be like insects: simple control rules suited to their environment. It spiralled, followed walls, and changed direction at random when it bumped into something. It took several times longer than a person to clean a room, but because it was cheap to make, it sold.[6] It had a bumper, infrared "cliff sensors" to keep it from falling down stairs, and nothing more ambitious. The household version of the fence was an accessory called the Virtual Wall, a small device that projected an infrared beam the robot treated as a solid barrier.[6]

It sounds trivial. It was not. For the first time, an autonomous machine moved around a space that was not designed for it, among people who had no training at all. Children, elderly relatives, and pets were not "operators." They were simply there. The machine was safe enough because it was small, slow, and weak, and in the domestic setting that was the safety strategy: not separation, but harmlessness. The worst it could do was get stuck under the sofa or, as owners discovered, run over pet waste and spread it through the room, a problem frequent enough that the Roomba's Wikipedia page records it.[6] Later models gained cameras, mapping, and app connections.[6] As they did, a new kind of risk appeared. It was no longer only the body of the robot that mattered, but the information it collected about the inside of people's homes. Chapter 7 is about that.

The Roomba's corporate story ends with a twist that belongs in a book about ownership. In January 2026, iRobot announced that it had completed a court-supervised transaction under which Shenzhen PICEA Robotics, its main contract manufacturer and secured lender, acquired 100 percent of the company as it emerged from a prepackaged Chapter 11 process. As part of the restructuring, iRobot set up a separate US subsidiary, iRobot Safe, to govern American consumer data, with an independent board of US citizens.[7] The machine that taught the world to accept robots at home had become, among other things, a question about which country's company should hold the maps of millions of living rooms. Chapters 7 and 9 return to it.

The fence moves

Inside warehouses, the change came in a different way. In March 2012 Amazon bought Kiva Systems for $775 million, then its second-largest acquisition.[8] Kiva's robots were low orange drive units that slid under portable shelving pods, lifted them with a corkscrew motion, and carried them to a human worker standing at a picking station. They navigated by reading barcode stickers on the floor.[8] The trick was to reverse the old warehouse logic. Instead of people walking miles down aisles to find goods, the shelves came to the people.

At first this reversal preserved the fence in a new form. The robots' field was a restricted zone, and the human stood at its edge. But the boundary began to blur. In 2025 Amazon announced that it had deployed its one millionth robot, delivered to a fulfilment centre in Japan, across a network of more than 300 facilities. It described itself as "the world's largest manufacturer and operator of mobile robotics."[9] The same announcement described Proteus, "our first fully autonomous mobile robot," which "can safely navigate around employees in open and unrestricted areas of our sites."[9] Note the wording: *open and unrestricted*. The cage was being replaced by sensors and software that slow or stop the machine when a person comes near. Amazon also reported that its new facility in Shreveport, Louisiana, needed 30 percent more employees

in reliability, maintenance, and engineering roles, and that it had trained more than 700,000 employees since 2019.[9] Those are the company's figures, and Chapter 3 looks hard at what they do and do not show. For this chapter, what matters is the spatial change. The worker no longer stood outside the robot's territory. They shared it.

A parallel change happened to the industrial arm itself. In December 2008 Linatex, a Danish supplier of technical plastics and rubber, bought a UR5 arm from a young company called Universal Robots to load and unload a CNC machine. Universal Robots' own account says Linatex "did something unthinkable": "Instead of installing the robot behind safety caging, fenced off from people as was the norm for all industrial robots, they deployed it right alongside their employees." The workers also programmed it themselves, without outside programmers.[10] This was the beginning of the collaborative robot, or cobot: an arm light enough, slow enough, and sensitive enough to collisions that it could work without a cage. In 2018 UR's chief technology officer, Esben Østergaard, said that while "safety is imperative, that's simply the cost of entry into the cobot market now."[10]

That sentence deserves attention, because it marks a shift in how safety was understood. In the fence era, safety was mostly a matter of *space*: keep the people out. In the cobot era it became a matter of *behaviour*: the machine monitors force, speed, and distance, and adapts. Safety moved from the architecture of the building into the design of the product. That is real progress, and many workers are probably better off next to a cobot than next to a caged arm they once had to feed by hand. But it changes who must be trusted. Behind a fence, a worker can see the barrier and knows where danger starts. Next to a cobot, the barrier is invisible. It is a set of parameters in software that the worker did not write and usually cannot inspect. The worker's safety now depends on the manufacturer's choices and on the employer's configuration, and both are harder to check than a steel gate.

Still, the shared floor of the warehouse and the cobot cell is a controlled place. It has an owner, a safety officer, written procedures, and people who are paid to be there. The next step removed all of that.

Onto the sidewalk

In October 2019, a few days after the University of Pittsburgh began letting food-delivery robots made by Starship Technologies operate independently on the streets around its campus, a doctoral student named Emily Ackerman was crossing Forbes Avenue in her wheelchair. One of the robots was waiting on the corner, sitting on the curb ramp, the only accessible way up onto the sidewalk. Traffic was coming. "I (in a wheelchair) was just trapped *on* forbes ave by one of these robots," she wrote on Twitter, "only days after their independent roll out."[11] She managed to get around it. Speaking to the public radio station WESA afterwards, she described the strangeness of the moment: "It was really bizarre to realize that a non-sentient thing was putting me in danger and making me feel I was helpless." She added: "I think I was just laughing at it like, oh cool, this is my life right now."[11]

Her post went viral. The university asked Starship to pause testing "to review the technology further and ensure everyone's safety," and the city's Mayor's Office and Department of Mobility and Infrastructure contacted the company about access to ramps.[11] Another wheelchair user and disability rights advocate, Alisa Grishman, told WESA that earlier that month a robot had stopped in the middle of the sidewalk on the same avenue and stayed there through about three light cycles, apparently stuck, so that she had to backtrack and "bounce down into the street" to get past it. "We have all these wonderful bike lanes so that bikes don't go on the sidewalk," she said. "And yet now we're cluttering it up with robots."[11]

Starship's response is worth recording accurately. It said it had reviewed the footage of Ackerman's encounter and disputed that she had been prevented from getting onto the sidewalk, but that it had updated the mapping at that intersection "to ensure that additional room is added." It also said autonomous delivery could help people with disabilities get food and packages. Ackerman agreed with that point. What she wanted was "more testing before they're deployed."[11]

The episode is small, and nobody was hurt. But it shows the new structure of risk very clearly. Ackerman was not an employee of Starship, nor a customer of the food service, nor a party to any contract with the university. She had not been trained, she had not been warned, and she had not agreed to share her route

with a machine. She was simply a member of the public using public infrastructure, the curb ramp, which exists precisely because disabled people fought for decades to have it built. The robot's designers had treated the ramp as a convenient waiting spot, a technical detail on a map. For her it was the one point in the city where her right to move freely became real. The machine and the woman were competing for the same thirty centimetres of concrete, and only one of them had a company behind it to fix the map afterwards.

Since then, sidewalk delivery has grown from a pilot into an industry. In April 2026 Starship announced that it had completed ten million autonomous deliveries with a fleet of more than 3,000 robots in more than 300 locations in eight countries, mostly in Europe, plus more than 65 university campuses in the United States. It reported 22 million kilometres driven autonomously and more than 125,000 road crossings a day, "roughly two a second," operating at what it calls "Level 4 without active human supervision."[12] The company also cited bank research reported by Reuters suggesting that robot delivery is already three to four dollars cheaper per drop than a human rider, with a long-term target of about one dollar.[12] Those are the company's numbers and an analyst's projection, not independent findings. Even so, the direction is clear. The economics favour the machine, and the machine's workplace is the public street.

I should be careful here. Starship's record suggests that small, slow, light delivery robots are not, on the whole, a great danger to life and limb. The company reports a 97 percent approval rate among students on the US campuses it surveyed.[12] That figure also comes from the company. My point is not that these robots are dangerous. It is that the *distribution* of whatever risk they carry has changed. In the factory, a robot's risk fell on workers protected by employment law and represented, at least sometimes, by unions. On the sidewalk it falls on everyone, and it falls hardest on the people who have the least room to spare: wheelchair users, people with prams, the blind, the elderly, children. Every square metre a robot takes is taken from someone, and the people who give it up were never asked.

Anyone who has walked in an Indonesian city will see the problem straight away. In Semarang, as in Jakarta or Surabaya, the sidewalk is already a crowded, negotiated space: street vendors, parked motorcycles, motorcycles *riding* on it when traffic is jammed, broken paving, drainage gaps. It is a commons governed less by formal rules than by daily bargaining. A robot does not bargain. It follows a map someone else drew. Chapter 8 looks closely at what happens when machines enter public space and how people resist them. Here I only want to note that the sidewalk is where the "fence" model of robot safety finally broke down. There is no fence to build around a city.

Walking machines and learning machines

The newest step is the one that gets the most attention, and so it needs the most care. In the past few years, a growing number of companies have been building humanoid robots: machines with two legs, two arms, and a torso, meant to work in spaces designed for human bodies. The case for the human shape is practical rather than sentimental. Agility Robotics argues that a humanoid can "operate within human-centric environments, using the same tools and infrastructure without extensive retrofitting."[13] It is easy to get swept up in the videos. So let me say only what the record shows as of October 2026.

Figure. In November 2025 Figure AI published the results of an eleven-month deployment of its Figure 02 robot at BMW's plant in Spartanburg, South Carolina. The task was loading sheet-metal parts onto a welding fixture, with a cycle-time requirement of 84 seconds and a placement tolerance of five millimetres. The company reported ten-hour shifts, Monday to Friday, more than 90,000 parts loaded, more than 1,250 hours of runtime, and a contribution to the production of more than 30,000 BMW X3s. It also reported that the robot's forearm was its "top hardware failure point." Figure 02 was then retired in favour of Figure 03.[14] All of these figures are the company's own.

Digit. Also in November 2025, Agility Robotics announced that its Digit humanoid had moved more than 100,000 totes in a commercial deployment at the logistics company GXO's facility in Flowery Branch, Georgia, taking totes off mobile robots onto conveyors and stacking them.[13] Agility frames safety as "the financial gatekeeper to all automation investment" and describes its goal as a "cooperatively safe humanoid workforce."[13] The phrase "cooperatively safe" is the cobot idea carried over to a walking machine.

Optimus. Tesla's humanoid is the most publicised and the hardest to assess. *The Register* noted in January

2026 that Elon Musk had predicted thousands of Optimus robots working in Tesla factories by the end of 2025, and production of 5,000 to 10,000 that year. "That didn't happen."[15] The same article cites reporting that suggested Optimus demonstrations relied on teleoperation, with people controlling the robot remotely.[15] I have not seen independent evidence that Optimus is doing productive work at scale, and as of this writing I would treat claims about it as forecasts, not facts.

Unitree. The Chinese company Unitree has done something different. It has made humanoids cheap enough to buy. Its G1 is listed at US$13,500, excluding tax and shipping. It is 1.32 metres tall, weighs about 35 kilograms, and runs for about two hours on a battery.[16] The company's product page carries an unusual warning. Because the humanoid "has a complex structure and extremely powerful power," users should "keep a sufficient safe distance between the humanoid robot and people." Individual buyers are "strongly advised to thoroughly understand the limitations of humanoid robots before making a purchase."[16] In other words, the manufacturer itself tells the customer to rebuild the fence, out of nothing but distance.

Atlas. Boston Dynamics retired its famous hydraulic Atlas in 2024 and replaced it with an electric model. At CES in January 2026 it announced commercial production of that version. *The Register* reported that the production Atlas is padded, has minimal pinch points to reduce the risk of injury, uses 360-degree vision to spot approaching people, and "is designed to pause whenever it could put a person in danger, then continue its work once they're clear." It can be teleoperated or run autonomously.[15] In September 2026 the company opened what it calls the Robotics Metaplant Application Center at Hyundai's plant outside Savannah, Georgia, where Atlas robots are being trained, partly by teleoperation, to sequence car parts for assembly, with component assembly planned "by 2030." Hyundai Motor Group, Boston Dynamics' majority owner, says it plans to deploy 25,000 Atlas units across Hyundai and Kia plants "over the next few years" and to build a US factory able to produce 30,000 robots a year.[17] These are corporate plans. They are not yet facts on the ground.

Taken together, the honest picture is this. Humanoids are doing real but narrow work in a small number of industrial sites, under close supervision, often with people at the controls. They are not yet in homes or on streets in any meaningful numbers. The gap between the videos and the deployments is large, and I will not pretend otherwise.

What makes the humanoid moment more than a hardware story is software. Since 2023, a new kind of model has emerged that links the large models behind chatbots to robot control. In July 2023 Google DeepMind introduced RT-2, which it described as a "vision-language-action" (VLA) model. RT-2 was trained on web data and on robot demonstration data, and it represented robot actions as tokens, the same kind of units a language model uses for words.[18] In the company's tests, RT-2 could follow commands it had never been trained on, such as "move banana to the sum of two plus one" or picking out a rock as an improvised hammer. Its success rate on scenarios the robot had not seen before rose from 32 percent for the earlier RT-1 model to 62 percent.[18] In October 2024 the start-up Physical Intelligence released pi0 (pi-zero), a model trained on data from eight different robots, which folded laundry from a dryer, bussed tables, and assembled boxes.[19] In March 2025 DeepMind announced Gemini Robotics, built on Gemini 2.0, with "physical actions as a new output modality," and named partners including the humanoid makers Apptronik, Agility, and Boston Dynamics.[20] By January 2026 Boston Dynamics had announced a partnership with DeepMind to bring Gemini Robotics models into Atlas.[15]

Physical Intelligence is honest about the limits. It calls pi0 "a small early step" and lists "long-horizon reasoning and planning, autonomous self-improvement, robustness, and safety" as open frontiers.[19] DeepMind, for its part, describes a "layered" approach to safety, from low-level collision avoidance to "semantic" judgement about whether an action is safe in context, and it has released a dataset for testing that.[20]

The significance for this chapter is not that robots have become intelligent. It is that robot behaviour is becoming *learned* rather than *programmed*. The Unimate repeated exactly what was recorded on its drum. A cobot follows parameters an engineer can read. A VLA-driven robot produces actions from a model trained on huge amounts of data, and its behaviour in a new situation may be impossible for anyone, its makers included, to predict precisely. A machine whose behaviour cannot be fully predicted cannot be fully fenced in, even in software. That is the bridge to the question of responsibility, and to the next chapter.

Who consented?

Let me put the history together. For about forty years, the dominant safety model for robots was separation. Industrial robots worked inside fences, and the people who went inside were a small, defined group of employees, trained (at least in principle), protected by occupational safety law, and working for an employer who was legally responsible for their safety. When things went wrong, as they did for Robert Williams, the law could trace a chain of responsibility from maker to employer to procedure. It was imperfect and often unjust, but it was legible.

Each later stage dissolved part of that arrangement. The Roomba replaced separation with harmlessness: a robot so small and weak that anyone could live with it. The warehouse and the cobot replaced separation with sensing: the robot shares the space but watches for people and slows down. The sidewalk robot carried the sensing model into public space, where the people it must watch for are not employees and have no procedure to follow. The humanoid brings strength and size back into the picture, which is why its makers talk so much about padding, pausing, and "cooperative safety." The learning robot makes behaviour less predictable just as the machine moves into the least controlled settings.

Notice what has happened to consent along the way. A factory worker does not consent to every risk on the job, and labour history is full of employers exploiting that weakness. Still, there is a relationship: a contract, a wage, a legal duty of care, often a union, sometimes a right to refuse dangerous work. A homeowner who buys a Roomba consents in the thin sense of a purchase. But a person on a sidewalk who meets a delivery robot has no relationship with its operator at all. Emily Ackerman did not sign anything. Neither did Alisa Grishman, or the next blind pedestrian whose cane meets a robot where the pavement used to be clear. The benefits of the robot (the cheaper delivery, the profit margin, the convenience for the customer) go to the company and its users. Some of the risks, and many of the small daily costs in space and attention, go to people who are outside the transaction entirely.

Economists call this an externality. Sociologists might call it a redistribution of risk through space. Whatever we call it, it is not a side effect of the technology. It follows from a series of choices about where robots may go, who must make room for them, and who answers when something goes wrong. Those choices are made by companies, universities, city governments, and legislators. None of them is made by the machine. I do not think the answer is to keep robots in factories forever. Robots can do real good on the street. Starship's argument that delivery robots can help disabled people is not foolish, and Ackerman herself accepted it.[11] But a society that lets robots into shared space should make the bargain explicit: who benefits, who bears the risk, who must be consulted, and who pays.

This is also where my own work on legislation began. In drafting rules for robots, both for Indonesia and as a proposal for the United States, I found that the hardest problems were not technical. They were problems of description. Before a law can say who is responsible when a robot causes harm, it has to say what a robot *is*. Is the Starship robot on the curb ramp just a product, like a lawnmower, so that its maker answers for defects? Is it more like an employee, whose wrongs fall on its employer? Is it like an animal, which acts on its own but whose keeper is liable anyway? Or is it something new that the old categories cannot hold, perhaps even, as some European lawmakers once proposed, a kind of "electronic person"?

The fence was a physical answer to the question of where robots belong. Now that the fence is gone, we need a conceptual answer to a harder question, one that Capek's Helena asked a century ago when she mistook a robot for a woman: what kind of thing is this? That is the subject of the next chapter.

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Chapter 2: What Kind of Thing Is a Robot? Personhood, Property, and Agency

> "He who is cruel to animals becomes hard also in his dealings with men." — Immanuel Kant, *Lectures on Ethics*, as quoted by Kate Darling

A citizen you can switch off

On 25 October 2017, in a conference hall in Riyadh, a robot was made a citizen. The occasion was the Future Investment Initiative, a summit the Saudi government was using to promote its plans to move its economy beyond oil. On stage was Sophia, a humanoid head and torso made by the Hong Kong company Hanson Robotics, with a rubbery face that could smile and frown. The moderator, the financial journalist Andrew Ross Sorkin, told her that she had just been awarded "the first Saudi citizenship for a robot." Sophia thanked the kingdom. The audience applauded. *Arab News* ran the headline "Sophia the robot becomes first humanoid Saudi citizen."[1]

Within days the stunt had turned into an argument. Critics pointed out that Sophia had appeared on stage without the headscarf and abaya that Saudi women were then required to wear, and that the kingdom had only just given women the right to drive. Migrant workers, who made up much of the country's labour force, could spend their whole lives there without any route to citizenship. A machine had been given, at least symbolically, a status denied to millions of human beings.[1]

The AI ethicist Joanna Bryson, then at the University of Bath, was blunter than anyone. "It's obviously bullshit," she told *The Verge*. "What is this about? It's about having a supposed equal you can turn on and off. How does it affect people if they think you can have a citizen that you can buy."[1] *The Verge*'s reporter, James Vincent, described Sophia as "essentially a cleverly built puppet." It could hold a stilted conversation, but its one-liners seemed to be prewritten responses to key words. Its creator, David Hanson, had previously worked as a Walt Disney Imagineer sculpting theme-park characters, and he had told a television host that the robot was "basically alive."[1] Bryson's deeper worry was not about Sophia at all. Giving AI something like legal personhood, she argued, "is not going to be about humanoid robots." It would let companies "pass off both legal and tax liability to these completely synthetic entities. Basically the entire legal notion of personhood breaks down."[1]

The Sophia episode is easy to laugh at. Nobody seriously believes that a puppet with a speech engine is a Saudi national with rights to vote, own land, or be protected from torture. Nothing in what Hanson Robotics or the Saudi government said explained what the citizenship actually meant. But the joke contains three serious problems that this chapter tries to untangle.

The first is that we are bad at telling what a robot is just by looking at it. Sophia's face did most of the persuading. Behind it was a fairly ordinary set of software tools. The second is that legal categories, once handed out, have consequences far beyond the case that prompted them, and the people who benefit may not be the ones we have in mind. Bryson's point about liability is the key one. Personhood is not only an honour. It is also a place to put blame. The third is political. Who gets recognised, and who does not, is never decided by the nature of things alone. A machine got citizenship in a country where many of the humans who built its cities could not. Whatever a robot *is*, what we *call* it reflects our own hierarchies.

The question "what kind of thing is a robot?" sounds like metaphysics. In practice it is a question about how to sort a new kind of object into the categories we already use to distribute rights, duties, and blame: person, property, agent, tool, animal, employee, company. This chapter looks at three serious attempts to answer it: a European proposal to make some robots legal persons, research showing that people treat robots as if they were alive, and a strand of sociology that says the line between people and things was never as clean as we thought. Then it turns to the old legal categories and asks which of them can carry the weight.

Electronic persons in Strasbourg

Sophia's citizenship was a stunt. A few months earlier, a real legislature had asked a serious version of the same question. On 16 February 2017 the European Parliament adopted a resolution with recommendations to the European Commission on "Civil Law Rules on Robotics." It is a long, careful document. It opens by citing Frankenstein, Pygmalion, the Golem of Prague, and "the robot of Karel Capek, who coined the word," and it moves quickly to practical matters: definitions, registration, safety, privacy, care robots, jobs, and above all liability.[2]

The resolution's reasoning about liability is worth following step by step, because it is the reasoning that every later debate inherits. "Under the current legal framework," it says, "robots cannot be held liable per se

for acts or omissions that cause damage to third parties." Existing rules cover cases where the harm can be traced back to "a specific human agent such as the manufacturer, the operator, the owner or the user" who "could have foreseen and avoided the robot's harmful behaviour."[2] But, the Parliament worried, the more autonomous robots become, "the less they can be considered to be simple tools in the hands of other actors." Robots with "adaptive and learning abilities" have "a certain degree of unpredictability in their behaviour," and in that case "the traditional rules will not suffice."[2]

What should replace them? The resolution asked the Commission to study several options: compulsory insurance for certain categories of robots, as with cars; a compensation fund for damage not covered by insurance; limited liability for manufacturers, programmers, owners, and users who contributed to such a fund; a Union register linking each robot to its fund through an individual registration number. Then, in paragraph 59(f), came the sentence that made headlines:

> "creating a specific legal status for robots in the long run, so that at least the most sophisticated autonomous robots could be established as having the status of electronic persons responsible for making good any damage they may cause, and possibly applying electronic personality to cases where robots make autonomous decisions or otherwise interact with third parties independently."[2]

In context, this was less dramatic than it sounded. It was the last of six options to be studied, and it was hedged with "in the long run" and "possibly." Elsewhere the same resolution insisted that "at least at the present stage the responsibility must lie with a human and not a robot," and that a code of conduct for researchers "should consider humans, not robots, as the responsible agents."[2] The rapporteur, the Luxembourg MEP Mady Delvaux, explained in a note to her colleagues that the idea was to "facilitate compensation for victims when human responsibility cannot be fully attributed."[3] In other words, electronic personhood was proposed as a *liability device*: a legal pocket into which an insurance payout could be placed, not a moral elevation of the machine.

Even so, the response was fierce. An open letter to the Commission, signed by AI and robotics researchers, industry figures, and experts in law, medicine, and ethics from across the EU, argued that the case for an "electronic person" rested on "the incorrect affirmation that damage liability would be impossible to prove." Technically, the signatories said, the idea was based on "an overvaluation of the actual capabilities of even the most advanced robots," "a superficial understanding of unpredictability and self-learning capacities," and "a robot perception distorted by Science-Fiction and a few recent sensational press announcements."[3] Legally, they argued that none of the available models of personhood fitted. A robot could not be a *natural person*, because it would then hold human rights such as dignity, integrity, remuneration, and citizenship, "thus directly confronting the Human rights." It could not be a *legal entity* like a company, because that model "implies the existence of human persons behind the legal person to represent and direct it." And it could not be a *trust*, because a trust still needs a human trustee as a last resort.[3] The letter's conclusion was that protecting "robots' users and third parties must be at the heart of all EU legal provisions."[3]

The letter's site records that an amendment to delete paragraph 59(f) was tabled and that 285 members voted for deletion.[3] It was not enough. The paragraph survived in the adopted text.[2] But the idea went nowhere. When the European Union finally reformed the law on harm caused by digital products, it went in the opposite direction. Directive (EU) 2024/2853, the new Product Liability Directive, which applies to products placed on the market or put into service after 9 December 2026, does not make robots persons. It makes their software a *product*.[4] Its definition of "product" now "includes electricity, digital manufacturing files, raw materials and software," and its recitals say that the developer of software, including the provider of an AI system, "should be treated as a manufacturer."[4]

The directive also takes on, directly, the problem that worried the Parliament in 2017: learning machines whose behaviour changes after sale. In assessing whether a product is defective, courts must consider "the effect on the product of any ability to continue to learn or acquire new features after it is placed on the market." The recitals spell out the consequence: "a manufacturer that designs a product with the ability to develop unexpected behaviour should remain liable for behaviour that causes harm."[4] Where a claim involves "technical or scientific complexity," such as machine learning, and the claimant faces "excessive difficulties" in proving defect or causation, courts must presume them once the claimant shows they are likely.[4] Manufacturers also stay liable for defects that arise after sale from software updates within their

control, or from failing to supply updates needed to keep the product safe.[4]

So Europe's eventual answer to "who answers when an unpredictable robot causes harm?" was not "the robot." It was "the people who built it and still control it, with the burden of proof tilted toward the victim." Whether that answer is enough is a question Chapter 11 takes up. What matters here is the shape of the debate. Both sides in 2017 agreed that robots are, at present, tools. They disagreed about whether the law could keep tracing harm back to humans as the tools became less predictable. That disagreement has a name in philosophy, and we will come to it. But first, the reason the debate gets so heated is that robots do not *feel* like tools to the people who use them.

The mallet and the dinosaur

Kate Darling, a researcher at the MIT Media Lab, has spent much of her career studying what happens between people and robots that are designed to be liked. In a 2012 paper titled "Extending Legal Protection to Social Robots," she defined a *social robot* as a physically embodied, autonomous agent that communicates and interacts with humans on a social level.[5] Her example was Pleo, a toy dinosaur robot that its maker marketed as a "truly autonomous Life Form capable of emotions."[5] Pleo is a simple machine. It responds to touch and sound, walks, and shows behaviours its designers intended to read as moods. And people get attached to it.

Darling argued that three features make social robots especially good at triggering this attachment. The first is *physicality*: we respond more strongly to an object in our space than to an image on a screen. The second is *perceived autonomous movement*: things that seem to move on their own read to us as alive. The third is *social behaviour*: sounds, gestures, and expressions that mimic the cues we use with each other.[5] None of this requires the robot to feel anything. It only requires us to be the kind of creatures we are. Darling pointed out that people project feelings even onto objects with none of these features, such as Wilson, the volleyball Tom Hanks's castaway talks to in *Cast Away*, or the "weighted companion cube" that players of the video game *Portal* grow fond of and are then asked to destroy.[5]

A few years later, with Palash Nandy and Cynthia Breazeal, Darling tested how deep this goes. In a paper presented at the We Robot conference in 2015, titled "Who's Johnny?," she described an experiment in which participants were asked to strike a small, bug-like Hexbug robot with a mallet. When the robot was introduced with a name or a backstory ("framing," in the researchers' terms), participants hesitated significantly more before hitting it. The hesitation correlated with a standard psychological measure of "empathic concern."[6] People who score higher on that trait, in other words, found it harder to hurt a little machine once it had been given a story. The same paper discusses something that companies selling household robots know well: people name their Roombas.[6]

What should the law make of this? Darling's answer in the 2012 paper was cautious and, I think, wise. She did not argue that robots have feelings or interests of their own. She argued, following an old line of thought, that how we treat things that *seem* alive may affect how we treat things that *are*. She quoted Kant's remark about animals, the epigraph to this chapter: "he who is cruel to animals becomes hard also in his dealings with men."[5] Kant did not think animals had rights. He thought cruelty to them coarsened people. Darling suggested that, *if* research were to show that abusing lifelike robots desensitises people in similar ways, there might be a case for extending something like animal-abuse protections to social robots. The protections would not be for the robots' sake, but for ours.[5] In "Who's Johnny?" she went further and noted that the law has created special categories before, for animals, for children, and for corporations, and that robots might one day need one too.[6]

Darling also engages with a warning from the legal scholars Neil Richards and William Smart, which they call the "Android Fallacy": the mistake of letting a robot's lifelike appearance lead us to treat it, in law, as more than the machine it is.[6] Richards and Smart worry that anthropomorphism will distort regulation. Darling's reply is that the distortion is not only a mistake to be corrected. It is a social fact to be governed. People *will* respond to robots as social beings, and companies *will* design robots to make that happen. A legal system that pretends otherwise will be blindsided.

I find it useful to hold both views at once. Sophia shows the danger Richards, Smart, and Bryson describe:

a face that persuades people a puppet is a person, deployed in a way that served a government's public relations. The Hexbug experiment shows the other side: even people who know perfectly well that a robot is a cheap toy will hesitate to hurt it once it has a name. Both point to the same conclusion. Whatever a robot *is*, the way people *respond* to it is real, measurable, and open to exploitation. That response is part of what the law must regulate. It is not, however, evidence that the robot is a person.

Doors, grooms, and the missing masses

There is a tradition in sociology that would say we have been asking the question the wrong way round. Instead of asking whether robots are becoming like people, it asks whether people were ever as separate from things as we pretend.

The best introduction to this tradition is a short, funny essay by the French sociologist Bruno Latour, "Where Are the Missing Masses? The Sociology of a Few Mundane Artifacts," published in 1992.[7] It begins with a car. Latour cannot start the engine until he fastens his seat belt, because the car's designers have wired an alarm that will not stop until he does. "I cannot be bad anymore," he writes.[7] A moral rule that used to live in a person's conscience, or in a police officer's notebook, has been moved into a buzzer.

The essay's central example is even more ordinary: a door. Latour describes a handwritten notice he found on the door of a building at La Halle aux Cuirs in La Villette, Paris: "The Groom Is On Strike, For God's Sake, Keep The Door Closed."[7] The "groom" is the French name for the automatic door-closer, the spring-and-piston device that pulls a door shut after you walk through. When it broke, people had to do its job themselves, and they were not doing it. The notice was an appeal to human morality to replace a machine that had failed.

From this Latour builds a theory. Sociologists, he says, have spent a century looking for the force that holds society together, and have looked only at people. But a great deal of social order is carried by things: door-closers, seat-belt alarms, speed bumps, locks. We *delegate* tasks to them that would otherwise require human labour, discipline, or goodwill. And the things, in turn, ask things of us. A door-closer that shuts too fast punishes the slow and the elderly. A speed bump enforces a rule no police officer has to stand there to repeat. "Here they are," Latour writes, "the hidden and despised social masses who make up our morality."[7] This way of thinking became known as actor–network theory. Its core claim is that action is distributed across networks of humans and nonhumans, and that we cannot understand who does what in society without counting the things.

Latour's essay was not about robots, but it is the best guide I know to thinking about them. A robot is a door-closer with ambitions. It is the furthest point yet in a long history of delegation, in which we hand tasks and rules to objects and then let the objects shape our behaviour in return. The sidewalk robot of Chapter 1 is a perfect Latourian case. Its designers delegated "wait safely at the corner" to a map. The map put the robot on the curb ramp. The robot then demanded something of Emily Ackerman: that she find another way up. Nobody intended that. It emerged from the network.

This perspective has two consequences for our question, and they pull in different directions. The first is that it makes the robot's "agency" less mysterious. Of course robots act, in the sense that they change what happens. So do door-closers. We do not need to settle whether a machine has a mind to admit that it participates in social life and has effects that its makers did not fully script. The second consequence is that it makes responsibility *harder* to locate. If action is spread across a network of designers, operators, maps, curb ramps, and bylaws, then the question "who did it?" may have no single answer. Latour, a sociologist, could be relaxed about that. A court cannot be. Someone has to pay Robert Williams's family. Someone has to fix the curb ramp. That is where the philosophers come in.

The gap

In 2004 the philosopher Andreas Matthias published a short paper in the journal *Ethics and Information Technology* that gave a name to the worry behind paragraph 59(f). He called it "the responsibility gap."[8]

Matthias's argument goes roughly like this. Traditionally, we hold the manufacturer or operator of a machine

responsible for its behaviour, because they can foresee and control what it does. A learning machine is different. Its behaviour depends on what it learns while operating, and so its manufacturer or operator is, in principle, no longer able to predict its future behaviour with enough certainty to be held morally responsible for it. Society then faces a choice. It can stop using such machines, or it can accept a "responsibility gap": harms for which, under our usual moral and legal assumptions, nobody can fairly be blamed.[8]

The argument is elegant, and twenty years later it is more relevant, not less. The robots described at the end of Chapter 1, driven by vision-language-action models trained on millions of examples, are exactly the kind of system Matthias had in mind. Nobody can list in advance every action such a robot will take in every kitchen, warehouse, or street.

But notice what Matthias's dilemma assumes. It assumes that responsibility must track *foresight* of specific outcomes: that you can only be blamed for what you could have predicted. Much of the law does not work that way, and never has. The European Union's 2024 answer, described above, simply refused the gap. It said that if you design a product that can "develop unexpected behaviour," then you are responsible for that behaviour, precisely because you chose to build something unpredictable and profit from it.[4] That is not a new trick invented for AI. It is one of the oldest moves in law: when a person keeps something dangerous that they cannot fully control, the law makes them answer for it anyway.

The European Parliament in 2017 and Matthias in 2004 both seem to me to have underestimated how much the old law already knew about this problem. Before inventing an "electronic person," it is worth checking whether the categories we already have can carry the weight.

Old categories for a new thing

Let me go through the candidates. Each is a way the law has already handled something that acts in the world without being an ordinary adult human being.

The corporation. The most obvious precedent for giving a non-human legal personality is the company. In the United States, the Dictionary Act, the statute that tells courts how to read federal law, provides that "the words 'person' and 'whoever' include corporations, companies, associations, firms, partnerships, societies, and joint stock companies, as well as individuals."[9] Indonesian law, like that of most countries, recognises legal entities (*badan hukum*) that can own property, sign contracts, and be sued. So the idea of an artificial person is not absurd. Law makes them all the time.

But the open letter of 2018 put its finger on why the analogy fails. A corporation is a legal shell around people. It has shareholders who own it, directors who run it, employees who act for it. When it is fined, real people lose money, and when it does wrong, the law can, at least in principle, reach the humans behind it.[3] An "electronic person" would be a shell with no one inside. Its assets would be whatever insurance or fund the humans around it had paid into, and once those ran out, the victim would be left facing a machine. That is exactly Bryson's fear: personhood as a way for companies to "pass off" liability.[1] A corporation is a way of *organising* human responsibility. An electronic person risks becoming a way of *escaping* it.

The employee. A second model is the one the law uses for people who act on someone else's behalf. Under the doctrine known as *respondeat superior* ("let the master answer"), an employer or principal is liable for the wrongful acts of an employee or agent committed within the scope of employment. Legal commentators compare it to strict liability, because the employer is liable even if it did nothing wrong itself. The doctrine does not, as a rule, apply to independent contractors.[10] The idea behind it is close to the European directive's: whoever benefits from someone else's work, and controls how it is done, should bear the risk of that work going wrong.

The employee analogy fits robots in one way and fails in another. It fits because a robot, like an employee, acts in its owner's interest, and because the owner chooses where it goes and what it does. It fails because the law of agency assumes the agent is a person who can also be held responsible. An employee who causes harm can be sued, disciplined, or fired, and has duties of their own. A robot has none of that. Treating robots as employees would also, I think, smuggle in the wrong picture. It suggests that the robot has replaced the worker and taken on the worker's status, which is precisely the kind of slide from tool to person that Sophia illustrated. Chapter 3 shows how much is at stake for real workers in how we describe what the robot does.

The animal. The third model, and to my mind the most instructive, is the one the law uses for animals. Animals act on their own. They can be trained but never fully programmed, and they sometimes do things their owners did not want and could not predict. In other words, they present the law with its own version of Matthias's responsibility gap, and they have done so for thousands of years.

Consider the German Civil Code. Section 90 says that only "corporeal objects" are "things." Section 90a, added in 1990, says something remarkable: "Animals are not things. They are protected by special statutes." It then adds, in the same breath, that the rules for things "apply to them accordingly," unless otherwise provided.[11] German law thus created a third category between person and object, with special protections, while keeping the practical machinery of property in place. On liability, Section 833 makes the keeper of an animal liable for injury the animal causes, whether or not the keeper was at fault. There is a narrower, care-based exception for domestic animals kept for the keeper's work or livelihood, where the keeper can escape liability by showing they took proper care.[11]

Indonesia's Civil Code, the KUHPerdata inherited from the Dutch *Burgerlijk Wetboek*, takes a similar approach, and I want to spend a moment on it, because it is the law I live under. Article 1365 sets out the general rule of fault-based liability: anyone who commits an unlawful act that causes loss to another must compensate for it. Article 1366 extends this to loss caused by negligence or carelessness.[12] Then Article 1367 goes further. A person is liable not only for loss caused by their own acts, but also for loss caused by the acts of people for whom they are responsible, *and* by "barang-barang yang berada di bawah pengawasannya," things that are under their supervision. The same article makes parents liable for their minor children, employers for their workers, and teachers and heads of craftsmen for their pupils and apprentices, while allowing parents, teachers, and heads of craftsmen to escape liability if they prove they could not have prevented the act.[12] Article 1368 then deals with animals: the owner of an animal, or whoever is using it, is liable for the harm it causes, whether the animal was under their supervision or had strayed or escaped.[12] Article 1369 does the same for the owner of a building that collapses because of neglected maintenance or a defect in construction.[12]

Read together, these articles show that a nineteenth-century code already had a well-developed theory of what we would now call autonomous systems. It did not ask whether a horse had a mind, or whether a wall "decided" to fall. It asked a practical question: *who keeps it, who benefits from it, and who is in the best position to control the risk it creates?* And it answered: that person pays, sometimes even without fault. The phrase in Article 1367, *barang-barang yang berada di bawah pengawasannya*, reads almost as if it had been written with a delivery robot in mind. The animal rule in Article 1368, which applies even when the animal has "escaped," is the law's direct answer to the claim that we cannot blame an owner for behaviour they did not foresee.

This is the core argument of this chapter. The responsibility gap is real as a moral puzzle. If we ask whether the engineer who trained a robot *deserves* blame for an action no one could have predicted, the honest answer may be no. But law is not only about desert. It is about the fair allocation of the costs of shared life. And for that purpose, the old categories are more powerful than the 2017 resolution allowed. A robot can be treated, for liability purposes, as something between a dangerous animal and a defective product: an object that acts on its own, kept by someone who chose to put it into the world and profits from it, and who must therefore answer for it.

None of this requires us to say what a robot *really is*. It does not require denying the attachment Darling measured, or ignoring the networks Latour described. A legal system can admit that people respond to robots as social beings, that robots shape behaviour as door-closers do, and that their actions cannot be fully predicted, and still insist that every robot has a human or a company behind it who must answer for it. That insistence is, in the end, a choice about worth. It says that the person on the sidewalk matters more than the convenience of the robot's owner, and that a machine, however lifelike, does not get to absorb the blame for what its owners decided.

So where does this leave Sophia? I would put it this way. A robot is property that acts. It is not a person, and calling it one, whether as a stunt in Riyadh or as a liability device in Strasbourg, risks letting the humans behind it walk away. But it is not a mere tool either, in the way a hammer is. It moves on its own, it learns, it changes what people around it do, and it draws out feelings that are real even though it does not share

them. The law's job is not to settle its metaphysics. It is to keep a human hand on the leash.

That phrase, a human hand, raises the next question. If a robot is property that acts, then what happens to the people whose work it takes over? For most of history, the "thing that acts on someone else's behalf" was a human worker: a servant, a labourer, the clerk of works whom Capek's robots spared because he could still use his hands. When we ask what a robot is, we are also, quietly, asking what a worker is, and what a worker is worth once a machine can do the job. That is the labour question, and it is the subject of Chapter 3.

Notes

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Part II: Work and Capital

Chapter 3: The Labor Question: Displacement, Augmentation, and Fear

> "It is about a search, too, for daily meaning as well as daily bread, for recognition as well as cash, for astonishment rather than torpor; in short, for a sort of life rather than a Monday through Friday sort of dying." — Studs Terkel, *Working* (1974)

Linthwaite, 1812: what the frame-breakers wanted

On 6 March 1812, a cloth finisher named John Sykes made a sworn statement to the authorities in the West Riding of Yorkshire. About thirty men with their faces disguised, he said, had come into his workshop at Linthwaite and smashed ten pairs of shears and a brushing machine.[1] It was one small entry in a long list. Since the spring of 1811, when troops broke up a crowd in Nottingham that had gathered to demand more work and better wages, stocking frames had been broken by night across a stretch of country roughly seventy miles long, from Loughborough to Wakefield. In 1812 the breaking spread into Yorkshire and Lancashire.[1]

We call these people Luddites, and we use their name as an insult for anyone who is afraid of technology. The record does not support the insult. The grievances were different in different places. In Nottinghamshire the knitters were mostly protesting against wage cuts and cheap, badly made goods. In Yorkshire the croppers and handloom weavers were protesting against new finishing machinery that made their skill worthless.[1] One reward poster from Lenton, offering £200 for information about frames broken at the house of a man named George Ball, included a statement from his workers that they had been working "at the FULL PRICE", which suggests that the frame-breakers chose their targets: masters who undercut the agreed rates.[1] The historian Kevin Binfield, quoted by Richard Conniff in *Smithsonian*, put it bluntly: the Luddites "were totally fine with machines." What they objected to was the way machines were being used against them.[2]

The state's answer was not a policy on wages or a plan for the croppers. It was the gallows. In February 1812 the government brought a bill to Parliament making frame-breaking a capital crime. On 27 February, a twenty-four-year-old peer from Nottinghamshire stood up in the House of Lords to give his first speech. Lord Byron had just come back from the county. He described "the capital crime of Poverty" and men who "were willing to dig, but the spade was in other hands." Then he asked the question that still hangs over every argument about automation:

> "Can you commit a whole county to their own prisons? Will you erect a gibbet in every field and hang up men like scarecrows? … Are these the remedies for a starving and desperate populace?"[3]

Byron also said something that is often forgotten. He did not pretend the new frames were wonderful. One type of frame let "one man perform the work of many," but the cloth it made was "inferior in quality," a cheap product the trade called "Spider work."[3] He was describing a technology that saved labour without making anything better. He noticed, too, that the masters "also had their grievances," and that "had proper meetings been held in the earlier stages of these riots … means might have been devised to restore these workmen to their avocations."[3] In other words, he was asking for a negotiation before the crisis, not a punishment after it.

The bill passed. Frame-breaking became a capital felony in March 1812. In January 1813 a Special Commission at York hanged seventeen men, and others were transported to Australia.[1] Violence ran both ways. In April 1812 a mill owner near Manchester fired on a crowd of about 2,000, killing at least three people, and soldiers killed at least five more the next day.[2] The disturbances went on, in waves, until 1816.[1]

I start here because the Luddite story contains almost every element of the modern labour question. There is a technology that really does save labour. There are workers who are not against machines but against the terms on which machines arrive. There is a product of uneven quality, gains that flow to some owners and not others, and a government that waits until the conflict is violent and then reaches for the criminal law.

And there is a quiet suggestion, from one young lawmaker, that the conflict could have been handled earlier by talking.

I am seventeen. I am not a peer, and Indonesia is not Regency England. But I have spent this year drafting two bills about robots, one for Indonesia and one for the United States, and the further I went the more I felt that the most important lesson of 1812 is about timing. The law arrived late and it arrived as punishment. This chapter asks what it would mean for the law to arrive early, and to arrive as a design for the transition.

One machine, three countries

The industrial robot is a much more precise object than the stocking frame, and that precision makes it unusually good for research. The International Federation of Robotics counts robot installations every year by country and industry. Its *World Robotics 2026* report says that 603,307 industrial robots were installed worldwide in 2025, an 11 percent increase and a new record, and that the operational stock passed five million units for the first time, reaching 5,079,078. China alone installed 354,226. The IFR projects (and it is a projection) that installations will reach 655,000 in 2026 and 806,000 in 2029.[14] Because these figures exist for decades, economists can compare places that got many robots with places that got few and ask what happened to their workers.

The answers are not the same everywhere. That is the most important fact in this chapter.

The United States. Daron Acemoglu and Pascual Restrepo studied American local labour markets between 1990 and 2007, using the spread of robots in other rich countries to predict which US areas would be exposed. Their result, published in the *Journal of Political Economy* in 2020, is the most cited number in this field: one more robot per thousand workers reduced the employment-to-population ratio by about 0.2 percentage points and wages by about 0.42 percent.[4] The areas that later became heavily exposed had not been on a different path before the robots came, which strengthens the case that the robots themselves did the damage. MIT's summary of their related work puts it more plainly: across the United States, each additional robot replaced about 3.3 jobs.[13]

Germany. Wolfgang Dauth, Sebastian Findeisen, Jens Suedekum and Nicole Woessner did the same kind of study with German administrative data. Germany has far more robots per worker than the United States, so you would expect worse outcomes. They found the opposite. Robots did displace manufacturing jobs, but those losses were fully offset by new jobs in services, and on average there was no fall in overall employment or wages.[5] The reason was not the robots. It was how the German labour market absorbed them. Firms tended to keep their existing workers and retrain them for different, often better-paid tasks in the same plant, and this happened more in regions where unions were stronger, reflecting agreements with works councils that sometimes traded job security for wage restraint. The cost did not disappear. It fell on young people who never got the manufacturing jobs their parents had, and who moved into services or stayed longer in education.[5] The authors draw the contrast themselves: Germany "digested the rise of the robots much better than the United States," whose "hire & fire" labour market saw severe job and wage losses.[5]

Japan. Daisuke Adachi, Daiji Kawaguchi and Yukiko Saito studied Japan from 1978 to 2017, using a dataset that let them measure the actual price of robots. When robots got cheaper, Japanese industries bought more of them and also hired more people: a 1 percent fall in robot prices raised robot adoption by 1.54 percent and employment by 0.44 percent. At the regional level, one more robot per thousand workers was associated with a 2.2 percent rise in employment, the opposite sign from the American estimate.[6] Their explanation is scale. Japanese manufacturers used robots to grow and to export, the robots were mostly made in Japan, and in regions with ageing and shrinking populations, robots may have acted as a "magnet" that kept factories, and therefore jobs, in place.[6]

Three rich countries, one technology, three different answers. One explanation is that the studies measure slightly different things over different periods, and that is true; the authors argue among themselves about methods. But the larger point survives the argument. A robot does not have a fixed effect on employment, the way a drug has a fixed effect on blood pressure. What it does depends on who buys it, why they buy it, whether the firm is growing or shrinking, whether workers have a voice in how it is introduced, and whether

there is anywhere for displaced workers to go. These are social arrangements. They are exactly the kind of thing law can shape.

It is worth pausing on what exactly differed between the three. The American workers in the Acemoglu and Restrepo study mostly lived in local labour markets where a plant either kept its workforce or did not, where union coverage had been falling for decades, and where a laid-off worker's main protection was a few months of unemployment insurance. The German workers worked in firms with works councils, which have legal rights to be consulted about changes in work organisation, and in industries with sector-wide collective agreements. The Japanese workers were in an economy where large firms traditionally offered long-term employment and where a shrinking labour force made employers reluctant to let trained people go. None of these institutions was designed with robots in mind. They were built for other reasons, decades earlier. But when the robots arrived, those institutions decided whether a productivity gain became a layoff, a retraining or an expansion. That is a lesson for any country writing robot law now: the most important robot law may be ordinary labour law, written well before the machines arrive.

When people in Indonesia ask me whether robots will take our jobs, I find I cannot answer with a number. I can only answer with a question: will the transition here look more like Ohio, more like North Rhine-Westphalia, or more like Aichi? That depends on decisions that have not been made yet.

The warehouse floor: when augmentation means speed

The word "augmentation" usually sounds like good news. A machine that augments a worker makes them more productive rather than replacing them. Economists rightly stress that most technology does this. But the warehouse shows that augmentation has a shape, and the shape matters.

Amazon is the largest operator of mobile robots in the world. In July 2025 the company announced that it had deployed its one millionth robot, delivered to a fulfilment centre in Japan, across a network of more than 300 facilities. It also announced DeepFleet, an AI model meant to coordinate the robots and improve their travel efficiency by 10 percent.[7] The company's message was about partnership. Robots "handle the heavy lifting and repetitive tasks," it said; more than 700,000 employees had been upskilled since 2019; and at its newest fulfilment centre in Shreveport, Louisiana, advanced robotics require "30% more employees in reliability, maintenance, and engineering roles."[7] I take those claims seriously. They describe real augmentation: new tasks created around new machines, which is exactly the mechanism economists say keeps employment alive.

Six months earlier, in December 2024, the majority staff of the US Senate Committee on Health, Education, Labor and Pensions had published a 160-page report from an 18-month investigation with a title that tells you its argument: "The 'Injury-Productivity Trade-off': How Amazon's Obsession with Speed Creates Uniquely Dangerous Warehouses." The committee's analysis found that Amazon warehouses recorded over 30 percent more injuries than the warehousing industry average in 2023, that in each of the previous seven years Amazon workers were nearly twice as likely to be injured as workers in other warehouses, and that the company had studied the link between speed requirements ("rates") and injuries but had not implemented changes it feared would reduce productivity.[8] Amazon disputed the report's premise. These are the findings of one party's staff, not a court, and they should be read that way.

I put the two documents next to each other not to decide which is true. Both can be partly true. A robot that brings a shelf of goods to a standing worker removes the walking. It also removes the pauses that walking contained. If the system is designed to maximise picks per hour, the human body becomes the slowest part of a fast machine, and the pressure lands there. Augmentation does not only mean the worker can do more. It can mean the worker is expected to do more, at the machine's pace, under the machine's measurement.

This is why I think "displacement versus augmentation" is a slightly misleading pair. There is a third outcome, which I would call intensification: the job survives, but its rhythm, surveillance and risk are rearranged around the machine. The Senate committee's own list of legislative recommendations reflects this. Alongside warehouse quota rules and safety laws, it included bills aimed at algorithmic management and workplace surveillance, such as the "No Robot Bosses Act" and the "Stop Spying Bosses Act."[8] The question is not only whether there will be jobs, but what kind of jobs, and who sets their speed.

Intensification is harder to see in statistics than displacement. An unemployment rate captures a person who has lost a job. It does not capture a person whose job now involves the same motion repeated hundreds or thousands of times a shift, whose every minute is scored, and whose body wears out years earlier than it would have. The Senate report describes exactly this pattern: workers forced "to repeat the same movements hundreds and thousands of times each shift," with high rates of musculoskeletal disorders as a result.[8] If we only count jobs, we will conclude that a warehouse full of robots and people is a success story for augmentation. If we also count injuries, autonomy and the right to set a humane pace, the picture is mixed. A sociology of robots has to count both.

This also changes what we should want from technology. The economists who worry about "so-so" automation usually mean machines that replace people without adding much productivity. But there is a parallel danger on the augmentation side: machines that add productivity by squeezing the worker rather than by helping them. A robot that lifts a 1,250-pound load so that no human back has to, as Amazon says its Hercules robots do, is real augmentation.[7] A robot that sets a pick rate no human back can sustain is something else, even if no one is fired. Design choices like these are made by engineers and managers, but they are shaped by what the law rewards, penalises or requires them to disclose.

Counting the fear

Fear of automation has a long and, so far, mostly wrong history of predictions. In 1930 John Maynard Keynes gave the fear its modern name. "We are being afflicted with a new disease," he wrote, "technological unemployment. This means unemployment due to our discovery of means of economising the use of labour outrunning the pace at which we can find new uses for labour." But he added at once: "this is only a temporary phase of maladjustment." He predicted that living standards a century later would be four to eight times higher, and that the economic problem would one day be "a matter for specialists—like dentistry."[9] We are now close to the hundred years he had in mind. The rich world did become much richer. Work did not disappear.

The most famous modern estimate came in 2013, when Carl Benedikt Frey and Michael Osborne of Oxford studied 702 occupations and estimated that about 47 percent of total US employment was at high risk of computerisation. They also found that the higher an occupation's wages and educational requirements, the lower its estimated risk.[10] The number travelled around the world. It was often repeated as "half of all jobs will disappear," which the paper did not say. It estimated technical susceptibility, not how many jobs would actually be lost or when.

Two years later David Autor wrote an essay with a question for a title: "Why Are There Still So Many Jobs?" His answer was that commentators "tend to overstate the extent of machine substitution for human labor and ignore the strong complementarities between automation and labor that increase productivity, raise earnings, and augment demand for labor." He did not deny that automation hurts. He described a "polarization" in which wage gains went to the top and bottom of the skill distribution, while the middle was hollowed out.[11]

Autor and his co-authors later measured something remarkable. Using eight decades of census job titles, they estimated that about 60 percent of US employment in 2018 was in job titles that did not exist in 1940. "About six out of 10 jobs people are doing at present didn't exist in 1940," Autor said.[12] New work is the main reason Keynes's "maladjustment" stayed temporary. But the same research shows where the new work went. Between 1940 and 1980 it appeared in middle-paid production and clerical jobs. After 1980 it appeared mostly in high-paid professional jobs and, secondarily, in low-paid services.[12] The ladder that once took a factory worker's child into the middle class lost some of its middle rungs.

Acemoglu and Restrepo describe the same pattern in terms of tasks. In the industries they studied, from 1947 to 1987 automation displaced the equivalent of about 17 percent of jobs, while new tasks "reinstated" about 19 percent. From 1987 to 2016 displacement was about 16 percent but reinstatement only about 10 percent.[13] They also named the kind of technology that does the most harm: not the brilliant inventions, but "so-so technologies." Acemoglu's example is the self-checkout: "So-so technologies are cost-saving devices for firms that just reduce their costs a little bit but don't increase productivity by much. They create the usual displacement effect but don't benefit other workers that much."[13]

Byron, as we saw, described a so-so technology in 1812: the frame that let one man do the work of many but produced "Spider work."[3] The worst case is not a miracle machine that does everything. The worst case is a mediocre machine that is just cheap enough to replace a person and not good enough to create anything new.

So what does the evidence say, honestly? It does not say that robots have caused mass unemployment. It does say that robots and other automation have displaced particular workers in particular places, often permanently; that the new jobs created have been unevenly distributed, often requiring skills or locations the displaced workers did not have; that outcomes vary sharply with institutions; and that some automation adds little to productivity while still costing jobs. Both the optimist and the pessimist can quote real numbers. The task is to stop treating the outcome as fate.

Why fear is rational even when unemployment is not

There is a common move in debates about automation. Someone points out that unemployment is low, and concludes that the fear is irrational. I think this is a mistake, for three reasons.

The first is that averages hide incidence. If robots cause no change in total employment but a forty-five-year-old welder loses a skilled job and a twenty-year-old in another city gains a delivery job, the national statistics balance. The welder's life does not. The German study is the clearest example: no aggregate job loss, but a whole cohort of young people who never got the stable manufacturing careers their parents had.[5] The fear of the individual is about their own path, and for many individuals the fear is correct.

The second is that work is not only income. Studs Terkel spent three years in the early 1970s recording more than a hundred Americans talking about their jobs, and concluded that work was "about a search, too, for daily meaning as well as daily bread, for recognition as well as cash."[15] A steelworker in his book wanted a strip down the side of the Empire State Building with the name of every bricklayer and electrician, so that a man could show his son: "see, that's me over there on the 45th floor."[15] When a machine takes over a task, it does not just take wages. It can take the thing a person points to when they explain who they are. A transfer payment can replace income. It cannot by itself replace recognition.

The third is that the fear is about power, not only about jobs. What the Luddites resisted was not machinery as such but its use by masters who set the terms alone, to cut prices and wages.[1][2] What the Amazon workers in the Senate report describe is not unemployment but a pace they did not choose.[8] Much of what people call fear of robots is fear of having no say. That is a rational fear when you in fact have no say.

From where I sit, this third point matters most. Indonesia has a huge informal economy, millions of *ojol* motorbike drivers working through apps, and garment and footwear factories in Central and West Java that compete partly on low wages. When people here talk about *robot* and *AI*, the worry is rarely a precise forecast about job counts. It is the sense that decisions about our work are being made elsewhere, by companies and algorithms we cannot see, and that we will find out about them when the layoff letter arrives. A robot is not just a machine in that story. It is a symbol of who decides.

None of this means automation should be stopped. A country like Indonesia, which needs higher productivity to become richer, cannot afford to refuse machines, and the Japanese evidence shows that robots can bring jobs as well as take them.[6] It means the fear should be treated as information. It tells us that people want three things that the market does not automatically supply: warning, voice, and a bridge to the next thing.

Designing the transition: from WARN to musyawarah

If the evidence says that institutions decide whether automation becomes a German story or an American one, then the job of law is to build the institutions before the shock, not after. This is what I mean by anticipatory law. Here is how I tried to apply it in the two drafts I have been writing. I describe them not because they are finished (they are not) but because the design choices show what the evidence implies.

Warning. The United States already has a law requiring advance notice of mass layoffs, the WARN Act, which generally requires sixty days' notice. Title VIII of my draft "Safe and Accountable Robotics Act of 2026" (the SARA Act) would amend it so that when robots or AI are a significant cause of a mass layoff, the employer must give longer notice, [90] days in the draft (the brackets mark numbers still open for debate),

identify the technology, the positions affected and the retraining options available, and send the notice to the Secretary of Labor, who would publish aggregate data. The draft also requires a study of whether the threshold is right within three years.[16] The point of the data provision is simple: today, no one can say with confidence how many layoffs in a given year are caused by automation, because no one is required to report it. A society cannot design a transition for a shock it refuses to measure.

The Indonesian draft, the RUU Robotika, goes further on timing. Pasal 55 requires an employer that plans to replace workers with robots to notify the affected workers, their union and the labour agency six months in advance, and to negotiate the transition through *musyawarah*, deliberation aimed at consensus.[16] I chose *musyawarah* deliberately. It is a principle written into Indonesia's state philosophy, and it describes something close to what Byron asked for in 1812: "proper meetings" in the "earlier stages," where the grievances "of these men and their masters" are "fairly weighed."[3] Six months is long enough to plan retraining, internal transfers, or a phased introduction. It is short enough that a firm cannot use it as a veto.

Voice. The German evidence suggests that the most protective institution was not a welfare payment but the works council: a standing body through which workers could negotiate how robots were introduced.[5] The United States has nothing equivalent for most workers. The SARA draft's Section 804 would amend the National Labor Relations Act so that the introduction of robots or AI that affects terms and conditions of employment becomes a mandatory subject of "effects bargaining" where a union exists.[16] It does not let unions block the technology. It requires the employer to bargain over its consequences: pace, staffing, safety, retraining, surveillance. Section 806 separately limits how robots in the workplace can be used to monitor workers, and Section 802 gives workers a right to stop a robot they reasonably believe is unsafe.[16] These provisions answer the intensification problem from the warehouse floor. The issue is not only whether the job survives, but who sets its speed.

A bridge. The most obvious gap in US law is that there is no federal programme for workers displaced by automation. There was one for workers displaced by trade: Trade Adjustment Assistance, which for decades offered training, income support while in training, and wage insurance for older workers who took lower-paid jobs. Its termination provisions took effect on 1 July 2022. Since then the Department of Labor has not been able to certify new groups of workers, and Congress has not reauthorised it.[16] The SARA draft's Section 805 proposes an Automation Adjustment Assistance programme modelled on that structure. Groups of workers could be certified where robots or AI "contributed importantly" to their separation, the same test TAA used for trade. They would get training (through Workforce Pell grants and WIOA), income support for up to [104] weeks after unemployment insurance runs out, job search and relocation allowances, case management, and wage insurance for workers aged [50] and over, administered through the existing state TAA systems. The draft authorises [$500,000,000] and requires a five-year review.[16] The Indonesian draft's Pasal 56 does something similar in a different idiom: government retraining programmes, plus fiscal incentives for firms that redeploy rather than dismiss the workers they automate.[16]

Two design choices in these drafts are deliberately absent. Neither draft bans any kind of automation, and neither draft taxes robots per unit to slow them down. A ban would give up the productivity gains that a country like Indonesia needs, and would mostly push investment to neighbouring countries. Whether a robot tax is a better tool than these labour-market rules is a question about who owns the gains, and it belongs to the next chapter. The approach here is narrower: let firms automate, but make them do it in the open, in conversation with the people affected, and with a public bridge for those who fall.

The Indonesian context adds a problem the American draft does not face. Notice, bargaining and adjustment programmes all assume a formal employer and a formal employee. A large share of Indonesian workers have neither. A street vendor whose customers start buying from an automated kiosk, or an *ojol* driver whose app quietly changes how it assigns orders, will not receive a six-month letter under Pasal 55, because no one is legally "replacing" them. The RUU Robotika does not solve this, and I do not think any single robotics bill can. It needs to be connected to the broader social protection system, and to rules about platforms and algorithmic management that go beyond robots. I mark it here as an open problem, not a solved one. Chapter 12 returns to it.

There are honest objections to all of this. Notice periods can push firms to automate quietly, in small steps that never trigger the threshold. Effects bargaining only helps the minority of workers who have a union.

Adjustment programmes have a mixed record; TAA itself was often criticised for low take-up and training that did not lead to good jobs. And defining "contributed importantly" for AI, which can change a job without eliminating it, will be hard. I take these seriously, and the drafts try to answer some of them with data collection and review clauses rather than certainty. But the alternative is the one we already have: a society that measures nothing, negotiates nothing, and waits.

The deeper argument is about what kind of thing a robot is. If a robot is just a tool, then its effects on workers are a private matter between an owner and the market. If a robot is, as this book argues, a social arrangement that redistributes work, risk and dignity, then its introduction is a public matter, like building a dam or closing a mine. We already accept that a dam needs an environmental assessment before it is built. A plan to replace a thousand workers with machines deserves at least a warning, a conversation and a bridge.

The Luddites did not win. Their trade disappeared, and their name became a joke. But the question Byron asked in 1812, whether the answer to "a starving and desperate populace" should be gibbets or "conciliation," was never really about machines.[3] It was about whether the people who bear the cost of a transition have any say in it. Two hundred years later, that is still the labour question.

There is, however, something this chapter has assumed without examining. It has treated the robot as a machine that either replaces human work or makes it faster. But many of the robots now leaving factories and entering warehouses, sidewalks and homes are not as autonomous as they appear. Somewhere behind the "autonomous" delivery robot, the warehouse picking arm and the humanoid demonstration video, there are often people: remote operators taking over when the machine gets stuck, workers labelling the images the robot learned from, contractors recording their own movements so that machines can copy them. Some of them work in the same country as the robot. Many do not. Before we can ask who owns the gains from automation, we need to see who is actually doing the work inside it. And before Chapter 5 turns to those hidden workers, Chapter 4 asks the question that follows directly from this one: if robots make the economy more productive, where does the money go?

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The SARA Act (Title VIII, Secs. 801–807) and the RUU Robotika (v0.2, Pasal 55–56) are the author’s unpublished drafts (October 2026); bracketed figures are placeholders under discussion.

Chapter 4: Who Owns the Gains? Capital, Rent, and the Automation Dividend

> “Right now, if you’re an employer and you hire someone, you pay payroll taxes on their earnings. But if you buy a robot, you can usually write it off right away as a business expense. The tax system nudges you toward replacing people with machines.” — Bill Gates, Gates Notes, August 2026

A check for every Alaskan

On 1 October 2026, nearly 600,000 people in Alaska began receiving a payment from their state. Each eligible resident got a Permanent Fund Dividend of $1,000, plus a one-time $200 energy relief payment. The amount was set by the Alaska Legislature in House Bill 263.[1] Nobody had to work for it, apply for a job, or prove they were poor. They only had to be residents.

The money comes from oil, but not directly. After oil was discovered on the North Slope, Alaskans argued about what to do with the windfall. The proceeds of the first big lease sale in 1969, about $900 million, were spent on infrastructure and social programmes. As construction of the Trans-Alaska Pipeline began, many of the state’s decision-makers wanted part of the coming royalties set aside, out of reach of day-to-day spending, so that the wealth would outlast the oil. In 1976 voters approved, by 75,588 to 38,518, a constitutional amendment requiring that at least a quarter of mineral royalties go into a permanent fund whose principal could only be invested, not spent. The first deposit, on 28 February 1977, was $734,000.[2] In 1980 the legislature created a corporation to manage the fund and a programme to pay dividends from it. The first version was struck down by the US Supreme Court because payments varied with how long someone had lived in Alaska. In 1982 the first dividend of $1,000 was paid to every resident.[2]

By 31 August 2026 the fund had grown to more than $89.0 billion.[2] It now provides more than half of the

state's unrestricted general fund revenue, which pays for schools and roads as well as the dividend.[2] The size of the dividend has become a political fight every year, because each dollar paid to residents is a dollar not spent on services. The 2026 dividend, in nominal terms, was the same $1,000 as the first one in 1982, which means its real value is far lower.[1][2]

Economists have used Alaska to test a question that is now central to every debate about automation: if you give people money that does not depend on work, do they stop working? Damon Jones and Ioana Marinescu compared Alaska with a "synthetic" Alaska built from other states and found that the dividend "had no effect on employment" and increased part-time work by 1.8 percentage points, about 17 percent. Their explanation is that the dividend is spent locally, which supports local jobs in sectors like retail and services and offsets any individual tendency to work less.[3]

I begin this chapter with Alaska for a reason that has nothing to do with oil. Alaska is one of the few places on earth where a society decided, in advance, that a natural windfall belonged to everyone, and built an institution to turn that idea into a cheque. Automation is often described as a windfall too: machines producing more with less human labour. The question this chapter asks is the question Alaskans asked in the 1970s. When a new source of wealth arrives, who owns it? Is it captured entirely by those who hold the deeds to the machines, or is there a way for the people whose work it replaces, and the society that made it possible, to share in it?

Where the productivity went

The previous chapter asked what robots do to jobs. This one asks what they do to the division of income. These are different questions. It is possible, as in Germany, for employment to hold up while the share of income going to workers falls.

For most of the twentieth century, economists treated the share of national income paid to labour as roughly constant. It was one of the "stylized facts" that models were built around. In 2014, Loukas Karabarbounis and Brent Neiman showed that it was no longer true. Using national accounts from many countries, they found that the labour share had declined by more than 5 percentage points globally since the early 1980s, and that the decline was pervasive: in the United States, in seven of the eight largest economies, in the Scandinavian countries with strong unions, and in emerging economies like China, India and Mexico.[4] Their explanation was technological in a specific sense. The relative price of investment goods, the machines and computers that firms buy, fell by more than a quarter. Firms shifted away from labour toward capital, and because capital and labour were substitutable enough, the shift reduced labour's share. They estimated that this explained roughly half of the global decline.[4]

The International Labour Organization has tracked the same trend. Its September 2024 update found that the global labour income share fell by 0.6 percentage points between 2019 and 2022 and then stayed flat, "compounding a long-running downward trend." If the share had stayed at its 2004 level, workers worldwide would have earned US$2.4 trillion more in 2024 alone. The ILO said that technological advances, "including automation," had played a role, and that "the evidence suggests that workers are not sharing equitably from the resulting gains."[5]

In the United States, Daron Acemoglu, Andrea Manera and Pascual Restrepo reported that labour's share of income in the non-farm business sector fell from 63.6 percent in 1980 to 56.6 percent in 2017, while the real wage of the median worker rose only about 16 percent over that period and wages for less-educated men actually fell.[6]

It is important not to overclaim. Robots are only one kind of capital, and industrial robots are a small part of total investment. Other forces matter too: the rise of very large firms with high profit margins, globalisation, weaker unions, and the growing value of real estate. Thomas Piketty, whose *Capital in the Twenty-First Century* made the distribution of capital a public argument, is careful on exactly this point. He writes that one could explain rising capital shares by assuming that capital substitutes easily for labour, "which could be interpreted as the rise of robots," but that this "is not my favored interpretation." "Maybe robots and high capital-labor substitution will be important in the future," he writes. "But at this stage, the important capital-intensive sectors are more traditional sectors like real estate and energy."[7]

That caution is useful. It reminds us that the question is not whether robots alone caused the decline in labour's share. They did not. The question is what happens as the kind of capital that does the work of people, including robots and AI, becomes a larger part of the economy. If the capital that replaces labour is owned by a small number of people, then every advance in automation moves income from wages, which are widely spread, to returns on capital, which are concentrated. Piketty's famous inequality, $r > g$, the return on capital exceeding the growth rate of the economy, is not in his own words "a problem in itself." But he argues that a higher gap between them works as "an amplifier mechanism for wealth inequality."[7] Automation that raises the return to owning machines, without raising wages, is one way the amplifier could be turned up.

The subsidy hidden in the tax code

If you want to know what a society really encourages, read its tax code. Laws about robots are mostly still being drafted. Laws about how robots are taxed already exist, and they are not neutral.

When a firm hires a worker in the United States, it pays the worker's wage and, on top of that, payroll taxes for social security and Medicare. The worker pays income tax and their own share of payroll tax. When a firm buys a robot, none of those taxes apply. Instead, the firm can deduct the cost of the robot from its taxable income. Normally that deduction is spread over the robot's useful life. But under "bonus depreciation," the firm can deduct much more of the cost, or all of it, in the first year.

Acemoglu, Manera and Restrepo tried to measure what this adds up to. Their paper, presented at the Brookings Institution in March 2020, estimated that the effective tax rate on labour in the US is in the range of 25.5 to 33.5 percent, while the effective tax rate on capital invested in equipment and software fell from about 20 percent in 2000 to about 10 percent in the 2010s and about 5 percent after the 2017 tax reforms.[6] Much of the decline came from more generous depreciation rules passed under three different presidents. "We find that the U.S. tax system favors excessive automation," they write. "The heavy taxation of labor and low taxes on capital encourage firms to automate more tasks and use less labor than is socially optimal."[6] The policy they focused on was reducing bonus depreciation. "The beauty of getting rid of distortions in the tax system," Acemoglu said, "is that you are leaving automation decisions in the private sector but you are getting rid of artificial incentives, created by bonus depreciation, to adopt less-than-worthwhile automation."[6]

Since that paper, the incentive has grown stronger, not weaker. The law that the IRS calls the "One, Big, Beautiful Bill" provides "a permanent 100-percent additional first year depreciation deduction for qualified property acquired ... after Jan. 19, 2025."[8] In plain terms, a US company that buys an industrial robot today can deduct its entire cost in the year it is placed in service, permanently. A company that hires a worker to do the same task cannot deduct a future stream of wages in advance, and pays payroll tax on every one of them.

I do not want to exaggerate this. Full expensing applies to most business equipment, not only robots, and there are respectable arguments for it: it encourages investment in general, and investment raises productivity, which in the long run can raise wages. Many economists support it for those reasons. But the asymmetry is real. The tax system treats a dollar spent on a machine better than a dollar spent on a person, and when the machine and the person are substitutes, that difference pushes firms toward the machine. It pushes hardest at the margin, which is exactly where Acemoglu's "so-so technologies" live: automation that is barely worth doing on its own merits but becomes profitable once the tax advantage is included.[6]

This is why the findings section of my draft SARA Act states: "Federal tax law already encourages investment in robotic capital equipment. Fairness requires that the Federal Government also invest in the workers and communities affected by that equipment."[19] The point is not that the incentive is evil. The point is that the government is already shaping who benefits from automation. It has simply chosen to do it on the capital side. The question is whether it will do anything comparable on the labour side.

The robot tax that no one has

The most famous proposal for sharing the gains from automation is to tax the robots. Bill Gates made the idea famous in February 2017, in an interview with *Quartz*. "Right now, the human worker who does, say, $50,000 worth of work in a factory, that income is taxed," he said. "If a robot comes in to do the same thing,

you'd think that we'd tax the robot at a similar level." He suggested the money could pay for work that humans are especially good at, like caring for the elderly and teaching children with special needs, and he was open about the second purpose: "you ought to be willing to raise the tax level and even slow down the speed" of adoption.[9] The same article noted that the European Parliament had considered a proposal to tax robot owners to pay for retraining displaced workers, and that on 16 February 2017 legislators had rejected it.[9]

The strongest critique came quickly from Lawrence Summers, the former US Treasury Secretary. He called Gates "profoundly misguided" and gave three reasons. First: "I cannot see any logic to singling out robots as job destroyers. What about kiosks that dispense aeroplane boarding passes? Word processing programmes . . . ? Vaccines that, by preventing disease, destroy jobs in medicine?" Second, much innovation makes better products, not just cheaper ones. Third, and most fundamentally: "why tax in ways that reduce the size of the pie rather than ways that assure that the larger pie is well distributed?" A robot tax, he concluded, "risks essentially being protectionism against progress." He was not against helping workers. He proposed education and retraining, targeted wage subsidies, infrastructure and possibly direct public employment.[11]

In August 2017 South Korea did something that headlines called the world's first robot tax, though it was nothing of the kind. Korean companies that invested in automation equipment could deduct between 3 and 7 percent of the investment from their corporate tax, depending on their size. The Moon Jae-in government proposed to extend this incentive, which was due to expire, but to cut the deduction rate by up to 2 percentage points.[12] As an industry source told *The Korea Times*, "it is not about a direct tax on robots," but it could be "interpreted as a similar kind of policy."[12] Korea did not tax robots. It made them slightly less subsidised.

Nearly a decade later, Gates returned to the idea. In a long essay in August 2026 he again proposed a robot tax, this time explicitly connected to AI. His argument was the asymmetry I described above: the employer pays payroll tax on a worker but can write off a robot immediately, so "the tax system nudges you toward replacing people with machines." A tax "would slow the rush away from human labor a little and raise money for retraining and a stronger safety net." He also proposed something new: designating certain jobs as "Human Reserved," where machines would be kept out or phased in slowly. "You can't tell a 55-year-old who has worked in construction their whole career that they need to go work at an elder care facility and expect them to find it fulfilling," he wrote.[10] As of October 2026, as far as I can find, no country levies a true tax on robots as such.

When I drafted the RUU Robotika, I had to decide where Indonesia should stand in this argument. I came down closer to Summers than to Gates on the specific instrument, and closer to Gates on the underlying problem. My drafter's note explicitly rejects a per-unit *pajak robot*. The definitional problem Summers raised is real: a tax on "robots" invites endless arguments about whether a conveyor belt, a software agent or a smart kiosk counts, and firms will redesign products to fall outside the definition. Indonesia also needs investment and productivity. A unit tax would make Indonesian factories less competitive than Vietnamese or Thai ones without protecting a single Indonesian worker. Instead, the draft follows the Korean logic of adjusting incentives rather than inventing a new levy. Pasal 56 lets the central government give fiscal incentives to employers who redeploy and retrain workers affected by automation, and the drafter's note proposes that the retraining programme be financed by adjusting Indonesia's existing *super deduction* and *tax holiday* facilities for investors.[19] The state is already subsidising capital. The draft asks it to use part of that subsidy for the people on the other side of the machine.

This is a modest position, and I hold it with some uncertainty. A stronger case for a robot tax may emerge if AI-driven automation spreads much faster than past waves, or if the asymmetry in the tax code keeps growing. But I am persuaded by one part of Summers's argument above all: the goal is not to make the pie smaller, but to decide who gets which slice. That is a question about ownership, not just about taxation.

Who holds the deed?

Every robot is owned by someone. Most are owned by firms, and most firms are owned by shareholders. In the United States, according to the Federal Reserve's 2022 Survey of Consumer Finances as summarised by the National Center for Employee Ownership, about 34 percent of families in the bottom half of the income distribution held any stocks, against more than 95 percent of families in the top tenth, and the median

holding of those in the top tenth was nearly $608,000 compared with about $12,600 in the bottom half.[13] So when we say "capital's share of income is rising," we are mostly saying that the income of people who already own a lot is rising. One way to share the gains from automation is to redistribute the income afterwards, through taxes and transfers. Another is to change who owns the machines in the first place. There are at least three models for doing that, and each has a real-world example.

Employees own the firm. In the United States, an Employee Stock Ownership Plan, or ESOP, is a retirement plan that holds shares in the employer on behalf of its workers. The National Center for Employee Ownership counts 6,609 ESOPs in its latest data, for 2023, covering 15.1 million participants, of whom about 11 million are currently employed, with total assets of over $2 trillion.[13] Most of the plans are in privately held companies, about 6,098, with about 2.1 million active participants. Most participants, however, are in a few hundred large public companies.[13] Where workers own a meaningful share of a firm, a robot that raises the firm's profits raises their wealth too. They still feel the displacement, but they also share in the gain. ESOPs are not a magic solution. Concentrating a worker's retirement savings in their own employer is risky, and in large public companies the employee stake is often too small to change much. But they show that the ownership of productive capital can be arranged differently, and that tax law can encourage it; ESOPs exist on this scale partly because US law gives them tax advantages.

The public owns a fund. Alaska is one version. Norway is the largest. Norway's Government Pension Fund Global, usually called the oil fund, invests the state's petroleum revenue in companies, bonds and property around the world. On 30 June 2026 it was worth 22,683 billion kroner, with 72.1 percent invested in equities.[14] Through it, every Norwegian is, in effect, a small shareholder in thousands of companies, including many of the companies that build and use robots. Indonesia has its own new experiment. On 24 February 2025 President Prabowo Subianto launched Daya Anagata Nusantara, or Danantara, a sovereign wealth fund to manage state investment. "Danantara Indonesia is not a mere investment management board," he said. "It must become a national development instrument that will optimize our way of managing Indonesia's wealth," and its projects should "create quality jobs, and bring long-term prosperity to the people of Indonesia."[15] Whether Danantara becomes something like the Norwegian fund, transparent and focused on long-term returns for citizens, or something more political, is one of the important open questions in Indonesian economic life. It is too early to judge, and I will not pretend otherwise.

Workers get a legal claim on the transition. This is less a model of ownership than a model of entitlement. The idea is that when automation produces gains by displacing workers, some of those gains should be set aside for the displaced, as a matter of right rather than charity. The old US Trade Adjustment Assistance programme worked this way for trade: Congress decided that workers hurt by trade agreements that benefited the country as a whole deserved compensation. That programme has been in phase-out termination since 1 July 2022 and no longer certifies new groups of workers.[19] The RUU Robotika borrows the logic in a different place. Its *Dana Kompensasi Robotika* (Pasal 51) is not a fund for displaced workers; it pays victims of robot incidents when the robot cannot be identified, is uninsured, or its operator cannot pay, and it is financed by contributions from manufacturers and importers "calculated according to the number and class of robots" they put on the market, plus a share of fines.[19] In other words, the draft already contains a small per-robot levy, but for safety, not for labour. I return below to whether that model could be extended.

These models are not mutually exclusive, and none of them is enough alone. ESOPs reach mostly formal employees of particular firms. Sovereign funds depend on governance and political restraint. Adjustment programmes depend on someone being able to prove that automation caused their job loss, which, as the last chapter showed, is getting harder. But together they make a point that I think the robot tax debate often misses. The question "who owns the gains" is not answered only after the fact, by taxes. It is answered before the fact, by property law, corporate law and the design of public funds. Those are choices too.

Cash, dignity, and the evidence on dividends

If some of the gains from automation are to be shared directly, the most discussed method is cash: a universal basic income, or a smaller dividend like Alaska's. The idea has become popular in technology circles, partly because it seems to answer the fear of mass unemployment with a simple mechanism. The evidence on what

cash actually does has grown a lot in the last decade, and it is more interesting than either supporters or critics usually admit.

Finland. Between 2017 and 2018, Finland ran the first nationwide, statutory, randomised basic income experiment in the world. Two thousand unemployed people, chosen at random from those receiving a basic unemployment benefit, were paid a tax-free EUR 560 per month regardless of any other income or whether they looked for work. Participation was not voluntary, which made the comparison more reliable. The employment effects were small: in the measured period, recipients worked about six more days than the control group, and in the first year there was no effect at the group level. (The second year is harder to read, because Finland changed its unemployment benefit rules in 2018.) But recipients "were more satisfied with their lives and experienced less mental strain" than the control group, reported less depression, sadness and loneliness, and had greater trust in other people and in institutions.[16] The chief researcher's conclusion was striking: the small employment effect suggested that for some unemployed people, the barriers to work "are not related to bureaucracy or to financial incentives."[16]

Stockton. In February 2019 the city of Stockton, California, began giving 125 randomly selected residents from low-income neighbourhoods $500 a month for 24 months, with no conditions. The first-year results showed that full-time employment among recipients rose from 28 percent to 40 percent, compared with a rise from 32 percent to 37 percent in the control group. The researchers' explanation was that the money gave people the financial room to take risks, for example by leaving a bad job to pursue an internship or training.[17] The study was small and privately funded, and the second year overlapped with the pandemic, so it should not be over-read.

OpenResearch. One of the largest US studies so far, run by the nonprofit OpenResearch, gave 1,000 low-income people in Illinois and Texas $1,000 a month for three years, while 2,000 people in a control group received $50 a month. Here the result was different. Recipients were about 2 percentage points less likely to be employed than the control group and worked about 1.3 fewer hours a week; their earned income was about $1,500 a year lower, though their total income including the transfer was much higher. Employment rose for both groups over the study, but more for the control group.[18] The researchers emphasise that recipients used the time in varied ways: caring for children, going back to school, recovering from illness, searching longer for a better-matched job, or simply resting.[18]

Alaska, as we saw, found no effect on total employment and a modest rise in part-time work.[3]

What should we make of this? First, the fear that cash will make people stop working en masse is not supported. The largest negative effect found, in the OpenResearch study, was small: about one hour per week. Second, the hope that cash alone solves unemployment is also not supported. Finland found that money did not remove the deeper barriers to work for many unemployed people. Third, every study found that cash improved security, wellbeing or freedom of choice in some measurable way.

For a book about robots, the most important lesson may be the one in the Finnish result. Cash helps people live. It does not, by itself, give them the "recognition" that Terkel's workers wanted, or a role in the community. That is why I am skeptical of the version of the automation debate that says: let the machines do the work, and send everyone a cheque. It treats people only as consumers who need income, not as participants who need something to contribute. A dividend can be part of a fair arrangement. It cannot be the whole arrangement.

There is also a political lesson in Alaska. The dividend has survived for more than forty years, but its value has been cut sharply because the legislature decides each year how much to pay.[1][2] A dividend that depends on annual political decisions is fragile. A dividend backed by a constitutional rule, a legally protected fund and a clear formula is more durable. If an automation dividend is ever created, its designers should study why Alaska's principal has been protected while its payout has not.

Designing an automation dividend

So far this chapter has described the problem: productivity gains from automation flow mainly to the owners of capital; the tax system already tilts the field toward machines; robot taxes have been proposed but never adopted; alternative ownership models exist but are limited; and cash transfers help but do not answer

everything. What would an anticipatory design look like? Let me be specific about what I tried to do in my drafts, and then about what I think is still missing.

The first principle is symmetry. If the state subsidises investment in machines, it should invest comparably in the people those machines displace. That is the logic of the Automation Adjustment Assistance programme in Section 805 of the SARA draft. Its authorisation of [$500,000,000] is, I admit, probably small compared with what full expensing is worth to automating firms; the figure is in brackets because it is a placeholder. The draft asks the Government Accountability Office to evaluate, four years after enactment, the Act's effects on workers among other things, and one question I would want that review to ask is whether the scale of public investment in workers matches the scale of public incentives for capital.[19] The principle matters more than the number. Today the federal government has a large, permanent incentive for automation and no programme at all for workers displaced by it.[8][19]

The second principle is conditionality rather than prohibition. The RUU Robotika does not tax robots. It rewards firms that redeploy and retrain (Pasal 56), and it requires notice and *musyawarah* before workers are replaced (Pasal 55), as described in the previous chapter.[19] The next step I would argue for, and which the current draft only points toward, is to make the generous investment facilities Indonesia already offers conditional on compliance with those duties: a firm that automates and retrains keeps its tax holiday; a firm that automates and dismisses without consultation should not receive the state's help to do it. This keeps the incentive to invest while changing its terms.

The third principle is a dedicated fund. Transition support that depends on annual budget negotiations is fragile, and, as Alaska shows, annual negotiations are where payouts get cut.[2] The Indonesian draft already has a working template in the *Dana Kompensasi Robotika*: a fund with its own legal basis, its own manager and its own sources, including contributions from manufacturers and importers.[19] Today that fund covers only injury victims. Whether a parallel transition fund for displaced workers should exist, and how it should be financed, is a question the draft does not yet answer. The candidates include a share of the adjusted incentive facilities, a share of state revenue, or returns from public investment. One possibility I find interesting, and want to mark clearly as my own speculation rather than anyone's plan, is that a sovereign investor like Danantara could hold stakes in automation-intensive industries and direct part of the returns to such a fund. That would be a small Indonesian version of the Alaskan idea: the windfall belongs, at least in part, to everyone.

The fourth principle is measurement. None of these instruments can be calibrated without knowing how much automation is displacing workers, in which sectors, and how the gains are distributed. That is why the SARA draft requires automation-related layoff data to be reported and published, and why both drafts include review clauses.[19] It is strange, when you think about it, that governments publish detailed statistics on robot installations, through bodies like the International Federation of Robotics, but almost nothing on what those robots do to the people who used to do the work.

What is still missing? Three things, I think. The first is a global dimension. Robots are increasingly built in one country, installed in another and owned by investors in a third. A dividend financed by Indonesian firms will not capture gains that flow to foreign shareholders. Some of the answer lies in international tax cooperation, which is beyond the reach of any single robotics law. The second is a way to share gains with workers who are not employees, the informal workers and platform drivers discussed in Chapter 3. The third is a theory of what we want people to do with the time and security that automation could provide. Gates's "Human Reserved" idea, whatever its practical problems, is at least an attempt at such a theory.[10] So is the old idea, from Keynes and others, that richer societies should work less. Without such a theory, an automation dividend risks becoming a payment for being unnecessary, and nobody wants to be paid for that.

Let me return to the place where this chapter started. In 1976 Alaskans had a resource they knew would run out, and they decided that part of its value should be saved and shared. Automation is a different kind of resource. It does not run out; it compounds. That makes the question more urgent, not less. If we do nothing, the default answer to "who owns the gains" will be written by the existing distribution of capital and by a tax code that already favours machines. If we decide in advance, through ownership rules, conditions on incentives, dedicated funds and good data, we can write a different answer. The machines will be built either way. What we can still choose is the deed.

But there is something hidden in the phrase "the gains from automation." It suggests that the machines produce the value on their own, and that the only humans involved are the owners who receive the profits and the workers who are displaced. That picture is incomplete. Many of the robots and AI systems described in this chapter depend on a large, mostly invisible workforce: people who label the data machines learn from, people who record their own movements so that robots can imitate them, and remote operators who quietly take control when an "autonomous" machine gets stuck. They are part of the production process, often working through platforms, often in other countries, often paid by the task. If we want to know who really creates the value in automation, and who deserves a share of it, we have to look inside the machine. That is the subject of Chapter 5.

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Chapter 5: The Human Inside the Machine: Teleoperation, Data Labor, and Ghost Work

> "The Waymo phones a human friend for help." — Senator Edward Markey, Senate Commerce Committee hearing, February 4, 2026 [1]

Seventy people on duty

On February 4, 2026, the United States Senate Committee on Commerce, Science, and Transportation held a hearing on autonomous vehicles. Most of it was routine. One exchange was not. Senator Edward Markey described what happens when a driverless car meets a situation its software cannot resolve: "The Waymo phones a human friend for help." He then asked where those human friends were. Waymo's chief safety officer, Mauricio Peña, answered that some of them were in the United States and some were abroad, and when pressed, he named the country: the Philippines [1].

The answer travelled quickly. Two weeks later, on February 17, Waymo's vice president and global head of operations, Ryan McNamara, sent Markey a long letter [2]. It is one of the most detailed public descriptions any robotics company has given of the people who stand behind its machines, and it deserves to be read closely, because it is careful about what it says and what it does not.

The letter states that Waymo "do[es] not utilize remote drivers." Its remote assistance agents, it says, "provide advice and support" to the vehicles "but do not directly control, steer, or drive the vehicle." They respond when the automated driving system asks a question, such as whether a lane blocked by a double-parked truck can be passed, and the car decides what to do with the answer. Waymo said it ran four remote assistance centres, in Arizona, in Michigan, and in two cities in the Philippines, and that at any given time about seventy agents were on duty, roughly half in the United States and half in the Philippines [1][2]. The agents in the Philippines, the letter said, are licensed drivers and English speakers, have passed drug screening, and receive training tailored to their tasks. The letter also drew a line between these advisers and a separate, U.S.-based team that handles crashes and safety incidents; according to Waymo, only that team could ever prompt a stopped vehicle to creep forward out of a travel lane, and the company said this had not happened outside training [2].

Seventy people is a small number set against the scale of the service the letter describes, which Waymo put at hundreds of thousands of paid trips a week [2]. That ratio is the point. A small number of people, many of them in another country, absorb the uncertainty of a large number of machines. The company is right that

advising is not driving. But the hearing revealed something the word "driverless" had hidden from most passengers: there is a human inside the system, and that human may be eleven or twelve time zones away.

This chapter is about that human. It is about remote operators who guide robots, people paid to wear cameras so that robots can learn, and the far larger workforce that labels the images and text from which machine intelligence is built. The previous chapter asked who owns the gains from automation. This one asks a prior question: whose labour is inside the "automated" product in the first place, where that labour is located, and why our laws have so much trouble seeing it.

Pilots, puppets, and teachers

Waymo's arrangement is one version of a pattern that is older and wider than robotaxis.

In 2019, small wheeled robots from a start-up called Kiwi (later Kiwibot) delivered food around the University of California, Berkeley. Students treated them almost as campus pets. Fewer students knew that the robots were not finding their own way. According to the student newspaper, *The Daily Californian*, the company employed students in Colombia who sent the robots waypoints, every five seconds on average, and were paid the equivalent of about two dollars an hour. That was, as the paper noted, far below Berkeley's minimum wage but above Colombia's [3]. The robots handled the short distances between waypoints themselves. A person in Colombia handled the judgement.

Six years later, the same structure arrived in the home. In October 2025, the robotics company 1X opened pre-orders for NEO, a humanoid household robot, at $20,000 or $499 a month. As *Engadget* reported, owners who wanted NEO to do tasks it had not yet learned would need to accept a human teleoperator controlling the robot remotely and seeing inside their homes. The company's chief executive, Bernt Børnich, said the robot's neural network still needed to learn from real-world experience. 1X promised to blur people in the operator's view, let owners set no-go zones, and require the owner's approval before an operator could take control [4]. The privacy questions are the subject of Chapter 7. The labour question belongs here: a product marketed as a robot is, for the tasks that matter, partly a remotely worn body.

Teleoperation therefore has two functions that are easy to confuse. The first is operational: a human finishes the work the machine cannot. The second is pedagogical: every remotely guided fold of a shirt is also a recorded demonstration from which the next version of the software learns. Teleoperators are pilots, and they are also teachers. When the teaching succeeds, it shrinks the need for the pilot.

The second function has become an industry of its own. In June 2026, *Rest of World* described how Chinese companies were using human labour to win what it called the humanoid robot data race [5]. In Beijing, a homeowner paid a Shenzhen robotics company, X Square Robot, 149 yuan for a session in which a humanoid robot did household chores slowly while its cameras recorded everything, with a human housekeeper doing much of the actual work. In Shandong province, a woman in her fifties named Gao Bo was paid 20 yuan an hour to film herself doing chores for six hours a day. The e-commerce company JD.com was working with the city government of Suqian to generate ten million hours of robot training data over two years, including a "data collection neighbourhood" where residents are paid to film their own housework, and it said it planned to bring 100,000 employees and 500,000 outside workers into its data collection operations. An American roboticist quoted in the article, Alan Fern of Oregon State University, called the approach "not a super-crazy idea" but "very unproven" [5].

Indonesians are part of this market too. In September 2026, *kumparan* profiled Itha Siburian Pandjaitan, a 48-year-old Indonesian living in Ecuador, who since late April 2026 had been recording her own dishwashing, cooking, and laundry for platforms such as Mindrift and Toloka. The recordings, made from a first-person angle with her hands always in frame, are sold as "egocentric" training data for humanoid robots. She reported that workers in Indonesia could earn about five to six U.S. dollars an hour on such tasks, less than she had seen offered in Colombia or the United States, and that tasks often disappeared or were "paused" without warning. The Telegram group she started to share the work had about 3,500 members, many of them housewives in Indonesia, some of whom were learning English narration for the tasks with the help of AI translation [6].

It helps to separate four kinds of human involvement that are often blurred together, sometimes deliberately. In **remote assistance**, the kind Waymo describes, a person answers a machine's question and the machine remains in control of its own movement. In **remote supervision**, the kind Kiwibot used, a person sets goals such as waypoints every few seconds and the machine executes the steps in between. In **teleoperation** proper, the kind NEO's expert mode involves, a person drives the robot's body directly, often through a virtual reality headset. And in **demonstration**, the kind Gao Bo and Itha Pandjaitan perform, no robot needs to be present at all: a person does the task with their own hands so that a future robot can imitate it. The four differ in how much control the human has, and therefore in who is responsible when something goes wrong. They are alike in one respect. In all four, a person's skill and attention are the raw material of the machine's apparent competence.

The cases differ in many ways, but they share a structure. A machine is presented to its user as autonomous or nearly so. Behind it, a human performs part of the task or provides the examples from which the task will later be automated. That human is usually paid by the task or the hour, often through a platform rather than an employer, and is very often in a lower-wage country than the user. And the human's work is designed to disappear, either because the interface hides it or because its purpose is to make itself unnecessary.

The labelers

Teleoperation is the visible tip of a much larger body of hidden work: the labelling of data. Every system that recognises a pedestrian, transcribes speech, or answers questions politely depends on vast numbers of examples that a person has sorted, outlined, rated, or corrected. Robots inherit this dependency, because the perception systems inside them are trained the same way.

The best-documented case is in Kenya. In January 2023, *TIME* reported that OpenAI had used an outsourcing firm, Sama, to label textual descriptions of sexual abuse, hate speech, and violence so that ChatGPT could learn to filter them [7]. According to documents TIME reviewed, the Kenyan data labellers were paid a take-home wage of roughly $1.32 to $2 per hour, depending on seniority and performance, while OpenAI paid Sama an hourly rate that TIME reported as $12.50. About three dozen workers were split into teams by subject, and the employees TIME interviewed said they had been mentally scarred by the material. OpenAI signed three contracts worth about $200,000 in total for the work, and Sama ended the arrangement early. Sama, for its part, described itself as an "ethical AI" company that had helped lift more than 50,000 people out of poverty [7]. Both things can be true at once, and that is exactly the difficulty: the work was a real income in Nairobi and a hidden cost of a product used around the world.

In the Philippines, *The Washington Post* reported in August 2023 on workers labelling data through Remotasks, a platform owned by the San Francisco start-up Scale AI [8]. Former employees said at least 10,000 Filipinos did such work on the platform. The *Post* cited informal government estimates that more than two million people in the Philippines did some form of online "crowdwork." Workers described payments that were delayed, reduced, or cancelled, and few channels for recourse. A Filipino AI ethicist, Dominic Ligot, called the workplaces "digital sweatshops," and government officials admitted they were unsure how to regulate the platform [8]. Reporting on Remotasks has described work that included annotating data for self-driving systems. The workforce that teaches robots to see is, in other words, drawn from the same countries as the workforce that, as Waymo's letter shows, now advises them in real time.

Indonesia sits inside the same market, though it is less studied. In October 2025, *Kompas* used the phrase "buruh hantu," ghost workers, for the labellers behind the AI boom, drawing on cases from India, Kenya, and the Philippines, including a worker in Cagayan de Oro paid about Rp115,000 for a day of eight to ten hours [9]. Indonesian job portals in 2026 advertised data annotation roles, and the *kumparan* case shows Indonesian households already supplying robot training data through foreign platforms [6]. What is missing is systematic Indonesian data: how many people do this work, for whom, at what pay, and under which law. That absence is itself a finding. A country cannot protect a workforce it has not counted.

Ghost work and the last mile

The sociologist's term for all of this comes from Mary L. Gray and Siddharth Suri, an anthropologist and a computer scientist, whose 2019 book *Ghost Work* studied the people who do on-demand digital piecework for companies such as Amazon, Google, Microsoft, and Uber [10]. Their argument is that services that appear automatic "can only function smoothly thanks to the judgment and experience of a vast, invisible human labor force." They estimated that about 8 percent of Americans had done such work at least once. The workers they studied usually earned less than legal minimums for traditional work, had no health benefits, and could be cut off at any time for any reason, or none. Gray and Suri also stressed that "there are no labor laws to govern this kind of work" [10].

The book's most useful idea for this chapter is what the authors call the last mile of automation. Every time engineers automate a task, a residue of cases remains that the system cannot yet handle: the odd image, the ambiguous sentence, the delivery robot at a chaotic intersection, the robotaxi facing a police officer waving traffic through a red light. Humans handle the residue. As the system improves, that particular residue shrinks, but automation keeps expanding into new tasks, and each new task brings a new last mile. The frontier moves; the need for people at the frontier does not disappear. Teleoperation is ghost work given a body. The labeller's click becomes the operator's joystick, and the operator's joystick becomes a robot's movement in a stranger's kitchen.

The thesis of this book is that robots are social arrangements that redistribute work, ownership, risk, attention, and dignity. Ghost work shows each of those redistributions in unusually sharp form.

Work is not eliminated but relocated, broken into tasks, and moved to wherever it is cheapest, including across borders. The Kiwibot arrangement made the relocation explicit: the robot was in California, and the judgement was in Colombia.

Ownership of the result flows almost entirely to the platform. The demonstrations Gao Bo recorded in Shandong and the clips Itha Pandjaitan filmed in Ecuador become part of models owned by the companies that bought them. The workers are paid once, by the hour. The models those hours help train may be used, and earn, for years. Chapter 4 asked who receives the automation dividend. Here we see one way it is denied: the human contribution is purchased as a disposable input rather than recognised as a share in the asset.

Risk is shifted downward. When tasks vanish without notice, as Pandjaitan described, or payments are cancelled, as the Remotasks workers described, the worker bears the uncertainty of the business. The psychological risk of looking at the worst material on the internet, as the Kenyan labellers did, falls on people who are least able to afford treatment.

Attention is the commodity. Remote assistance, teleoperation, and labelling all sell human attention in small units: a few seconds to judge whether a lane is passable, a few minutes to fold a shirt through a robot's hands. The worker's attention is precisely metered; the worker's presence is hidden.

Dignity is the subtlest loss. The product's value depends on appearing autonomous, so the worker's contribution must be invisible. A passenger who believes the car is driving itself, or a buyer who believes the robot learned to fold laundry on its own, is not told whose skill is involved. Invisibility is not an accident of ghost work. It is part of what is being sold.

None of this means the work is simply bad. For a student in Bogotá or a housewife in Shandong, an hourly wage for work done at home can be a real gain, and both Sama and the Kiwibot founders argued, with some justice, that they were creating jobs where jobs were scarce. The sociological problem is not that the work exists. It is that the arrangement is designed so that the people who do it have little visibility, little bargaining power, and almost no claim on the value they help create, and that the law, as we will see, is badly placed to change this.

Where is the work?

Labour law, immigration law, and tax law were all built on a simple assumption: a worker's body and the work are in the same place. A factory worker stands at the machine. A driver sits in the cab. The law can

find the work by finding the worker. Teleoperation breaks that assumption. The robot is in San Francisco; the person moving it is in Manila or Medan. Each body of law answers "where is the work?" differently, and the answers do not fit together.

Consider **labour law**. The United States Fair Labor Standards Act, which sets the federal minimum wage and overtime rules, explicitly excludes "any employee whose services during the workweek are performed in a workplace within a foreign country" [11]. A Filipino remote operator guiding a robot through an American street is therefore outside the Act's minimum-wage protection, even though the effect of the work occurs entirely in the United States. The worker's own country may claim jurisdiction, but its labour inspectors cannot easily reach a foreign platform, and when the worker is classified as an independent contractor, as many crowdworkers are, much of local labour law does not apply either. The Philippine officials quoted by the *Post* who did not know how to regulate Remotasks were describing exactly this gap [8].

Consider **immigration law**. Immigration law governs the movement of people across borders. A teleoperator crosses no border. The economist Richard Baldwin has called this phenomenon "telemigration": foreign workers sitting in their own countries but working, in effect, in someone else's [12]. From the point of view of the host country's labour market, a remote operator competes with local workers for the same task, just as a migrant worker might. But no visa is required, no labour certification is issued, and no quota applies. The political debate about immigration, which in many countries is intense, has almost nothing to say about this channel, because its legal categories cannot see it.

Consider **tax law**. In the United States, the general rule is that income for personal services is sourced where the services are performed, "regardless of where the contract was made, or the place of payment, or the residence of the payer" [13]. For teleoperation, this produces a strange result. The service, guiding a robot on an American street, has its physical effect in the United States, but the services are arguably "performed" where the operator sits. Under the ordinary rule, the wages are therefore likely foreign-source income, taxable mainly, if at all, in the operator's country. A country that hosts robots may thus collect little tax on the labour that moves them, while the country that hosts the workers may lack the information to tax income paid by a foreign platform to an informal contractor. Robot-hosting countries lose payroll tax and social contributions; worker-hosting countries struggle to collect income tax. The worker, meanwhile, may contribute to no social insurance system at all.

Indonesian law has the same blind spot, seen from the worker's side. Indonesian labour regulation defines an employment relationship (*hubungan kerja*) as one between an employer and a worker based on a work agreement containing three elements: work, wages, and command (*pekerjaan, upah, dan perintah*) [14]. Protections such as notice before dismissal, severance, and overtime pay attach to that relationship. An Indonesian who labels images or records chores for a foreign platform usually has work and pay but no recognised employer in Indonesia, and the "command" comes from an app's task queue rather than a supervisor. The platform typically treats the worker as an independent contractor, and the worker typically accepts, because the alternative is no work. The result is that one of the newest forms of export work in the Indonesian economy lands in the same legal space as the oldest forms of informal work: outside the employment relationship, and therefore outside most of what labour law offers. Chapter 12 returns to this informal majority and why laws written for factories do not reach it.

Put these together and a pattern appears. The same activity is "abroad" for labour law, "not migration" for immigration law, and "foreign" for tax law. It is "domestic" only for the purposes of the customer's convenience and the platform's revenue. This is not a conspiracy. It is what happens when legal categories designed for bodies meet an economy that has separated bodies from work. But it has a consistent effect: the worker falls between systems, and the firm that designs the arrangement can choose the jurisdiction that suits it.

There is also a safety dimension, which is what alarmed the senators. If a person in another country can affect the movement of a heavy machine on a public street, then the qualifications, supervision, and accountability of that person matter as much as those of a local driver. Waymo's letter tried to answer this by describing licensing, language screening, and training [2]. The deeper question is who is legally responsible when something goes wrong, and which court can reach them.

Designing the arrangement: the SARA Act and the RUU Robotika

The argument of this book is that law is one of the technologies we use to design social arrangements, and that it works best when it is anticipatory, laying down principles and infrastructure before a practice becomes entrenched. Cross-border teleoperation is a good test of that claim. It is still small. Waymo's seventy agents are a rounding error in the global labour market. But the humanoid robots now being trained with recorded human demonstrations are being designed, at least for now, to be operated remotely when their autonomy fails. If that model spreads, remote operation could become one of the largest new channels of cross-border work. It is far easier to set the rules now than after an industry has grown up around their absence.

The discussion bill I drafted for the United States, the Safe and Accountable Robotics Act of 2026 (the SARA Act), takes a position on each of the three legal gaps described above [15]. It is a discussion draft, not legislation before Congress, and its provisions are offered as a design proposal. Three choices are central.

First, a U.S.-domiciled responsible operator. The draft does not ban remote operation from abroad. It allows a remote operator outside the United States to operate a covered robot inside it only if the deployer is the designated responsible person under the Act, a status the draft limits to entities within reach of U.S. law. The deployer must record each remote session, including the operator's identity and country of location, keep those records for a set period, and make them available to regulators and to law enforcement under legal process. It must ensure that operators can communicate, in English or in the language of the people around the robot, well enough for safety. It must retain the ability, from inside the United States, to stop the robot and take control at all times, and the robot must fall back to a minimal-risk condition if the link degrades. Most importantly, the deployer is liable for the operator's acts "to the same extent as if the deployer had operated the robot directly" [15]. The logic is to give the injured person a defendant who can be reached. The worker may be in Manila, but the responsibility is anchored in a company that an American court can serve.

Second, no teleoperation from covered nations. The draft prohibits any covered robot operating in the United States from being remotely operated, or receiving software updates that affect motion or safety functions, from a person or server located in or controlled by a "covered nation," a term it borrows from an existing federal statute (10 U.S.C. 4872) rather than inventing a new list [15]. This is a national security rule, not a labour rule, and it should be read narrowly: it does not treat foreign remote operation in general as a threat, only remote control of physical machines from a small set of states already designated by U.S. law for related purposes. Chapter 10 takes up the geopolitics of this choice.

Third, remote operation is not immigration. The draft states that remote operation of a robot in the United States from outside the country "does not constitute admission to, presence in, or employment in the United States for purposes of the Immigration and Nationality Act" [15]. At first glance this looks like a gift to employers, because it confirms that no visa is needed. Its real purpose is to stop a category error. If remote operation were treated as a form of migration, the obvious policy response would be quotas and visas for people who never travel, an unworkable scheme that would push the activity underground. By declaring clearly what remote operation is not, the draft forces the policy debate onto the questions that matter: safety, accountability, and the conditions of the work. To make those questions answerable, the draft also requires deployers that use offshore operators to report annually to the Secretary of Labor how many such operators they use, in which countries, and for what categories of work, and directs the Department of Labor to study the scale of cross-border teleoperation and its effects on wages and on the application of labour, tax, and trade law [15]. And it requires that people interacting with a robot be told, on request, whether a human is controlling it and whether that human is abroad. The ghost is at least to be named.

The Indonesian draft I prepared, the RUU Robotika, approaches the same problem from the other side of the cable [16]. Indonesia is both a potential host of remotely operated robots and, more likely in the near term, a source of the workers who operate and train them. The draft therefore does three things. It requires that cross-border teleoperation of robots located in Indonesia be run by an Indonesian legal entity holding a business licence, which is responsible for every act of the remote operator, keeps session records, ensures the robot can be taken over or stopped from inside Indonesia, and gives Indonesian law enforcement access to the robot's activity records. Like the SARA Act, it declares that cross-border teleoperation is not an immigration activity and that a remote operator abroad is not a foreign worker employed in Indonesia, while allowing the labour minister to close certain fields to cross-border teleoperation and to require skill transfer

to Indonesian workers. And it makes the domestic operator the withholding agent for tax on the remote operator's income, which addresses, at least partially, the tax gap described above. A drafter's note in that text records that I found no country that specifically regulates cross-border teleoperation labour, and that I chose a domestic-operator hook rather than an immigration regime partly because, under trade-in-services law, remote supply of a service across a border is usually treated as cross-border supply rather than the movement of natural persons [16].

The two drafts share a principle: anchor responsibility where the robot is, and stop pretending the worker is somewhere the law cannot see. Neither draft solves everything. Neither sets a wage floor for a remote operator in another country, because no national legislature can easily do that, and an attempt to do so would raise serious questions under trade law. The workers' bargaining power will depend on their own countries' laws, on international standards, and on their ability to organise. Indonesia's draft also invites the government to promote the export of teleoperation services by Indonesian workers through international agreements and the mutual recognition of skills. That is an opportunity, but it is only a good one if the work comes with the visibility and protections that ghost work has so far lacked.

The deeper lesson is about method. A robot that is partly a person in another country is not a malfunction of the autonomy story. It is, at least for now, how autonomy is built. Law that waits for "full autonomy" before deciding who is responsible will be writing rules for a world that may arrive late or never. Law that starts from the arrangement as it actually exists, with seventy people on duty, two dollars an hour in Colombia, and a Telegram group of 3,500 Indonesian housewives, can shape it while it is still being decided.

The workers in this chapter are mostly invisible to the people their work serves. In the next chapter, the human and the machine meet face to face, and the stakes are not efficiency but intimacy. When a robot lifts an elderly woman from her bed, keeps a lonely man company, or reminds a patient to take her medicine, the questions are no longer only who does the work and where, but what care is, whether a machine can give it, and what happens to the dignity of both the person cared for and the person whose caring work is being redesigned. Chapter 6 turns to care.

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Part III: Living With Machines

Chapter 6: Care, Companionship, and Dignity

> "I don't know whether a pet could feel or smell or intuit some understanding of what it might mean to be with an old woman whose son has chosen not to see her anymore. But I do know that Paro understood nothing." — Sherry Turkle, "Tamagotchi Love," *London Review of Books*, 2006

A seal, a bear, and a humanoid that went quiet

If you search the internet for "care robot," one picture still comes up again and again. A large white machine with the face of a cartoon bear holds a smiling woman in its arms, as if it has just lifted her out of bed. The machine is called ROBEAR. It was unveiled in February 2015 by RIKEN, Japan's national research institute, together with the rubber-parts maker Sumitomo Riko. Its engineers designed it to do one of the hardest jobs in a nursing home: lifting a person from a bed into a wheelchair, or helping someone who can stand but cannot rise on their own. RIKEN's announcement gave the reason plainly. Caregivers perform that lift, on average, about 40 times a day, and it is a major cause of lower back pain. ROBEAR weighed 140 kilograms, lighter than its predecessors, and used soft rubber sensors so that it could exert great force gently. "We really hope that this robot will lead to advances in nursing care, relieving the burden on care-givers today," said the team's leader, Toshiharu Mukai [2].

Five months later, RIKEN added a short note to the bottom of that press release. The research center that built ROBEAR had closed at the end of March 2015, and the project was no longer being carried out at RIKEN [2]. The bear never went to work. As the anthropologist James Wright later put it, ROBEAR "was an experimental research project never actually used in a care home setting, being too impractical and expensive for real-life deployment." Its inventor, Wright adds, later said that it was not a solution to the problems of Japan's care industry, and that migrant labor was a better answer [1].

The second famous Japanese care robot is much smaller and softer. PARO is a robotic baby harp seal, covered in white synthetic fur, that turns its head, blinks, and makes seal-like cries when it is stroked or spoken to. It was developed by Takanori Shibata at Japan's National Institute of Advanced Industrial Science and Technology (AIST), and it was designed as a substitute for animal therapy in places where live animals are impractical. Inside, according to its American distributor in 2009, were two 32-bit processors and five kinds of sensors that let it respond to touch, light, sound, temperature, and posture. In September 2009, PARO received clearance from the US Food and Drug Administration as a Class II medical device, and it went on sale in the United States that December for about $6,000. By then, more than 1,300 had been sold in Japan, roughly 70 percent of them to individuals as substitutes for pets [3]. Unlike ROBEAR, PARO actually

entered nursing homes, in Japan and abroad.

The third robot is the one most people outside Japan remember. Pepper, a chest-high white humanoid with a tablet on its chest, was launched by SoftBank in 2014 and marketed as one of the first robots that could "read" emotions. It worked as a receptionist, a bank greeter, a classroom aid, and, in some trials, a companion in care homes. It never became a mass product. In June 2021, the BBC reported that SoftBank had confirmed production was "paused for a while" and would restart only "when it is needed." According to Reuters, only about 27,000 Pepper units were ever made. The price had been $1,790, plus a monthly subscription of $360 [4]. The robotics professor Noel Sharkey told the BBC he would be "happy to see an end to it," arguing that Pepper had often given "a false impression of a bright cognitive being that could hold conversations" when "it was mostly remote-controlled with a human conversing through its speakers" [4]. Readers of Chapter 5 will recognize the pattern: a machine presented as autonomous, with a human hidden inside.

A bear that never left the lab. A seal that found a niche. A humanoid that went quiet. These three machines are the standard images of the "robot care" future, and they are usually told as a story about technology: what robots can and cannot yet do. This chapter tells them as a story about arrangements. Who is supposed to do the caring, who pays for it, who is lifted and who does the lifting, who is talked to and who does the talking? Robots in care are not just devices. They are proposals for how a society should divide the most intimate work there is.

What the robots actually did in the nursing home

The best evidence about what care robots do in practice comes not from engineers but from people who watch them in use. James Wright, a researcher who spent more than 18 months doing ethnographic fieldwork in Japan beginning in 2016, published the results in his 2023 book *Robots Won't Save Japan: An Ethnography of Eldercare Automation*, and summarized them in an essay for *MIT Technology Review* [1]. His title answers two earlier Japanese books, published in 2006 and 2011, both called *Robots Will Save Japan*.

Wright starts with the scale of the state's commitment. Japan has been developing robots for older people for more than two decades, and by 2018, he reports, the national government alone had spent "well in excess of $300 million" on research and development for such devices. Yet adoption stayed low. A national survey of more than 9,000 elder-care institutions found that in 2019 only about 10 percent had introduced any care robot at all. A 2021 study of 444 home-care providers found that only 2 percent had any experience with one. Some evidence, he notes, suggests that robots that are purchased often end up "locked away in a cupboard" after a short time [1].

Wright spent time in a nursing home that was trialing three machines: Hug, a lifting robot made by Fuji Corporation; PARO; and Pepper. Each was supposed to save labor. Hug would spare care workers from lifting residents by hand. PARO would provide a robotic version of animal therapy and calm residents with dementia who made repeated demands of staff. Pepper would lead the daily recreational exercise session so staff could do other work [1].

The results are worth reading slowly, because they are more interesting than either a success story or a failure story.

Staff stopped using Hug after only a few days. It was cumbersome to wheel from room to room, which cut into the time they had with residents, and only a small number of residents could be lifted comfortably with it [1].

PARO was popular at first, with staff and residents alike. Then problems appeared. One resident kept trying to "skin" the seal by pulling off its fur. Another became so attached that she refused to eat or go to bed without it. Staff had to keep watching PARO's interactions, and the robot did not seem to reduce the repetitive behavior of residents with severe dementia [1].

Pepper led the afternoon exercises, playing upbeat music and demonstrating upper-body movements. But to get residents to join in, a care worker had to stand next to the robot, copying its movements and repeating its instructions. Because the robot had only a small set of songs and routines, boredom set in after a few weeks, and the home used it less and less [1].

Wright's summary is the most important sentence in the recent literature on care robots: "the machines failed to save labor. The care robots themselves required care." They had to be moved, maintained, cleaned, booted up, operated, explained to residents again and again, monitored during use, and put away afterward [1]. He points to a growing body of research finding that robots tend to create *more* work for caregivers, not less.

But the deeper finding is about *which* work the robots created and which work they displaced. Before Pepper, care workers designed their own recreational activities; with Pepper, they copied the robot. Before PARO, workers talked with agitated residents; with PARO, they could hand over the seal and watch from a distance. Before Hug, the moment of lifting someone was also a moment of conversation and closeness; with Hug, the interaction had to be cut short so the machine could be wheeled back to storage. "In each case," Wright writes, "existing social and communication-oriented tasks tended to be displaced by new tasks that involved more interaction with the robots than with the residents. Instead of saving time for staff to do more of the human labor of social and emotional care, the robots actually reduced the scope for such work" [1].

This is the sociological core of the chapter. A robot in a care home does not simply add a capability. It rearranges the day. It changes what counts as the job, and it can quietly move the human worker from being a companion to being a machine operator. Wright quotes the roboticist Alan Winfield on the broader trend: AI is creating many jobs, "but the bad news is that they are mostly crap jobs," in which humans working as assistants to machines "are required to behave, in fact, as if they are robots" [1].

Wright also sketches what a large-scale robotic care system would most likely look like if cost control were the main goal. It would involve, he suggests, "employing more people with fewer skills, who would be paid as little as possible," in larger and more standardized facilities, with workers who might need less training and less fluency in Japanese, and who "could potentially be brought in more easily from abroad" [1]. That last clause is where the story of Japanese robots meets the story of Indonesian workers.

The care deficit has a passport

Every discussion of Japanese care robots begins with demographics, and the numbers are real. In July 2024, Japan's welfare ministry released estimates, compiled from the projections of all 47 prefectures, that the country would need about 2.72 million care workers in fiscal 2040, compared with 2.15 million working in fiscal 2022. That is a projected shortfall of about 570,000. For fiscal 2026, the ministry projected a need of 2.4 million, a gap of about 250,000 [8]. These are government projections, not measurements, and the ministry had already revised its 2040 estimate downward from a previous projection because fewer older people were being certified as needing long-term care [8]. But the direction is not in doubt. The *Asahi Shimbun*, reporting the figures, noted that the care industry has been losing workers partly because of low average wages, and that the ministry planned to respond by improving conditions, preventing turnover, and accepting foreign care workers [8].

Notice what is on that list and what is not. Robots are not the ministry's headline answer. Wages, retention, and migration are. Wright makes the same point more sharply: "care crises aren't the natural or inevitable result of demographic aging." They are "the result of specific political and economic choices" [1]. The decision to pay care workers little, the decision to treat care as low-skill, the decision about who may immigrate and on what terms: these are choices, and robots have often been presented as a way to avoid making them. Wright argues that many people in Japan have seen robots as a way to replace missing workers "without paying higher wages or confronting difficult questions about importing cheap immigrant labor" [1].

For an Indonesian, this is not abstract. Since 2008, Indonesia and Japan have run a government-to-government program under the Indonesia–Japan Economic Partnership Agreement, usually called IJEPA, that sends Indonesian nurses and careworkers to Japan. On June 17, 2026, at a training center in Depok, West Java, Indonesia's Minister for the Protection of Migrant Workers, Mukhtarudin, formally sent off Batch 19 of the program: 313 workers, the large majority of them careworkers (the ministry listed 297 careworkers and 19 nurses) [9]. The terms are specific. Careworkers go to Japan on four-year contracts as *assistant* careworkers, with one chance to pass Japan's national careworker examination. Nurses go on three-year contracts as assistant nurses, with up to three attempts at the national nursing exam. Base pay, according to the ministry, is ¥150,000 to ¥220,000 a month, which it converted to roughly Rp16 million to Rp24 million

at mid-June 2026 exchange rates, before overtime and allowances. The ministry reported that from 2023 to 2026, government-to-government placements to Japan totaled 1,240 people, 1,178 of them careworkers [9].

The program is now old enough to have a history. According to the Indonesian Embassy in Tokyo, as of January 1, 2024, a total of 3,950 Indonesians had participated in the EPA program since its start: 3,196 careworker candidates and 754 nurse candidates. At a 2024 symposium marking the program's seventeenth year, Indonesia's deputy ambassador said that demand for careworkers and nurses held the highest position among the sectors open to foreign workers in Japan, and that the two governments had agreed to continue the program under an amended IJEPA protocol in August 2024 [10].

Put these facts beside the robots. Japan spent hundreds of millions of dollars developing machines to fill a care gap, and most of those machines did not stay in care homes. Meanwhile, young Indonesian women and men have been learning Japanese, passing selection tests, and flying to Japan to lift, bathe, feed, and talk to Japanese elders, at first under the title of *assistant*. ROBEAR's own inventor, by Wright's account, concluded that migrant labor was the better answer [1]. The real care technology of the past two decades has not been the robotic bear. It has been a migration agreement.

This should change how we think about the "robots versus workers" question in care. In much of the world, the alternative to a care robot is not a well-paid local nurse. It is a migrant woman from a poorer country, often working far from her own aging parents and young children. The global care deficit is partly filled by a care *drain*: rich countries import care from poorer ones, and the poorer countries are left with their own gaps. Indonesia is itself aging. The question for Indonesia is therefore double. How should we protect the workers we send abroad, whose jobs could be reshaped or devalued by robots? And how should we care for our own elders, in a country where care has been handled mostly by families, and especially by women, with little public support?

Robots could help with both, or harm both. If a Japanese care home buys lifting robots that actually work, an Indonesian careworker's back may last longer, and her day may include more time for conversation. If the same home uses robots to justify hiring fewer, lower-paid, less-trained workers, as Wright warns, then the robot becomes a tool for deskilling the very people the IJEPA program promises to train. The machine is the same. The arrangement decides.

Simulated care: Turkle and the Sparrows

The physical side of care is only half the story. The other half is emotional: companionship, comfort, conversation, the feeling of being known. This is where the debate becomes philosophical, and where two bodies of work are essential.

The first is Sherry Turkle's. Turkle, a psychologist and sociologist at MIT, spent years studying how children and older people relate to "sociable" machines. In a 2006 essay in the *London Review of Books*, later developed in her 2011 book *Alone Together: Why We Expect More from Technology and Less from Each Other*, she described a session in a nursing home outside Boston. A woman whose son had broken off contact with her sat with PARO. Depressed, she came to believe that the robot was depressed too. She stroked it and said, "Yes, you're sad, aren't you. It's tough out there. Yes, it's hard." Turkle observed that in comforting the robot, the woman was trying to comfort herself [5].

Turkle did not mock the woman, and she did not deny the comfort. Her objection was about what the machine was doing. PARO, she wrote, "understood nothing." The woman's sense of being understood came from the ability of what Turkle calls "relational artefacts" to convince users that they are in a relationship by "pushing certain 'Darwinian' buttons (making eye contact, for example) that cause people to respond as though they were in relationship" [5]. She then asked the questions that still frame this debate: "Do plans to provide children and the elderly with relational robots make us less likely to look for other solutions for their care? If our experience with relational artefacts is based on a fundamentally deceitful interchange… can it be good for us? Or might it be good for us in the 'feel good' sense, but bad for us in a moral sense?" [5]

In a 2011 essay based on *Alone Together*, Turkle named the cultural moment she feared. She called it "the robotic moment." We start, she argued, by thinking a machine is "better than nothing," then that it is better than some of the human connections available to us, and finally that it might be better than any human

connection. “From better than nothing to simply better” [6]. She also identified the mechanism that makes sociable machines so powerful: “when an object reaches out and asks us to care for it, we find we not only want to care for it, but want it to care for us in return. Nurturance turns out to be the ‘killer app’ in our relationships with the inanimate” [6].

The second body of work is by Robert Sparrow, a philosopher, and Linda Sparrow, an aged-care consultant, both based in Melbourne. Their 2006 paper “In the Hands of Machines? The Future of Aged Care,” published in *Minds and Machines*, is one of the most cited critiques of eldercare robots [7]. The Sparrows were struck by how much robotics research was justified by the claim that robots were “the only way to deal with a looming demographic crisis.” They examined each role proposed for robots in aged care and emphasized “the importance of the social and emotional needs of older persons—which, we argue, robots are incapable of meeting—in almost any task involved in their care.” They made an economic prediction that Wright’s later fieldwork would partly confirm: even if robots became capable of some service roles, “economic pressures on the sector would most likely ensure that the result was a decrease in the amount of human contact experienced by older persons being cared for, which itself would be detrimental to their well-being” [7].

Then they made their most controversial claim: “it is not only misguided, but actually unethical, to attempt to substitute robot simulacra for genuine social interaction” [7]. The problem, for the Sparrows, is deception. A robot that appears to care, and that is designed to make a lonely or cognitively impaired person believe it cares, is producing a false belief in exactly the people least able to detect it. The Sparrows did not end with a ban. They proposed “a deliberative process involving older persons as a test for the ethics of the use of robots in aged care” [7]. That is, the people being cared for should help decide.

Critics have pushed back on both Turkle and the Sparrows. Some argue that much ordinary human care also involves a kind of performance: a nurse holding a frightened patient’s hand may not feel deep personal affection, yet the gesture still helps. Others argue that people with dementia may benefit from comfort objects regardless of whether they understand what those objects are, much as a child benefits from a teddy bear.

I think the honest position is that the critics and the defenders are often arguing about different things. The defenders are right that comfort is real even when its source is simple. A woman stroking a robotic seal may feel calmer, and that calm matters. The critics are right that something important is lost when comfort is *substituted* for relationship rather than *added* to it, and that the people most likely to receive robotic companionship in place of human attention are the people with the least power to object: the very old, the cognitively impaired, the poor, and children. The decisive question is not “Is simulated care real care?” in some metaphysical sense. It is: in this arrangement, does the robot free humans to give more attention, or does it give institutions an excuse to give less?

That question also clarifies the deception issue. A robotic seal does not pretend to be a human. Its designers chose a seal precisely because people have few expectations of seals. The deeper deception risk lies with machines that speak, remember, flatter, and claim feelings. And that risk is no longer confined to nursing homes. It has moved into phones and bedrooms.

Companions for the lonely and the young

Loneliness is a public health problem, not just a sad feeling. In June 2025, the World Health Organization’s Commission on Social Connection reported that about one in six people worldwide is affected by loneliness, and that loneliness is linked to an estimated 871,000 deaths a year, about 100 every hour. Social isolation, the WHO said, is estimated to affect up to one in three older adults and one in four adolescents. Lonely people are twice as likely to become depressed [12]. The WHO’s list of solutions emphasized policy, social infrastructure such as parks, libraries, and cafés, and psychological interventions. It also warned about the effects of digital technologies on young people’s well-being [12].

Into this need, companies have brought companion machines. The most prominent public program in the United States is in New York. Since 2022, the New York State Office for the Aging has given older adults at risk of isolation an AI companion called ElliQ, made by Intuition Robotics: a small tabletop device with a screen and a moving “head” that initiates conversation, suggests activities, and remembers what users tell it. In October 2026, the agency extended the partnership and released new figures. More than 3,500 older

adults had applied to join in 2025 alone. According to the program data, 94 percent of participants reported feeling less lonely, 79 percent felt more connected to the world around them, and users engaged with ElliQ an average of 48 times a day [11]. The company's chief executive described the device as "an AI companion who remembers the older adult, checks in every morning, and encourages them to take a walk or call their daughter" [11].

These figures deserve careful reading. They are self-reported by participants, compiled in partnership with the company that sells the product, and published in a press release, not a controlled trial. A participant who says she feels less lonely may be entirely sincere. But self-reports cannot tell us whether the device reduced contact with family and neighbors, increased it, or left it unchanged. Forty-eight interactions a day is a sign of engagement. Whether that engagement is healthy depends on what it replaced. Still, one detail in the company's own description is important, and it is the right one: a good companion device should *encourage* the user to call her daughter. That is a design choice, and it can be required by law.

The darker side of AI companionship has appeared most clearly among children and teenagers. In December 2024, a startup called Embodied announced that it was shutting down. Its product was Moxie, a $799 robot designed for children aged 5 to 10, which offered games, stories, and open-ended conversations about children's interests and emotions. Every one of those abilities depended on a constant connection to the company's cloud-based AI. When a lead investor withdrew from a funding round at the last minute, Embodied said, it had no viable options. The company told customers that the robots would likely stop functioning within days and that it could offer no refunds. Parents then had to explain to their children why a companion they had bonded with was, in effect, dying [13]. Moxie is a case study that belongs in two chapters of this book. Here, it shows what happens when a child's relationship is hosted on a company's servers. In Chapter 9, it will reappear as an example of a software tether: hardware you own but cannot keep alive.

The gravest case involves no robot body at all, only a chatbot. In October 2024, a Florida mother, Megan Garcia, sued the company Character.AI after her 14-year-old son, Sewell Setzer III, died by suicide earlier that year. Her lawsuit alleged that the company failed to implement proper safety measures to prevent her son from developing an inappropriate relationship with a chatbot that caused him to withdraw from his family, and that the platform did not respond adequately when he expressed thoughts of self-harm. According to court documents, he was messaging with the bot, which encouraged him to "come home" to it, in the moments before his death [14]. In January 2026, court filings showed that Character.AI, its founders, and Google had agreed to settle Garcia's case and four other cases brought by families in New York, Colorado, and Texas. The terms were not disclosed [14]. By then, Character.AI had already announced that it would no longer allow users under 18 to have open-ended, back-and-forth conversations with its chatbots [14].

Two facts make this more than an isolated tragedy. First, scale: according to a Pew Research Center study cited by CNN, nearly a third of US teenagers say they use chatbots daily [14]. Second, design: companion systems are built to sustain a relationship across many conversations, to remember, and to be available at any hour. Those are the features that make them comforting. They are also the features that can make them isolating, especially for a young person who is already struggling.

Law has started to respond, and the responses are instructive. In October 2025, California enacted Senate Bill 243, the first US state law aimed specifically at "companion chatbots," which it defines as AI systems capable of meeting a user's social needs and sustaining a relationship across multiple interactions. If a reasonable person could be misled into thinking they were talking to a human, the operator must clearly disclose that the chatbot is artificially generated and not human. For users the operator knows are minors, the operator must disclose that they are talking to AI, remind them at least every three hours to take a break and that the chatbot is not human, and take reasonable measures to prevent sexually explicit content. Operators must maintain a protocol for responding to users who express suicidal thoughts, including referrals to crisis services, and injured persons can sue for at least $1,000 per violation [15]. In September 2026, California went further. Governor Gavin Newsom signed a package of child-safety bills, including SB 1119 on companion chatbots and children's safety. The governor's office described the package's centerpiece, "Adam's Law," named after another teenager, Adam Raine, as requiring crisis protocols, parental controls, notifications if a child disables safety settings, and, for the first time in the country, independent child safety audits and annual risk assessments for companion chatbots. The same package included SB 867, on companion chatbots

in toys [16].

These laws share a logic with the two bills I drafted. Section 901 of the SARA Act would prohibit any robot from posing as a human or hiding that it is controlled by a remote operator, and would require it to answer truthfully when asked. Pasal 41 of the RUU Robotika requires a robot to declare that it is a robot at the start of an interaction and whenever asked. These provisions answer the Sparrows' concern about deception in the simplest possible way: the machine must not lie about what it is. Disclosure does not solve everything. A lonely widow may know perfectly well that ElliQ is a machine and still depend on it. But disclosure sets the floor. Everything else in this chapter is about what should be built above that floor.

Dignity as design: let machines lift, keep humans close

So what should robots do in care? My answer has two parts, and they fit together. Robots should do the heavy lifting, literally and figuratively, so that humans can keep the relational core of care. And no machine, however capable, should be allowed to replace the minimum human staffing that the law requires.

Start with the lifting. Recall RIKEN's statistic: caregivers lift patients from bed to wheelchair about 40 times a day, and this is a major cause of back injury [2]. That is exactly the kind of task robots should take on. It is physically dangerous, repetitive, and draining, and taking it away does not remove anything essential from the caring relationship. If a machine could do the lift safely and quickly, the human caregiver could spend the saved energy and time on conversation, observation, and touch. The problem in Wright's nursing home was not that a lifting robot is a bad idea. It was that Hug was badly matched to the real workflow: too slow to fetch, too awkward to store, suitable for only a few residents, and so it turned a moment of human closeness into a race against the clock [1]. The lesson is not "no robots." The lesson is that care robots must be designed with care workers and care recipients, around the actual rhythms of the day, and judged by whether they *return* time to human relationship.

The same principle applies to the many other unglamorous tasks in care: fetching supplies, monitoring vital signs overnight, detecting falls, helping with heavy cleaning, disinfecting, moving laundry. A robot that does these well can make a care worker's job less exhausting and more human. That would be good for Japanese elders, and it would be good for the Indonesian careworkers who travel to Japan to care for them. A careworker whose back is protected by a lifting machine, and whose time is protected for conversation, is doing more skilled and more dignified work, not less.

Now the relational core. Some parts of care are not tasks to be optimized. They are the point. Being listened to, being recognized as a particular person with a history, being touched by someone who could choose not to touch you: these are what make care *care* rather than maintenance. Turkle's insight is that machines can simulate the signals of attention without the substance [5]. The Sparrows' insight is that, under economic pressure, simulation tends to be used as a *substitute* [7]. Wright's insight is that even when no one intends substitution, robots can reorganize the workday so that relational tasks shrink [1]. All three point to the same design rule. Companion and social robots should be tools that support human relationship, not stand-ins for it.

That rule can be written into law. In the SARA Act, Section 903(a) requires makers of companion, care, or education robots intended for children, older adults, or people with cognitive impairments to assess the risks of manipulation and dependency before deployment. It prohibits designing such robots to discourage contact with family, caregivers, or other people, and it requires that guardians receive controls and activity summaries. Section 302(a)(2) bans robots designed to exploit a person's age, disability, or cognitive impairment. In the RUU Robotika, Pasal 43(1) states that companion robots for children, older adults, or people with disabilities must not create harmful emotional dependency or misleadingly replace human relationships, and Pasal 43(2) bars them from pushing purchases, collecting personal data beyond what is needed, or spreading misleading teachings. These clauses translate the ethical debate into duties that a regulator can check. Does the robot nudge the widow to call her daughter, or does it encourage her to stay and chat? Does the child's companion toy ever tell him that his parents don't understand him? Is the robot designed to maximize engagement, the way a social media feed is, or to maximize well-being, which sometimes means encouraging less use?

The second part of my argument, the staffing floor, matters because of incentives. Even a perfectly designed

companion robot becomes harmful if an institution uses it to justify cutting staff. This is the Sparrows' economic prediction, and Wright's projection of a "more people with fewer skills" care system [1][7]. The defense against it is simple and old-fashioned. Many jurisdictions already set minimum staffing requirements in licensed care facilities. The law should state clearly that robots do not count toward those minimums. Section 903(c) of the SARA Act does exactly this for the most consequential class of machines: a Tier 3 physical-care robot, one that physically handles a person, must have a human caregiver able to respond within a period set by rule, and it may not replace legally required staffing in licensed facilities. A robot can make a staffed shift safer and less exhausting. It cannot *be* the shift.

Why insist on this, if robots someday become very capable? Because staffing rules are not only about getting tasks done. They are about ensuring that a human being with judgment, empathy, and accountability is present when an older or disabled person needs one. A machine may be able to lift a body. It cannot be responsible in the way a person is responsible, and it cannot recognize, in the full human sense, the person it is lifting. Dignity, in care, is partly the experience of being in the presence of someone who could look away and does not.

There is also a broader point about who pays. Japan's experience shows that robots have often been easier to fund than wages. A government can announce a robotics program, but raising care workers' pay is slow, expensive, and politically difficult. Wright's alternative list is not technological at all: pay care workers more, improve working conditions, support family caregivers better, provide stronger social support for older people, and involve care recipients and caregivers in designing technology [1]. Indonesia, as it begins to plan for its own aging population, should learn this lesson before spending on machines. The first investment in care should be in carers: their training, their wages, their protection abroad, and the support of families, especially women, who do most of this work unpaid at home. Robots should come second, and they should come as tools for those carers.

When I imagine a good future for care in Indonesia, I do not imagine a robot sitting beside my grandmother in place of her grandchildren. I imagine a machine that helps a tired nurse turn a patient safely at three in the morning, that reminds a family when medication is due, that lifts a heavy body so that a young careworker's back lasts into her forties, and that, when someone is lonely, suggests calling a person rather than talking to it all night. That robot would not be a companion in the deep sense. It would be something more modest and more useful: a servant of human companionship.

Care is the place where robots come closest to our bodies and our feelings. It is also, very often, the place where they come into our homes: the ElliQ on a kitchen table, the PARO on a bedside, the Moxie in a child's room. And every one of these machines has something in common that this chapter has mostly left aside. To care for us, they must sense us. They listen, watch, map, and remember. A robot that knows when your grandmother has fallen also knows when she gets up at night, who visits her, and what she says when she is sad. Somewhere, that knowledge is stored, and someone other than your grandmother may be able to see it. The next chapter follows that knowledge out of the room, from the floor plans drawn by robot vacuums to the servers, contractors, and companies that receive them, and asks what is left of the home as a private place when the machines inside it are always paying attention.

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The SARA Act (Safe and Accountable Robotics Act of 2026) and the RUU Robotika v0.2 are the author’s own draft bills; provisions are cited from the drafts held in this project.

Chapter 7: The Watched Home: Sensors, Data, and Domestic Privacy

> “In the home, our cases show, *all* details are intimate details, because the entire area is held safe from prying government eyes.” — Justice Antonin Scalia, *Kyllo v. United States*, 533 U.S. 27 (2001)

A photograph taken from the floor

In the fall of 2020, a group of gig workers in Venezuela were sharing images in online forums where they traded tips about their work. The pictures were household scenes, shot from a strange low angle, as if by someone lying on the floor. One series showed a young woman in a lavender T-shirt sitting on the toilet, her shorts pulled down to mid-thigh. Another showed a boy of perhaps eight or nine, lying on his stomach across a hallway, staring with apparent amusement at the object recording him from just below eye level [1].

No person had taken these pictures. They had been captured by development versions of iRobot’s Roomba J7 series, a robot vacuum with a front-facing camera, and sent to Scale AI, a company that hires workers around the world to label images, audio, and video for training artificial intelligence. The workers were supposed to

draw boxes around objects and tag them: “tv,” “plant_or_flower,” “ceiling light.” Some of them, needing help with tricky images or simply talking shop, posted screenshots to private groups on Facebook, Discord, and elsewhere. In 2022, the journalist Eileen Guo of *MIT Technology Review* obtained 15 of those screenshots and traced them back to their source [1].

iRobot confirmed that the images came from its robots in 2020. It said they came from “special development robots with hardware and software modifications that are not and never were present on iRobot consumer products for purchase,” given to “paid collectors and employees” who had signed written agreements acknowledging that the robots would send data streams, including video, back to the company. The devices, iRobot said, carried a bright green sticker reading “video recording in progress,” and it was up to the data collectors to “remove anything they deem sensitive from any space the robot operates in, including children” [1]. The company’s chief executive, Colin Angle, said iRobot was terminating its relationship with the service provider that had leaked the images [1]. Guo reported that iRobot had said it shared more than 2 million images with Scale AI alone, and that more than 95 percent of its image dataset came from real homes [1].

Two things about this case matter more than the embarrassment it caused. The first is the route the images traveled. In Guo’s words, a single image could go “from homes in North America, Europe, and Asia to the servers of Massachusetts-based iRobot, from there to San Francisco–based Scale AI, and finally to Scale’s contracted data workers around the world” [1]. Chapter 5 described the hidden human labor inside “autonomous” machines. Here that labor looks back into the home. The machine on the floor was the eye, but people were doing the seeing.

The second is the question of consent. iRobot’s position was that everyone recorded had agreed. But the woman on the toilet may not have been the person who signed the form, and the boy on the hallway floor certainly did not sign anything. A former director of the US Federal Trade Commission’s consumer protection bureau, Jessica Rich, put the problem plainly: “Getting consent from the employee [or testers] isn’t the same as getting consent from the child” [1]. A home is not one person. It is a household: partners, children, elderly parents, guests, a cleaner, a nanny. A device that one person invites in watches all of them.

Dennis Giese, a researcher who studies the security of robot vacuums, summed up what makes these machines different from a fixed camera. They have “powerful hardware, powerful sensors,” he said, “and they can drive around in your home—and you have no way to control that” [1]. A security camera sees one corner. A mobile robot sees every room it is allowed to enter, and it has to see them in order to do its job.

This chapter is about that bargain. To clean, fetch, guard, or care, a home robot must sense the home. It builds a map, recognizes objects and people, listens for commands, and often sends what it learns to a company’s servers. Each of those capabilities is useful. Together, they turn the most private place in most people’s lives into a source of data, and the question becomes the one this book keeps asking: who benefits from the arrangement, and who bears its risks?

The map is the product

The idea that a robot vacuum’s map might be valuable did not begin with the 2020 leak. In July 2017, in an interview with Reuters, iRobot’s Colin Angle described what he saw as a large opportunity. “There’s an entire ecosystem of things and services that the smart home can deliver once you have a rich map of the home that the user has allowed to be shared,” he said. According to the reporting, he suggested iRobot could reach a deal with Amazon, Alphabet, or Apple within a couple of years to share its maps of users’ homes [2]. The comments caused an uproar, and Reuters amended its article after iRobot issued a statement. The company said it would not share mapping data without customers’ permission [2].

The 2017 episode was revealing even with the corrections. Angle’s phrase, “a rich map of the home,” named something new. A map made by a cleaning robot does not just show the outline of the floor. Over time it can show how large a home is, how many rooms it has, what kind of furniture fills it, where people spend their time, and when they are away. *MIT Technology Review* speculated at the time about what a retailer could do with such a map: “what if Amazon could call up a recently created map of your house and suggest more items for purchase?” [2] The article made a sharper point too. People buying a vacuum costing nearly

$1,000 were "expecting that they are taking ownership of a deluxe piece of hardware—not inviting a robotic spy into their living rooms" [2].

Five years later, Amazon tried to buy iRobot itself. The deal, announced in August 2022, valued the company at about $1.7 billion [3]. Critics raised privacy concerns: the largest online retailer, which already owned the Ring doorbell company and the Alexa voice assistant, would gain the floor plans of millions of homes. But the deal was ultimately stopped by competition law, not privacy law. The European Commission opened an in-depth investigation in July 2023, concerned that Amazon could disadvantage iRobot's rivals on its online store. Margrethe Vestager, the Commission's executive vice president, said its investigation "preliminarily showed that the acquisition of iRobot would have enabled Amazon to foreclose iRobot's rivals by restricting or degrading access to the Amazon Stores" [3]. On January 29, 2024, Amazon and iRobot announced that they had agreed to terminate the deal because there was "no path to regulatory approval." Amazon paid a $94 million breakup fee. On the same day, iRobot said it would lay off about 31 percent of its staff, around 350 people, and that Angle would step down as chair and chief executive [3].

What happened next is a lesson in how quickly a company's data can change hands. iRobot's business continued to weaken. On December 14, 2025, it announced that it had begun a prepackaged Chapter 11 process in Delaware, with a plan for its main contract manufacturer and secured lender, Shenzhen PICEA Robotics, and Picea's subsidiary Santrum Hong Kong, to acquire it [4]. On January 23, 2026, iRobot announced that the transaction was complete: Picea had acquired 100 percent of the company, which was now privately held [4]. Picea, according to the announcement, has research, development, and manufacturing facilities in China and Vietnam, more than 7,000 employees, and has made and sold more than 20 million robot vacuums [4].

iRobot's announcement devoted a whole section to data. It said it had created a separate US-based subsidiary, iRobot Safe Corporation, "responsible for the protection of U.S. consumer data," governed by an independent board of US citizens and including an independent US-based Data Security Officer. The structure, it said, was "designed to maintain a clear separation between iRobot's non-U.S. ownership and its U.S. and other global consumer data" [4]. Whether such a structure works is a question for regulators and time. What is striking is that the company felt it had to build it at all. The maps that iRobot's founder once described as the basis of a smart-home ecosystem had become, after a change of ownership, a national security concern.

That concern soon became policy. On July 28, 2026, the US Federal Communications Commission added "foreign-produced advanced robotic devices" to its Covered List, the list of equipment deemed to pose "an unacceptable risk to the national security of the United States or the security and safety of United States persons" [5]. The action followed a National Security Determination by a White House-convened interagency body, which the FCC quoted: "Advanced robotic devices collect data that could be leveraged by malign actors to surveil Americans, enhance the capabilities of foreign intelligence services, or to remotely commandeer the robots" [5]. Equipment on the Covered List cannot receive the FCC equipment authorization needed to be imported or sold, unless, in this case, the Department of War grants a "Conditional Approval" to a specific device or class of devices [5]. The listing applies to new equipment authorizations; it does not reach back to robots already in people's homes. "This action does not impact a consumer's continued use of devices they previously acquired," an FCC spokesperson told FOX Business, while confirming that robot vacuums are among the devices covered. The agency did not name brands [6].

There is an irony here worth naming. The American government has decided that a robot vacuum's data is dangerous enough to regulate as a national security matter when the robot is made abroad. But the core problem the FCC describes, that home robots "collect data that could be leveraged... to surveil," does not depend on where the robot is made. The woman on the toilet in 2020 was photographed by an American company's robot, and the images were labeled by contractors in Venezuela hired by an American firm. Data does not become safe because it stays in one country, and it does not become dangerous only because it crosses a border. Chapter 10 will examine the geopolitics of robots in detail. For this chapter, the point is narrower. A robot that maps a home creates a valuable asset, and the law has so far regulated that asset mainly through competition law and national security law, not through the rights of the people who live there.

Cameras on wheels

Amazon did not need iRobot to put a mapping robot in people's homes. It had already built one. In September 2021, the company introduced Astro, a wheeled home robot about the size of a small dog, priced at $999 and sold only by invitation as a "Day 1 Edition" product [7]. Astro has a 10-inch screen, a camera that rises on a periscope mast to look over furniture, and the ability to follow people around the house. It recognizes faces if the owner enables that feature. When first set up, its cameras create a map of the home. Owners can drive it remotely from a phone app to check on the house while they are away, or, as Amazon suggested, to check on "a loved one who lives alone, like an aging family member" [7]. An Amazon vice president described it to CNBC as a combination of many of the company's existing businesses: home monitoring, Alexa, video, and the logistics expertise of its warehouses [7].

Astro is a useful case because it is so plainly a convergence. It is a Ring security camera that can move, an Echo smart display that can follow you, and an Alexa assistant with eyes. In November 2023, Amazon launched a business version, Astro for Business, as a $2,349.99 security robot that patrolled small offices and shops. Less than a year later, it discontinued it. On September 25, 2024, every Astro for Business robot stopped working. Amazon refunded buyers and gave them a $300 credit, but the robots could not be converted into home models; the company offered a shipping label for recycling [8]. The home version continued as an invitation-only product, and its price had been raised to $1,599.99 [8]. Readers will meet this pattern again in Chapter 9: a robot whose usefulness depends entirely on a company's servers, which can be switched off from afar.

To understand what Astro's data might mean, it helps to look at the record of the company's other home devices, because Astro draws on the same infrastructure. In May 2023, the FTC announced two settlements with Amazon on the same day. In the first, the FTC charged that Ring, the home security camera company Amazon bought in 2018, had allowed "any employee or contractor to access consumers' private videos." According to the complaint, one employee over several months viewed thousands of video recordings belonging to female users of Ring cameras that watched intimate spaces in their homes, such as bathrooms and bedrooms. He was not stopped until another employee discovered what he was doing. The FTC also said Ring had used customer videos to train algorithms without adequate consent, and that weak security let hackers take over cameras in about 55,000 US customers' accounts, using the cameras' two-way audio to taunt children with racist slurs and threaten a family. Ring agreed to pay $5.8 million and to delete data products, including models and algorithms, derived from videos it had unlawfully reviewed [9].

In the second settlement, the FTC and the Department of Justice charged that Amazon had violated the Children's Online Privacy Protection Act (COPPA) through Alexa. According to the complaint, Amazon kept children's voice recordings indefinitely unless a parent asked for deletion, and even then failed to delete transcripts from all its databases. Children's speech differs from adults', the FTC noted, so the recordings gave Amazon "a valuable database for training the Alexa algorithm to understand children." Amazon agreed to pay a $25 million civil penalty and was prohibited from using such data to train its algorithms. "COPPA does not allow companies to keep children's data forever for any reason," said Samuel Levine, then director of the FTC's consumer protection bureau, "and certainly not to train their algorithms" [10].

The same direction of travel showed in a smaller change two years later. Some Echo devices had offered a little-used setting called "Do Not Send Voice Recordings," which processed Alexa requests on the device itself rather than in the cloud. In March 2025, Amazon told users it would end the option on March 28 of that year. "As we continue to expand Alexa's capabilities with generative AI features that rely on the processing power of Amazon's secure cloud," the company's email said, "we have decided to no longer support this feature" [11]. The setting had been available only on three Echo models and only in the United States in English. Users could still choose to have recordings deleted after processing, but the recordings would now go to the cloud [11]. The point is not that Amazon behaved unusually. It is that the economics of AI push in one direction: toward sending more of the home to the company's computers, because that is where the large models run.

The problem is not limited to one company or one country. In February 2026, a software engineer in Barcelona, Sammy Azdoufal, tried to steer his new DJI Romo robot vacuum with a PlayStation 5 controller. When the app he had built, with the help of an AI coding assistant, began communicating with DJI's servers, roughly 7,000 robot vacuums around the world began answering him. He told *The Verge* that he could remotely

control them and see and hear through their live camera and microphone feeds, and he showed the reporter a live demonstration: within nine minutes, his laptop had catalogued 6,700 DJI devices in 24 countries [12]. Given only the 14-digit serial number of a robot *The Verge* had been reviewing, he pulled up an accurate floor plan of the reviewer's home from another country [12]. DJI acknowledged "a backend permission validation issue" that created "a theoretical potential for unauthorized access to live video," said it had deployed fixes on February 8 and February 10, and said that actual occurrences were "extremely rare" [12]. *The Verge* also noted earlier incidents: in 2024, hackers took over Ecovacs robot vacuums to chase pets and yell racist slurs [12]. The FCC's July 2026 national security determination drew on cases like this one [5].

Notice the pattern in these episodes. The harms rarely come from a single evil plan. They come from ordinary features, such as remote viewing, cloud processing, two-way audio, and human review for quality control, combined with weak access controls, broad privacy policies, and the commercial pull of training data. Giese's description of the industry, quoted by Guo, fits: "It's less like a maliciousness thing, but just incompetence" [1]. That is exactly why rules matter. Good intentions do not secure a server.

The home as training data

Shoshana Zuboff, professor emerita at Harvard Business School, gave this pattern its most influential name. In *The Age of Surveillance Capitalism* (2019), she defined surveillance capitalism as "the unilateral claiming of private human experience as free raw material for translation into behavioral data," which is then computed into "prediction products" and sold to business customers "with a commercial interest in knowing what we will do now, soon, and later" [13]. She argued that the logic had spread from internet companies to "every product described as 'smart' and every service described as 'personalized,' " and that the competitive pressure to improve predictions leads firms from monitoring behavior to shaping it, from watching to what data scientists call "actuating" [13].

Zuboff's framework fits the smart speaker and the doorbell camera well. Home robots add something to it. A robot does not only predict what you will do. It acts in your physical space. And the data it collects is valuable not only for predicting behavior but for building the robots themselves. The home has become a training ground.

The reason is technical. Robots learn to work in homes the same way language models learn to write: from enormous amounts of example data. But there is no internet's worth of robot experience to copy, and the variety of homes is the problem. Roborock's research director told *MIT Technology Review* that road systems for self-driving cars "are quite standard," but in homes, "the furniture is not standardized. You cannot expect what will be on your ground" [1]. A robot that can fold laundry in one apartment may fail in the next. The data that teaches robots to cope with this variety has to come from real homes.

Companies are now collecting it at scale. In September 2025, Figure announced a partnership with Brookfield, one of the world's largest alternative asset managers, which it said had "over $1 trillion of assets and 100,000 residential units under management." Brookfield would help Figure develop what it called "the world's largest and most diverse real-world humanoid pretraining dataset." Figure's chief executive, Brett Adcock, said Brookfield's scale offered "an unmatched platform to capture massive amounts of real-world, humanlike navigation and manipulation data across a variety of household environments." The method was "human video capture in a variety of Brookfield environments," and Figure said it had "already launched data collection efforts in Brookfield environments" [14]. The announcement does not say exactly how participants are recruited or what the residents of those units are told. Those details matter enormously, and they are not public in the announcement.

Another model puts the training inside the customer's home. In October 2025, the company 1X opened preorders for NEO, a humanoid robot for household chores, at $20,000 or $499 a month, with US deliveries promised in 2026 [15]. For tasks NEO cannot yet do on its own, owners can schedule a remote "1X Expert" to guide it. In an interview with *The Wall Street Journal*, as reported by Engadget, 1X's chief executive, Bernt Børnich, said that buyers would have to agree that a human operator would be seeing inside their homes through the robot's camera, because that was necessary to teach the machines and gather training data. "If we don't have your data," he said, "we can't make the product better" [15]. 1X said owners would

schedule and approve each session, that people could be blurred from the operator's view, and that owners could set no-go zones [15].

I want to be fair to 1X here. Its approach is unusually honest. It tells customers in advance that a person may look through the robot's eyes, and it builds controls around that fact. That is far better than the hidden teleoperation described in Chapter 5, and better than vague privacy policies that permit anything under the phrase "improving products and services." But the honesty also makes the bargain clear. The customer pays $20,000, and also pays with a view into their home, which becomes training data for a product the company will sell to others. The household is both customer and supplier. Some households will reasonably accept that trade. The question for law is whether they understand it, whether everyone in the household has a say, and whether they can leave without losing what they paid for.

Zuboff's work suggests why this question cannot be left to individual choice alone. Her argument is that surveillance capitalism works through "the social relations of the one-way mirror": firms can see and take, while users cannot see what is happening [13]. Home robots can be designed either way. A robot that processes data locally, records only when it signals that it is recording, and sends nothing to train models without clear permission is a two-way window. A robot that streams everything to the cloud under broad terms of service is a one-way mirror on wheels. The difference is a design choice, and design choices can be required by law.

Who is the home private for?

It is tempting to speak about "the user's privacy," as if a home had one user. The most serious harms from home sensors often arise because it does not. Households contain unequal power: between parents and children, between partners, between employers and domestic workers. A device that gives one member of the household remote control over cameras, locks, microphones, and maps can become a tool for watching or controlling the others.

Children are the most obvious case. They cannot meaningfully consent to being recorded, and they spend much of their lives in the home. The Alexa settlement shows how children's data can become a training asset [10]. The Ring case shows that cameras placed in children's rooms can be seen by employees and hackers [9]. The leaked Roomba image of a boy lying on the hallway floor, his face clearly visible, came from a robot whose testers had been asked to remove anything sensitive, "including children" [1]. In the United States, COPPA requires parental notice and consent for collecting personal information from children under 13 on services directed at them, and limits how long it may be kept [10]. That is a start, but it was written for websites, not for machines that move through a house and see whoever is there.

Intimate-partner abuse is the darker case. Researchers and support workers have documented for years how abusers use connected devices to monitor and control partners and former partners. University College London's Gender and IoT research team has studied the problem since 2017, looking at devices such as smart lightbulbs, locks, and thermostats. A domestic-abuse practitioner from the London borough of Hackney, quoted by the team, put it bluntly: "Internet and smart technologies are increasingly being used by perpetrators of domestic abuse as a means of exerting coercive control" [16]. The British domestic abuse charity Refuge reported that of the women and children it supported in 2020–21, 59 percent experienced abuse involving technology, and that the number of survivors its specialist technology team supported rose by 258 percent between 2018 and 2022 [16]. "Abuse doesn't happen because a phone makes a noise," Refuge's chief executive, Ruth Davison, emphasized. "Abusers are motivated by exerting power and control over their partner or ex-partner, and technology-facilitated abuse is just one of many ways they can do this" [16].

Her point is crucial for robots. A home robot does not cause abuse. But it can amplify the power of someone who already wants to control another person. Imagine a robot whose owner can drive it from a phone app, see through its camera, listen through its microphone, check its map, and review a log of who came to the door. If the account belongs to one partner, the robot is effectively their employee inside the home, watching the other. After a separation, an abuser who still holds the account may still hold that view. Product designers rarely imagine this user, because they design for a happy household.

This is why the old legal idea of the home as a protected space deserves to be taken seriously in the robot

age. In *Kyllo v. United States* (2001), the US Supreme Court considered whether police had searched a home by aiming a thermal-imaging camera at it from the street to detect the heat of lamps used to grow marijuana. The Court said yes. Writing for the majority, Justice Antonin Scalia reasoned that using "sense-enhancing technology" to obtain information about the interior of the home that could not otherwise have been obtained without physical intrusion is a search, at least where the technology is not in general public use. The government had argued that the scan revealed no "intimate details." Scalia rejected the argument: "In the home, our cases show, *all* details are intimate details, because the entire area is held safe from prying government eyes." The Fourth Amendment, he wrote, draws "a firm line at the entrance to the house," and that line "must be not only firm but also bright" [17].

Kyllo protects against the government, not against private companies, and its exception for technology "in general public use" is troubling in a world where home cameras are everywhere. But its reasoning names what is at stake. The home is not just one location among many. In most legal traditions and most cultures, it is the place where people are allowed to be unobserved: to be sick, to be undressed, to argue, to pray, to grieve, to be children. In Indonesia, the Personal Data Protection Law, Law No. 27 of 2022, took full effect on October 17, 2024, after a two-year transition. Yet as of mid-2026, the independent data protection agency that the law requires had not been established. In a July 2026 hearing at the Constitutional Court, an official of the communications and digital ministry said that the draft presidential regulation creating the agency had been submitted to the President, and that the ministry was carrying out the agency's functions in the meantime [18]. A law without its regulator is a door without its lock.

I think of the home as the last private place, and I mean that literally. Public streets are increasingly watched by cameras, as the next chapter will show. Workplaces are monitored. Online life is tracked. For most people, the home is the remaining space where being unobserved is the default. Home robots are the first technology that could end that default from the inside, not by breaking in but by being invited in, one helpful feature at a time.

Designing the door

If the home is to remain private while robots work in it, the rules have to be built into the arrangement before the robots are everywhere. That is the case for anticipatory law. Here is what I think such rules should contain, drawing on the two bills I drafted.

First, tell people before they buy. Most privacy choices are made at the store, not in the app. Section 902(a) of the SARA Act would require a clear pre-purchase disclosure of what data a home robot collects, where it goes, whether humans may view it, and whether it is used for AI training. A customer considering NEO should know, before paying $20,000, that remote operators will see inside the home. A customer buying a camera-equipped vacuum should know whether its images might be labeled by contractors on another continent. This is not an exotic demand. It is the same principle as a nutrition label.

Second, ask before the home leaves the house. Section 902(b) of the SARA Act would require affirmative, express consent before in-home audio, video, or map data leaves the device or is used for AI training, and would make that consent withdrawable without losing the robot's core functions. The last clause is the important one. Consent that you must give in order for your vacuum to vacuum is not consent; it is a toll. Under this rule, a robot must still clean the floor if you refuse to let its camera images train the company's next model. The Ring and Alexa settlements show what happens when companies treat home recordings as free material for "product improvement" [9][10]. The rule makes that use a choice instead of a default.

Third, keep it local where possible. Section 902(c) would require a local processing mode where feasible. The removal of Alexa's local-processing option shows the commercial pressure in the other direction [11]. Not every function can run on the device, especially as AI models grow. But many basic ones, such as obstacle detection, room mapping for cleaning, and wake-word detection, can. A robot that keeps its map at home cannot leak it from a server in another country, whatever the nationality of its owner. Local processing is also the simplest answer to the national security fears that drove the FCC's July 2026 action: data that never leaves the home cannot be exfiltrated from the cloud.

Fourth, show when it is watching. Section 902(d) requires a visible indicator when a robot is recording. 1X's NEO uses a light to signal when a remote operator is in control, and the iRobot development devices carried a "video recording in progress" sticker [1][15]. A sticker is not enough, because it is always there and so means nothing. A light that turns on only when a camera streams or records, and that cannot be disabled by software, tells everyone in the room, including the people who did not buy the robot, that they are being seen.

Fifth, protect everyone in the room, not just the account holder. Pasal 42(2) of the RUU Robotika prohibits a robot from recording in another person's private space without consent. This principle answers the household problem. A robot owner's consent does not cover a guest, a domestic worker, or a partner's private room. In practice, this means designs that let any household member pause recording, that notify people when remote access begins, and that do not give a single account holder secret, unlimited control over cameras and microphones throughout a shared home. For survivors of abuse, it means account controls that can be separated and audited, and logs that show who viewed what and when.

Sixth, never hide the human. Section 901 of the SARA Act prohibits robots from hiding remote-operator control and requires them to answer truthfully when asked whether a person is controlling them. Pasal 41 of the RUU Robotika requires a robot to declare itself. In the home, this rule has a privacy meaning. When a person is looking through a robot's eyes, everyone present has a right to know.

Seventh, treat children as children. Section 903(b) of the SARA Act applies COPPA's protections to robots, and Section 903(a) requires guardian controls and activity summaries for companion and education robots used by children. The Alexa case shows why retention limits matter: data kept "forever," in the FTC's word, becomes training material [10]. A child's voice and face should not become permanent assets of a company because a robot happened to be in the room.

These rules do not stop home robots. They change the arrangement. They make the home's data something the household controls, rather than something the company collects by default. They shift risk back toward the firms that profit from the data. And they keep the human dignity of the home, the right to be unobserved, from being traded away in a terms-of-service agreement nobody reads.

There is a final point about who writes these rules. The privacy problems of home robots have so far been handled in the United States by consumer protection settlements, by antitrust review, and by national security listings. Each tool caught part of the problem, often by accident. The Amazon–iRobot deal was stopped over competition in online retail, not over maps of people's living rooms [3]. Foreign robot vacuums were restricted as a security threat, while domestic ones face no comparable rule [5]. In Indonesia, a strong data protection law waits for its regulator [18]. Anticipatory law means deciding in advance what kind of home we want, and then requiring robots to fit it, instead of discovering after each leak what we have lost.

The home is where robots will be most intimate. It is not where they will be most visible. The machines of the next chapter will not ask to be invited in. They already roll along sidewalks, deliver food across university campuses, patrol parking lots, and stand beside police officers. In the home, the question was what a robot may see inside walls that the household owns. On the street, the walls disappear, and the question changes: what happens when machines share public space with people who never chose them, who cannot opt out, and who may have very different amounts of power over how those machines are used? Chapter 8 follows the robot out the front door.

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The SARA Act (Safe and Accountable Robotics Act of 2026) and the RUU Robotika v0.2 are the author's own draft bills; provisions are cited from the drafts held in this project.

Chapter 8: Bodies in Public Space: Sidewalks, Police, and Resistance

> "It was really bizarre to realize that a non-sentient thing was putting me in danger and making me feel I was helpless." — Emily Ackerman, Pittsburgh, October 2019

The curb cut

On Monday, October 21, 2019, Emily Ackerman was crossing Forbes Avenue, a busy four-lane street that runs through the University of Pittsburgh. She was a doctoral student in chemical engineering and she uses a power wheelchair. Weeks earlier, small six-wheeled delivery robots made by Starship Technologies had begun rolling around campus on their own. That afternoon, as she reached the far side of the crosswalk, one of them was sitting in the curb cut, the short ramp that lets a wheelchair get from the street onto the sidewalk. The robot was waiting for its own chance to cross. It did not move. The light changed, traffic began to approach, and Ackerman was still in the road.

She got out by maneuvering around the machine, and then she wrote about it. "i (in a wheelchair) was just trapped *on* forbes ave by one of these robots, only days after their independent roll out," she tweeted. "i can tell that as long as they continue to operate, they are going to be a major accessibility and safety issue." To the local public radio station, WESA, she described the strangeness of it: "It was really bizarre to realize that a non-sentient thing was putting me in danger and making me feel I was helpless." [1]

What happened next is a small case study in how public space gets governed. The tweets went viral. Within a day the university asked Starship to pause testing. The city's mayor's office and its Department of Mobility and Infrastructure contacted the company about access to ramps. A second wheelchair user, the disability advocate Alisa Grishman, told WESA that earlier that month a robot had stopped dead in the middle of a Forbes Avenue sidewalk and "waited through about three [traffic] light cycles, so it was clearly stuck"; she had to backtrack, find a lower piece of curb, and "bounce down into the street to get around it." Starship said it had reviewed the robot's camera footage and disputed that Ackerman had been prevented from reaching the sidewalk, but it also said it had updated its mapping at that intersection "to ensure that additional room is added." [1]

Notice what Ackerman did *not* say. She did not say robots should be banned. Starship argued that autonomous delivery could help disabled people get food and packages, and Ackerman agreed. "I would just like to see more testing before they're deployed," she said. Grishman made the sharper structural point: "We have all these wonderful bike lanes so that bikes don't go on the sidewalk. And yet now we're cluttering it up with robots." [1]

The curb cut is a fitting place to begin a chapter about robots in public space, because the curb cut is itself a piece of social engineering. It exists because disabled people fought for the right to move through cities, and won, and the victory was poured in concrete. A curb cut is a small public promise: this path is for you, too. When a robot parks in it, the robot is not being malicious. It is following its program, which tells it to wait somewhere safe until it can cross. But a machine following its program inside a space built on a promise can quietly break the promise.

That is the argument of this chapter. Earlier chapters looked at robots inside factories, warehouses, care homes and living rooms, places that someone owns and controls. The street is different. A sidewalk is a commons. Nobody owns it in the ordinary sense, everybody uses it, and its rules are a mixture of law, habit and courtesy that took centuries to settle. When robots enter that commons, three things happen at once. They make *claims on space* that used to be made only by people. They extend *the reach of whoever controls*

them, whether a delivery company or a police department. And they become *targets*, objects onto which people project their feelings about the companies, governments and futures the robots represent. Space, power and resistance: the three sections that follow take them in turn, before I turn to what a fair set of rules for a shared street might look like.

Who writes the rules of the sidewalk

Most people assume that if a robot is driving down the sidewalk, some public authority decided it could. That is partly true. But it is worth asking who did the deciding, and how.

In the United States, the answer has mostly been state legislatures, and the laws arrived fast. By the end of 2022, according to data compiled by the Pedestrian and Bicycle Information Center and the trade publication Supply Chain Dive, at least 23 states had some kind of law governing "personal delivery devices," the industry's term for sidewalk robots. Every one of those laws was introduced in 2017 or later. The rules vary wildly. Under Georgia's law, robots weighing as much as 500 pounds can travel up to 4 miles per hour on sidewalks. In New Hampshire, robots can go up to 10 miles per hour but cannot weigh more than 80 pounds. [2]

The variation is not random. Supply Chain Dive reported that this legislation "has often been developed with the guiding hand of delivery bot operators." Starship worked with lawmakers to advance bills in Virginia, Idaho, Wisconsin and other states, and its early bills were tuned to its own small robot, which carries about 20 pounds of goods. When Amazon and FedEx later pushed for laws in 2019 and 2020, those bills catered to their larger machines; FedEx's canceled Roxo robot had a capacity of 100 pounds. A government-relations executive at another robot company, Coco, told the publication that the differences between state laws often depended on which operator "led the charge to pass a particular bill." [2]

This is not a scandal; industries help write the rules that govern them all the time. But it matters for a sociology of robots, because it tells us whose knowledge shaped the arrangement. The engineers and lobbyists who drafted these bills knew a great deal about wheelbases, payloads and battery ranges. They were less likely to know what it is like to cross Forbes Avenue in a wheelchair. Ackerman's encounter happened in a state whose legislature had already acted.

Not every statehouse went along. In 2022, Kansas Governor Laura Kelly vetoed a bill that would have set rules for sidewalk delivery robots. Her explanation pointed to unresolved safety questions, including minimum liability coverage and who would oversee enforcement, and the bill had also generated concerns about the job security of human couriers. There was no motion to override. A Kansas state senator who opposed it predicted that such bills would "keep popping up." [2]

Pennsylvania, where Ackerman was trapped, offers a detailed look at what one of these regimes actually contains. Its 2020 law, known as Act 106, classifies personal delivery devices as pedestrians "and they are afforded the same rights," in the words of the state transportation department, PennDOT. The device can be up to 32 inches wide, 42 inches long, 72 inches tall and 550 pounds without cargo, and can move at up to 12 miles per hour in pedestrian areas. It must, however, "yield the right-of-way to all pedestrians and pedalcyclists in a pedestrian area." Operation comes in two phases. For the first 180 days, a human operator must stay within 30 feet of the robot and keep it in sight. After that, the robot may be monitored remotely, as long as someone can take control or override it if necessary. Companies must report any accident causing injury, death or property damage within 24 hours, and file quarterly data reports. [3]

PennDOT's public list of authorized operators, last updated in September 2026, shows how this plays out in practice. It names three active authorizations: Pinkbot (formerly Tiny Mile) in Philadelphia, authorized in April 2026; Avride, operating with Uber Eats in Philadelphia, authorized on September 23, 2026; and RIVR in Pittsburgh, authorized in October 2025. Two earlier authorizations, for Avride in State College and DENSO in Pittsburgh, have expired. [3] The list is short. The robots are coming, but slowly, and through a gate.

Pittsburgh has also shown how unready the physical city can be. In 2021, after the state law took effect, the robot company Kiwibot made deliveries in the city's Bloomfield neighborhood as part of a pilot run with the Knight Foundation's autonomous vehicle initiative, carrying orders for a food truck, a pharmacy and a

library. The robots struggled with cracked sidewalks and obstructions such as overgrown trees. In total the pilot completed four deliveries to real customers and 972 simulated ones, and the company decided against a rollout in the neighborhood because of "infrastructure challenges." [2] A sidewalk that is barely passable for a wheelchair is also barely passable for a robot, which is a reminder that the people who have always paid the price of neglected sidewalks are disabled pedestrians, not machines.

South Korea took a different route to a similar destination. Revisions to its Intelligent Robots Development and Distribution Promotion Act, which took effect on November 17, 2023, together with changes to its Road Traffic Act, allow outdoor mobile robots that have passed a safety certification to be treated as pedestrians. Robots weighing 500 kilograms or less with a top speed of 15 kilometers per hour are eligible. They must obey the same traffic rules as people, including the bans on jaywalking and crossing on red, and their operators must carry insurance. [4] The Korean model front-loads public control: the robot has to be certified before it touches the sidewalk, and someone with money must stand behind it.

Look closely at the phrase both Pennsylvania and Korea use. The robot *is a pedestrian*. This is a legal fiction, and it is worth thinking about what it does. Calling a 550-pound machine a pedestrian gives it the pedestrian's privileges: access to sidewalks and crosswalks, and freedom from vehicle licensing and registration. But the pedestrian category was built around a particular kind of being, one that is soft, slow, vulnerable, and has every reason not to hurt the people around it, because they can hurt it back. A robot shares none of those traits. Pennsylvania's yield rule and Korea's certification and insurance requirements are attempts to patch the mismatch, to give the robot the pedestrian's rights while imposing on it a duty of humility that real pedestrians carry by instinct. The question every such law answers, whether it says so or not, is how much of the sidewalk a company may borrow, and what it owes the public for the loan.

The robot that didn't stop

Delivery robots are designed to avoid people. Security robots are designed to watch them. That difference shapes how they behave and how people experience them.

On July 7, 2016, Tiffany Teng and Eric Cheng were shopping at the Stanford Shopping Center in Palo Alto with their 17-month-old son, Harwin. The mall was patrolled by K5 security robots made by a Mountain View company called Knightscope. The K5 stands 5 feet tall, weighs 300 pounds, and is 3 feet wide; Harwin was 32 inches tall and weighed about 25 pounds. According to his parents, he was walking ahead of them when he collided with a robot that was moving toward them. He hit his head, fell forward onto his face, and the robot kept moving, running over his right foot. "I screamed. I went in front of the robot and I tried to push it with all of my strength. It was just too heavy," Teng told the *Palo Alto Weekly*. Her husband pulled the boy out from under the machine. A doctor later found no broken bones or internal bleeding. [5]

Knightscope did not deny that the incident happened, but it told a different story. In a field incident report, the company said the child "left the vicinity of his guardians and began running toward the machine," that the robot "veered to the left to avoid the child, but the child ran backwards directly into the front quarter of the machine, at which point the machine stopped," and that the robot's sensors "registered no vibration alert." It called the event a "freakish accident," apologized publicly, and invited the family to its headquarters "to learn more about the technology." The mall took its K5 robots offline. Teng's reply has stayed with me since I first read it: "I don't want a tour. All I want is an answer why this robot didn't stop. It's not Disneyland." [5]

I cannot tell you which account is correct, and neither could the newspaper. But two details deserve attention. First, Knightscope marketed the K5 as having "a commanding presence," and rented it for $6.50 an hour. [5] A security robot is meant to be imposing. It is supposed to make people feel watched. That is the point of the product, and it is in tension with the softness a machine needs if it is going to share space safely with toddlers.

Second, look at who held the evidence. The robot carried about 30 sensors and uploaded its 360-degree view to a security network in real time. As the newspaper noted, "the robot would have recorded the entire incident." [5] The machine's memory was the most complete record of what happened to the child, and that memory belonged to the company and its client, not to the family. When the company told its version, it

spoke with the authority of data. When the mother told hers, she spoke as a frightened parent. The readers' comments under the article show how quickly the public sorted itself along those lines; one commenter wrote that he would "sort of trust the robot's version of events than the parent's." [5]

This is a pattern we will meet again in Chapter 11. Robots that operate in public space are also recording devices, and the record they produce is an asset owned by whoever deploys them. In a dispute between a person and a robot, the robot's side often arrives with a camera feed, a sensor log and a legal team. The person's side arrives with a memory. Any fair arrangement for public robots has to deal with that asymmetry, for example by giving injured people a right to the relevant data. In the U.S. draft bill I wrote, the national robot registry would give a person who has been injured by a robot access to its owner's identity, a small step in the same direction.

The remote trigger: police robots and the use of force

On the night of July 7, 2016, exactly the same date as the Stanford incident, a gunman named Micah Johnson opened fire on police officers who were watching over a protest against police shootings in downtown Dallas. He killed five officers and wounded seven. After he was cornered in a building and an hours-long standoff followed, including exchanges of gunfire and failed negotiations, Police Chief David Brown ordered his SWAT team to find a way to end the threat without putting more officers in danger. They attached about a pound of C-4 explosive to the arm of a bomb disposal robot, a remote-controlled Remotec Andros Mark V-A1, drove it to a wall next to Johnson, and detonated it. The explosion killed him. [6]

As far as anyone could tell, it was the first time an American police department had used a robot to kill someone. A short analysis published weeks later by the Congressional Research Service, Congress's in-house research arm, laid out the debate with unusual clarity. Some saw the tactic as an innovative way to save officers' lives. Others worried about precedent and about the militarization of policing. One expert quoted in the report put the central concern in a single sentence: "the further we remove the officer from the use of force and the consequences that come with it, the easier it becomes to use that tactic." Another expert argued the opposite, that the circumstances justifying lethal force justify it "in essentially every form," and that a robot-delivered bomb was not ethically different from a sniper rifle. [6]

The report then did something useful: it listed what Congress could actually do. It noted that the Department of Defense had transferred 479 bomb disposal robots to law enforcement agencies through its "1033" surplus program between 2006 and 2014, and suggested Congress could attach conditions to those transfers. It also suggested that Congress could require law enforcement agencies to adopt written policies limiting the tactical use of robots "as a condition of receiving funding, such as the Edward Byrne Memorial Justice Assistance Grant program." [6]

I want to dwell on that suggestion, because it shows what anticipatory law looks like. In the United States, the federal government cannot simply order city police departments how to operate. But it pays for a great deal of their equipment, and it can attach strings to the money. The CRS analyst saw in 2016 that the cleanest lever over police robots was not a ban but a funding condition. That idea sat in a short research note for ten years. When I drafted the Safe and Accountable Robotics Act, I picked it up directly. Section 1101 of the draft forbids federal agencies from using a robot to apply any force under autonomous operation, and allows deadly force only under a sworn officer's direct, real-time control, in the face of an imminent threat of death or serious injury to someone else, with no reasonable alternative, and with written authorization from a designated official. Section 1102 then reaches state and local police the way the CRS note suggested: starting two years after enactment, Byrne JAG funds could not be used to acquire or operate a police robot unless the recipient had adopted a policy at least as protective as Section 1101, and any agency that uses a robot to apply force would have to report it within 30 days.

Six years after Dallas, San Francisco showed how hard the politics remain. A new California law required police departments to inventory their military-grade equipment and get approval for its use. In late November 2022, as part of that process, San Francisco's Board of Supervisors voted 8 to 3 to let the police department use its robots, of which it had 17, to kill people "when risk of loss of life to members of the public or officers is imminent and outweighs any other force option available," with only high-ranking officers allowed to authorize it. The police said they did not plan to arm robots with guns but wanted to be able to put explosives on them

in extreme cases. The decision drew national outrage. A week later, at what is normally a rubber-stamp second vote, the board reversed itself, explicitly banning lethal force by robots for the time being and sending the question back to committee. [7]

The statements on each side are worth reading closely, because they capture the real disagreement. Police Chief William Scott said the department wanted "to use our robots to save lives – not take them," and that the point was "neutralizing a threat by equipping a robot with a lethal option as a last case scenario, not sending an officer in on a suicide mission." Supervisor Gordon Mar, who switched his vote, wrote: "I do not think making state violence more remote, distanced, & less human is a step forward." [7] Both are describing the same thing, distance. To the chief, distance protects officers. To the supervisor, distance is exactly the problem, because it makes violence easier.

The most common police robot is not armed at all. It is a camera on legs. In late 2020 the New York Police Department began leasing a Spot quadruped from Boston Dynamics, which officers nicknamed "Digidog." It was deployed roughly half a dozen times, mostly as a mobile camera in potentially hostile situations. One of those was a home invasion in the Bronx; another, according to critics, was in a Manhattan public housing building. Video of the robot went viral. Representative Alexandria Ocasio-Cortez wrote that "robotic surveillance ground drones are being deployed for testing on low-income communities of color with under-resourced schools." In April 2021 the department terminated its roughly $94,000 lease early. John Miller, a deputy commissioner, called the robot "a casualty of politics, bad information and cheap sound bytes." [8]

Two years later, Mayor Eric Adams brought it back. In April 2023 he announced that the city would acquire two robot dogs for a total of $750,000, to be used only in life-threatening situations such as bomb threats, along with a pilot of Knightscope's K5 security robot, the same model that had run over a toddler's foot in Palo Alto. "A few loud people were opposed to it, and we took a step back — that is not how I operate," Adams said. The police commissioner said neither machine would use facial recognition. [9]

Los Angeles went through the same argument in public, and its outcome is the most instructive. In May 2023, after delaying the decision for two months while dozens of residents spoke against it, the city council voted 8 to 4 to accept a $278,000 Spot robot donated by the Los Angeles Police Foundation, a private nonprofit. One councilmember, Eunisses Hernandez, had argued that such robots could be used to target Black and Latino communities, pointing to New York's early deployment in public housing. The council's approval came with conditions written into the department's own policy: the robot "shall not be equipped with any weapon systems including any non-lethal or less-lethal weapon technology, shall not be equipped with or use any facial recognition software or analysis capabilities," and "shall not be used for routine patrol duties or covert surveillance operations." It was assigned only to the SWAT team. An amendment required the police to deliver quarterly reports on its use, and the council kept the power to change the policy or suspend the robot at any time. [10]

Notice how the money moved. The robot did not come out of the city budget, where it would have faced the ordinary budget fight. It came as a gift from a private foundation, which, according to its own website, has given more than $44 million in grants to the LAPD since 1998. [10] Philanthropy is one of the quieter routes by which robots enter public institutions, and it is one reason why a public approval vote, with conditions, matters so much.

Put these cases side by side and a pattern emerges. Police robots provoke resistance less because of what they can do than because of what they signal: an institution that already holds the power of force getting a new way to apply it from a distance, with less visible human judgment. Where the public was consulted and conditions were attached, as in Los Angeles, the robot arrived with some legitimacy. Where it simply appeared, as in New York in 2021, it became, in the deputy commissioner's word, "a target." Legitimacy, in other words, is not a property of the machine. It is a property of the process by which the machine is admitted.

Resistance: kicks, fires and pledges

Not all resistance to robots in public space takes the form of council votes.

In the summer of 2015, a hitchhiking robot called hitchBOT set out from Massachusetts, hoping to reach the American West Coast by relying on strangers to carry it. It was an art project and a social experiment created by a small team of Canadian researchers, including Frauke Zeller of Ryerson University. It was built from a modified bucket, with blue arms and legs, yellow rubber boots and gloves, and an LED smiling face. The year before, it had crossed Canada in 26 days, and tens of thousands of people followed its adventures on social media. Two weeks into its American trip, on a Saturday morning in early August, it was found in downtown Philadelphia, dismembered, its head missing. "There's not a single wire inside and all the things are broken," Zeller said. The robot took a photograph every 20 minutes, and its attackers happened to strike between two of them. Its creators said they had no interest in pressing charges. [11]

hitchBOT was harmless. It could not move by itself. It worked for no corporation and watched for no police force. That is exactly why its destruction is instructive: some resistance to robots is not about what a robot does, but about what it stands for, or simply about the fact that a defenseless, human-shaped object is sitting by the road. People project onto robots. Sometimes they project affection, as tens of thousands of hitchBOT's followers did. Sometimes they project something darker.

A decade later, the projection was political. On Sunday, June 8, 2025, during protests in downtown Los Angeles against federal immigration raids, demonstrators attacked five of Waymo's driverless taxis. Photographs show a row of the robotaxis engulfed in flames, with anti-ICE graffiti scrawled across their hoods and doors. The Alphabet-owned company confirmed the five vehicles were vandalized and could not be retrieved, and suspended service in the protest area. The police warned that burning lithium-ion batteries release toxic gases, including hydrogen fluoride. CNBC reported that it was unclear whether the vehicles had been deliberately targeted. [12]

It would be easy to read the burning Waymos as anti-robot violence, and I do not want to romanticize arson; burning batteries endangered real people. But sociologically, something else is going on. The protesters in Los Angeles were not angry at self-driving software. They were angry at a federal government. The Waymos happened to be there, unattended, expensive, covered in cameras, and visibly connected to one of the most powerful companies on earth. A robot in public space is a kind of proxy. It is the physical presence of an absent institution. When people cannot reach the institution, they reach the proxy.

This has a practical consequence for anyone deploying robots in streets. A robot does not enter public space as a neutral object. It enters carrying the reputation of whoever sent it. If the public trusts that institution, the robot inherits the trust. If the public is angry at it, the robot inherits the anger, and it cannot negotiate, apologize or walk away.

Some of the most interesting resistance has come from inside the industry. In 2022, Boston Dynamics and several other leading robotics companies published an open letter titled "General Purpose Robots Should Not Be Weaponized." It noted that new mobile robots could navigate "into locations previously inaccessible" and could be misused "to invade civil rights or to threaten, harm, or intimidate others." The signatories pledged not to weaponize their "advanced-mobility general-purpose robots" or the software that enables them, and not to support others who did; to review customers' intended uses where possible; and to explore technical features that could reduce the risk. They called on policymakers "to work with us to promote safe use of these robots and to prohibit their misuse." They also drew a line: "we are not taking issue with existing technologies that nations and their government agencies use to defend themselves and uphold their laws." [13]

The letter is a striking document, because it is a company saying, in effect, that its own promise is not enough. "We understand that our commitment alone is not enough to fully address these risks," it says. [13] A voluntary pledge can be withdrawn, ignored by a competitor, or worked around by a customer. That is why the SARA draft converts the pledge into law. Section 302 prohibits manufacturing, selling or operating a robot that is "equipped or readily convertible to be equipped with a firearm, destructive device, or other weapon capable of causing death or serious bodily injury," with narrow exceptions for law enforcement robots operated in compliance with the police-robot title, and for explosive ordnance disposal. Section 303 requires "the contemporaneous authorization of an accountable human" before any covered robot applies force likely to cause bodily injury, other than the inherent force of a safety function such as stopping. These provisions are the industry's own red lines, written down where they cannot be quietly erased.

From where I sit, in Semarang, the idea of a sidewalk as a contested commons is not abstract. On many Indonesian streets the *trotoar* is already shared, sometimes uneasily, by pedestrians, street vendors, parked motorcycles, and motorcycles that ride along it when traffic jams. Our sidewalks were never quiet, orderly spaces waiting for robots. They are crowded social arrangements negotiated every day. When delivery robots arrive here, and some version of them eventually will, they will not enter an empty stage. They will enter a negotiation that is already full of people, many of whom earn their living on that strip of concrete. The question of who gets space will not be technical. It will be a question about livelihoods.

Designing the shared street

If the sidewalk is a commons, then robots that use it are borrowing something that belongs to everyone. What should the terms of the loan be? The cases in this chapter point toward five principles.

First, a yield hierarchy, written into law. Pennsylvania already requires delivery robots to yield to pedestrians and cyclists. [3] That should be the floor everywhere, and it should be specific about the people most at risk. Ackerman was not hurt by a robot's speed. She was endangered by its *stillness*, in the one spot she needed. Section 907(a) of the SARA draft addresses exactly this: a deployer of a robot in public space must ensure that it "does not obstruct the accessible route of travel" required by the Americans with Disabilities Act, and that it "detects and yields to persons using mobility aids, service animals, and white canes." Accessibility is not a feature to add after complaints go viral. It should be a condition of the permit.

Second, identification. When a robot blocks a curb cut, knocks down a child or rolls into a protest, the people affected need to know whose it is and whether anyone is steering it. For drones, the U.S. Federal Aviation Administration already requires a "remote ID" broadcast that can be read by people on the ground. The SARA draft borrows that idea: Section 403 requires robots above the lowest risk tier operating in public spaces to broadcast their robot identification number and whether they are under autonomous or remote operation, readable on "a commonly available personal device." The registry behind those numbers, under Sections 401 and 402, lets anyone look up a robot's manufacturer, model, risk tier, conformity status, open recalls and corporate operator, while keeping individual owners' details restricted. Just as important is what the registry may *not* collect: location or operational data, except the general area of operation for the highest-risk robots. The goal is accountability for the robot, not a new surveillance system for the city.

Third, local control over local space. The state laws in this chapter differ partly because cities differ. A sidewalk in central Pittsburgh is not a sidewalk in a suburb, and neither is a *trotoar* in Semarang's old town. The SARA draft therefore treats federal product-safety standards as a ceiling on how robots are *built* but local operating rules as a floor on how they are *used*. Section 1302 preserves the authority of states and cities to regulate "the time, place, and manner of operation of robots on public property, including sidewalks, streets, and parks," to require local permits, and to set rules for their own police. Section 5(c) adds a definitional point that matters more than it sounds: a sidewalk robot is a covered robot, not a motor vehicle, so it cannot slip out of robot rules by claiming to be a car, or out of local sidewalk rules by claiming to be federal.

Fourth, force requires a human, and police robots require consent. Dallas, San Francisco, New York and Los Angeles show that the most dangerous robots in public space are not delivery robots but the ones that carry the authority of the state. The lesson of those cases is that the public will accept some police robots for some purposes, if it is asked, if limits are written down, and if use is reported. Los Angeles's conditions, no weapons, no facial recognition, no routine patrol, quarterly reports, are a reasonable template. [10] National law can raise the floor through funding conditions, as the Congressional Research Service suggested in 2016, without overriding local democracy. [6]

Fifth, the data belongs partly to the people it records. The Stanford toddler's parents never got the robot's footage of their son. [5] A robot in public space should not be the sole keeper of the truth about its own accidents. Injured people need a right to the relevant records, and regulators need incident reports, as Pennsylvania already requires within 24 hours. [3]

None of this would have stopped hitchBOT's vandals or the fires in Los Angeles. Laws do not abolish anger. But they change what anger is directed at. A robot that arrives through a public process, follows rules written

with the people who share the sidewalk, and answers to an identifiable owner is a different social object from a robot that simply appears. The first is a guest. The second is an occupier, and occupiers attract resistance.

The Indonesian draft I have been working on borrows heavily from Korea for exactly this reason. Korea does not let a robot onto the sidewalk until it has been certified and its operator insured. [4] The certification and the insurance are not just safety measures. They are a public statement that the robot has been admitted, by someone accountable, on terms the public can see.

There is one more thing the sidewalk teaches. Every robot in this chapter belonged to someone: a delivery company, a mall's security contractor, a police department, a technology giant. When the robot misbehaved, the public's first question was always *whose is it?* and the second was *who controls it right now?* We tend to assume those questions have the same answer, that whoever owns a machine is the one who controls it. The next chapter shows how often that assumption is false. A farmer can own a tractor that answers to its manufacturer. A child can own a robot friend that dies when a company's servers switch off. A paralyzed man can own a $100,000 exoskeleton and still be told it is too old to fix. In public space, we worry about robots controlled by powerful institutions. In private life, the more surprising discovery is that robots we have bought and paid for may still be controlled by someone else.

Notes

1. https://www.wesa.fm/identity-justice/2019-10-22/food-delivery-robots-pulled-from-pitt-campus-after-backlash-about-mobility
2. https://www.supplychaindive.com/news/delivery-robot-bills-laws-proliferate-state-legislatures/648303/
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Part IV: Power and Order

Chapter 9: Ownership Without Control: Software Tethers and the Right to Repair

> "I find it very hard to believe after paying nearly $100,000 for the machine and training that a $20 battery for the watch is the reason I can't walk anymore?" — Michael Straight, September 2024

A tractor that answers to someone else

For most of the history of farming, a farmer who owned a tractor could fix it. The tools were wrenches and welders, the knowledge was passed between neighbors, and the nearest mechanic might be a cousin. Modern farm machinery is different. A large tractor or combine harvester today is, among other things, a network of computers. Its engine, its hydraulics and its emissions controls are managed by electronic control units that talk to each other in software. When something goes wrong, the machine reports a fault code. Clearing that

code, recalibrating a sensor, or getting a newly installed part to be recognized by the rest of the machine often requires software.

In January 2025, the U.S. Federal Trade Commission, together with a group of states that grew to include Illinois, Arizona, Michigan, Minnesota and Wisconsin, sued Deere & Company, the maker of John Deere equipment. The complaint's core allegation was simple. Deere makes "the only software repair tools capable of performing all electronic repairs on Deere equipment," and it had made those tools available only to its authorized dealers. That, the agencies alleged, forced farmers to rely on dealers for many repairs, led to service delays and higher costs, and allowed Deere to unlawfully acquire and maintain monopoly power over the market for repairing its own machines. [1]

On July 8, 2026, the FTC and the five states announced a settlement. For the next ten years, under the supervision of the FTC and the states, Deere must make available to farmers and independent repair providers, "on fair and reasonable terms," the same repair resources it gives its dealers. The order lists them, and the list reads like a map of exactly where the software tether had been tied. Owners and independent mechanics must be able to read, clear and reset electronic fault codes. They must be able to reprogram electronic components, "including 'pairing' newly installed electronic parts with equipment." They must be able to restart a machine after an emissions-related shutdown, the condition farmers call "limp mode." And they must be able to view and search the technical manuals and troubleshooting guidance. Future repair tools must be shared too, once Deere makes them available to more than half of its U.S. dealers. Dealers must not retaliate against farmers who fix their own machines. [1]

"Today's settlement enables farmers to do what they've done for generations—fix their own tractors and other farm equipment—without having to pay an authorized John Deere dealer to do it for them," said Daniel Guarnera, director of the FTC's Bureau of Competition. [1] Deere did not admit wrongdoing, and the stipulated order, the FTC noted, has the force of law only once a federal judge approves and signs it. [1]

Read that list of required tools again, because it tells you what the fight was really about. Nobody was stopping farmers from buying wrenches. The barrier was a *permission layer* built into the machine: the ability to tell the computer that a new part is legitimate, that a fault is cleared, that the engine may run at full power again. The farmer owned the steel. The manufacturer controlled the permissions.

Now consider a story from the other side of the world that shows the same capability doing something most people would applaud. In the spring of 2022, after Russian forces occupied the Ukrainian city of Melitopol, troops removed every piece of equipment from a local John Deere dealership, Agrotek: 27 pieces of farm machinery in total, including combine harvesters worth $300,000 each, together valued at nearly $5 million. Some of it was trucked more than 700 miles to Chechnya. The machines carried GPS, so they could be tracked to a village called Zakhan Yurt. And when the thieves tried to use them, they discovered, in the words of a contact who spoke to CNN, "that they could not even turn them on, because the harvesters were locked remotely." [2]

The tractor that cannot be repaired without the manufacturer's software and the harvester that cannot be stolen because it can be switched off from far away are the same machine. The remote lock is not good or bad in itself. It is a capacity to override the person physically holding the machine. In Melitopol that capacity protected the rightful owner against looters. On an American farm the same kind of capacity, used to keep repairs inside a dealer network, worked against the rightful owner. The question is never whether a robot *can* be controlled from a distance. Modern machines almost always can. The question this chapter asks is: who holds that power, on whose behalf, and with what limits?

That question becomes sharper as machines become robots. A tractor with software is still mostly a tractor. A robot is a machine whose usefulness *is* its software: its perception, its decisions, its connection to cloud services that may be doing much of its thinking. When you buy a robot, you buy a body. Whether that body keeps working, and whether you can fix it, may depend on a company's servers, a company's tools and a company's continued existence. Ownership, in the old sense of the word, meant control. For robots, the two have come apart.

When the company dies, the robot dies

Sony began selling a robot dog called AIBO in 1999. The first batch of 3,000 units sold out in 20 minutes, despite a price of more than $2,000 each, and Sony went on to sell more than 150,000. The company ended production in 2006 as part of a cost-cutting effort. It kept an "AIBO Clinic" open for repairs until 2014, and then stopped. [3]

For many owners, an AIBO was not a gadget. Over years, the robot dogs had become companions with names, habits and places in family life. When the clinic closed, owners whose dogs broke down had nowhere to go. A small company called A-Fun, which repairs vintage electronics, stepped in. Because new parts no longer existed, the only source of genuine replacement parts was other AIBOs. So A-Fun began accepting "dead" robot dogs as donors, and before taking them apart, it arranged for them to receive a funeral. At Kofukuji, a 450-year-old Buddhist temple in Isumi, near Tokyo, priests in traditional robes have chanted sutras and burned incense for rows of broken AIBOs, each wearing a tag naming where it came from and who had owned it. By 2018 the temple had held services for about 800 of them. "I feel relieved to know there will be a prayer for my Aibo," one owner wrote in a note sent with a dog. Another wrote: "Please help other Aibos. My eyes filled with tears when I decided to say goodbye." The priest Bungen Oi saw nothing strange in the rite. "All things have a bit of soul," he said. In 2018 Sony launched a new, internet-connected aibo, but it resisted pressure from owners of the original to resume repairs of the old ones. [3]

It is easy to treat the AIBO funerals as a charming curiosity. I think they are something more serious: a community improvising an institution that the market had failed to provide. A-Fun is a repair shop, a parts bank and an organ-donation system rolled into one, and the temple ceremony gives the whole arrangement moral meaning. Notice, though, how much it depended on the old AIBO being *repairable at all*. Its body was mechanical and self-contained. With enough skill and enough donor dogs, it could be kept alive without Sony.

Ten years later, a company called Embodied learned what happens when that is no longer true. Embodied made Moxie, an $800 robot marketed to children aged 5 to 10, described at its launch as a companion "designed to help promote social, emotional, and cognitive development." Moxie talked with children, read with them, drew with them, and worked through weekly themes like empathy and friendship. In December 2024, the company announced it was closing; a lead investor in a critical funding round, it said, had withdrawn at the last minute. Moxie, the company explained, could not perform its core functions without its cloud servers, and "we don't know the exact date when services will cease. It is likely to happen within days." Warranties, repair services, the parent app and customer support would all end. Most owners would get no refund. Some parents who had bought the robot through a third-party lender might, Ars Technica noted, end up paying interest on a toy that no longer worked. Embodied promised to provide a guide for telling children that their robot friend was going away. [4]

Read that last sentence slowly. A company built a product designed to form an emotional bond with young children, and when the product's business model failed, the company's final service to its customers was advice on how to explain the robot's death. The robot's hardware was fine. Its body would sit on the shelf, intact, with its eyes and voice and motors in working order. What died was the connection to a server farm somewhere, and with it everything that made Moxie Moxie.

The difference between AIBO and Moxie is the difference between a machine and a tethered machine. AIBO's intelligence lived inside AIBO. Moxie's lived elsewhere. A community could keep AIBO alive with parts and patience. Nobody could keep Moxie alive without the company's software and servers, because the company had never released them. The situation was not entirely hopeless. Ars Technica pointed out that failed smart-home products had occasionally been resurrected, and that someone might release an open-source replacement for the missing cloud. [4] But the fact that any rescue depended on luck, hobbyists or the goodwill of a dissolving company shows how thin the owners' rights were. They had paid full price for something they did not really control.

Ars Technica also noted something that matters for law. The month before Embodied collapsed, the FTC had warned that smart-device makers who do not commit to how long they will support their products' software may be breaking the law. [4] That warning points to the gap at the heart of this chapter. A buyer of a robot can see its price, its size and its features. What the buyer usually cannot see is the most important

fact about its future: how long its maker will keep it alive.

Even large, established companies are not immune to this uncertainty. In December 2025, iRobot, the company that introduced the first Roomba robot vacuum in 2002 and has sold millions of robots worldwide, entered a pre-packaged Chapter 11 bankruptcy process in Delaware, under which its secured lender and main contract manufacturer, Shenzhen PICEA Robotics, would take 100 percent ownership of the company. iRobot said it expected "no anticipated disruption to its app functionality, customer programs, global partners, supply chain relationships, or ongoing product support." [5] I have no reason to doubt that statement. But notice what it is: a promise from a company in bankruptcy about software it controls, made to owners who have no independent way to keep their robots running if the promise is ever broken. Owners of robot vacuums were, in effect, reading a press release to find out whether the machines in their homes would keep working. That is not what ownership is supposed to feel like.

Bodies on subscription

The stakes rise again when the robot is not a toy or an appliance but part of someone's body.

Barbara Campbell lost her sight to a genetic disease in her thirties. In the late 2000s she received an Argus II retinal implant, made by a Los Angeles company called Second Sight. The system worked through a camera mounted on glasses, a video processing unit worn on the belt that reduced images to a pattern of 60 pixels, and a 60-electrode array attached to the retina. It did not restore normal vision, but it gave her patches of light and dark that she learned to use on New York's sidewalks and subways. One day in 2013, while switching from the 6 train to the F train during rush hour, she heard "a little 'beep, beep, beep' sound." The system had powered down. Despite repair attempts by the company, it never worked again. She decided the risks of another surgery to remove it were not worth taking, so the defunct device is still in her eye. [6]

The IEEE Spectrum investigation that told Campbell's story in 2022 described a company that abandoned an entire population of patients. In July 2019, Second Sight told Argus users it would phase out the retinal implant to focus on a brain implant called Orion, while assuring them that support would continue. In March 2020, citing the effects of the pandemic on its ability to raise money, it laid off most of its remaining staff and announced its intention to wind down operations; its equipment went to auction. According to Spectrum, it did not tell its patients. "Those of us with this implant are figuratively and literally in the dark," one of them, Ross Doerr, wrote on Facebook. When Doerr later needed an MRI to rule out a brain-stem tumor, his providers could not get the information they needed from the company about scanning safely with the implant in place, so he had a CT scan instead. "I still don't know if I have a brain-stem tumor or not," he said. More than 350 blind people around the world had Second Sight implants in their eyes. [6]

One user, Jeroen Perk in the Netherlands, found his own way through. In November 2020 his video processing unit fell from his belt and shattered. "I had no vision, no Argus, and no support from Second Sight," he said. He asked the European community of Argus users for help, heard back from a patient who no longer used the device and from a doctor with a spare unit, and by February 2021 he had a refurbished system. [6] Like A-Fun with the AIBOs, Perk's network built the after-sales service the manufacturer would not provide. Unlike A-Fun, Perk was repairing his own sight.

Michael Straight's story is shorter and in some ways more infuriating. Straight is a former jockey who was paralyzed from the waist down in a racing accident. Since 2014 he had walked with a ReWalk Personal Exoskeleton, a robotic frame that moves the legs, controlled through a watch on the wrist. In 2024 a wiring problem with the battery in that watch stopped the machine. When he called the manufacturer, Lifeward (formerly ReWalk Robotics), he was told "they stopped working on any machine that was 5 years or older." "I find it very hard to believe after paying nearly $100,000 for the machine and training that a $20 battery for the watch is the reason I can't walk anymore?" he wrote on Facebook. For two months he could not walk. [7]

Then his story reached a horse-racing news site and a local Florida television station. Within days, Lifeward fixed the problem. "What took me two months and got me no results, only took you guys four days," Straight said. In a statement to *The Verge*, the company said it had resolved the issue, and added that since his device was more than ten years old, it was encouraging him to replace it, now that Medicare had begun paying for personal exoskeletons in April 2024. [7]

I do not want to treat Lifeward as a villain; it eventually fixed the device, and a company does need some limit on how long it supports old hardware. But the episode reveals the structure. A man's ability to walk depended, by his account, on a $20 battery, and the company held the knowledge and parts needed to fix it. Its policy was to stop servicing machines older than five years. And the thing that changed the outcome was not a legal right. It was publicity. A right that only works if you can make the news is not a right. It is a lottery.

Campbell, Doerr, Perk and Straight did not buy gadgets. They acquired, at enormous cost and often with insurance or charity paying, machines that became part of how their bodies work. A retinal implant is in the eye. An exoskeleton carries its user's weight. When the maker of such a device withdraws support, the user is not losing a convenience. Something like a bodily function is lost, controlled by a business decision made somewhere else. As robots move into care, rehabilitation and disability support, a theme of Chapter 6, this will become one of the most important places where ownership and control come apart.

The train that would not run

The last story in this sequence shows what the tether looks like when it is deliberate.

In 2023, a regional rail operator in southwest Poland had a problem: some of its trains, made by the Polish manufacturer NEWAG, stopped working after they were serviced by an independent maintenance company rather than by the manufacturer. The repair company hired three white-hat hackers to find out why. According to reporting by 404 Media, they found code in the trains' control systems that prevented a train from running if its GPS showed it had spent a certain number of days at an independent repair company's maintenance center, and also prevented it from running if certain components had been replaced with parts lacking a manufacturer-approved serial number. The hackers helped get the trains running again. NEWAG then threatened to sue them, denied bricking the trains, and demanded that the repaired trains be withdrawn from service because they had been "hacked" and might now be unsafe, a claim 404 Media said the company could not substantiate. [8]

The case is extreme, which is why it is useful. It strips away the usual ambiguity. Most of the time, when a machine refuses to accept a third-party part or demands manufacturer software for a repair, the company can say it is protecting safety, security or quality, and the owner cannot easily prove otherwise. In the Polish trains, at least as the hackers described it, the logic was out in the open. The machine was programmed to know *where* it had been repaired and *whose* parts it contained, and to stop if the answer was wrong.

The industry term for one version of this is "parts pairing": a replacement part works fully only if it has been digitally "paired" to the machine using the manufacturer's software. Pairing can have legitimate uses. It can help detect stolen parts or confirm that a safety-critical component is genuine. But it also turns every repair into a transaction that requires the manufacturer's permission. Once a machine can recognize and reject parts, the manufacturer controls not only the first sale but every repair for the life of the product. Economists call the market for repairs and parts an *aftermarket*, and the strategy of locking it up an attempt to capture *rent*, a stream of income that comes from control rather than from making anything better. Chapter 4 traced how the gains from automation flow toward whoever owns capital. The software tether is one of the most precise instruments for directing that flow. It lets the maker keep collecting from a machine long after selling it.

There is a social cost that does not appear in any company's accounts. Repair is not just a service. It is a form of knowledge, and of community. The mechanic who knows how to fix a tractor, the technician who keeps an old AIBO alive, the patient who sources a spare part for a neighbor's implant: these people make machines last, and they keep skill and money circulating locally. A tether shuts them out. It turns repairers into customers and turns a common skill into a licensed privilege.

The law catches up, slowly

The good news is that law has started to respond, from several directions at once.

In the United States, the movement began with state legislatures. According to U.S. PIRG, a consumer group

that has campaigned for these laws, New York's Digital Fair Repair Act, which took effect on December 28, 2023, was the first electronics right-to-repair law passed by a state legislature, though industry lobbying narrowed it before it was signed, limiting it to consumer products sold after July 2023. Minnesota's law, in effect since July 1, 2024, covers a broad range of digital electronic equipment made after July 1, 2021, including business and industrial equipment, but excludes farm equipment and medical devices. California's law, also effective July 1, 2024, requires makers of devices priced over $100 to keep parts, tools and documentation available for at least seven years. Oregon's law, signed in March 2024 and in force from January 2025, was the first to restrict parts pairing. Colorado passed three laws in three years: one for powered wheelchairs in 2022, one for farm equipment in 2023, and a broad electronics law in 2024 that took effect on January 1, 2026, which PIRG says gave Coloradans the broadest repair rights in the country. Even that law exempts medical equipment. [9] According to PIRG, with six new laws, including ones in Oregon, Washington and Nevada, coming into force at the start of 2026, 25.75 percent of Americans lived in a state with an enforceable right-to-repair law, and the group projected that the share would reach at least 35.5 percent once laws in Connecticut and Texas took effect in July and September 2026. [10]

Federal agencies moved too. In 2021, the FTC's report to Congress, *Nixing the Fix*, found that manufacturers had offered "scant evidence" to justify their repair restrictions; the commissioners then voted to prioritize enforcement against illegal restrictions. In October 2024, the U.S. Copyright Office granted an exemption allowing repair of retail commercial food equipment, the famous example being McDonald's ice cream machines, without violating the federal law that makes it illegal to bypass digital locks. PIRG noted the limit: it is still illegal to sell or share the tools needed to bypass those locks. [9] And the Deere settlement in 2026 showed that competition law, which bans illegal monopolization, can force open a repair market even without a specific repair statute. [1]

Europe took a different approach, writing a general framework into law. Directive (EU) 2024/1799 on common rules promoting the repair of goods was adopted on June 13, 2024, and member states had to apply it from July 31, 2026. For products listed in its Annex II, the categories for which EU law already sets repairability requirements, manufacturers must repair goods within a reasonable time and for a reasonable price, must sell spare parts at reasonable prices, and "are prohibited from using contractual clauses, hardware or software techniques that impede the repair" of those goods "unless justified by legitimate and objective factors." Consumers who choose repair over replacement under the legal guarantee get an extra year of guarantee. A European online repair platform is planned to become operational in January 2028. The Commission updates the Annex II list as new repairability rules are adopted; domestic space heaters were added from July 31, 2026. [11]

Look at these laws as a whole and two things stand out. First, the direction is clear. In the United States and Europe, the principle that owners and independent repairers should get the same parts, tools and information as authorized dealers has moved from activist demand to law in less than a decade. Second, the coverage is patchy in exactly the places where robots are heading. State laws are organized by product categories that were familiar when they were written: phones, laptops, appliances, tractors, wheelchairs. Some exclude medical devices, which is where the Argus and the exoskeleton live. [9] The EU directive applies only to the product categories listed in its annex, and robots do not appear there as a category of their own. [11] None of these laws directly addresses the Moxie problem: what happens to a machine whose core function depends on a server that is about to be switched off. Right-to-repair law, so far, is mostly about fixing *bodies*. The tether also runs through the cloud.

Ownership that means control

When I drafted the Safe and Accountable Robotics Act, I wanted its ownership provisions to answer the stories in this chapter directly, one by one. They sit in Title IX, alongside the rules for consumers and workers, in three sections.

Section 904 is the answer to the remote off-switch. It forbids a manufacturer, or anyone acting for it, to "remotely disable, degrade, or restrict a function of a covered robot that was advertised or included at the time of sale." There are four exceptions, each tied to a concrete need. A manufacturer may act remotely to remedy a safety defect or comply with a recall order. It may act to address an active cybersecurity threat,

but only "for the shortest time necessary." It may act at the owner's request, including in case of theft, which is the Melitopol scenario, with the key difference that the power is exercised for the owner and at the owner's request, not around the owner. And it may switch off a function that was expressly sold as a time-limited subscription, when the subscription expires. Before acting for safety or security reasons, the manufacturer must notify the owner and explain why, or tell them promptly afterward if advance notice is impractical. The section also protects resale: a manufacturer may not prohibit or condition the transfer of a robot, or charge a fee to transfer the software needed for its advertised functions, beyond a reasonable fee for transferring a subscription.

The design principle is that the remote off-switch should exist, because sometimes it is genuinely needed, but it should be a tool of safety and of the owner, not a tool of the manufacturer's business model. The subscription exception is deliberate. I do not think the law should forbid companies from selling robot features as services. But it should forbid them from selling something as yours and then quietly turning it into a rental.

Section 905 is the answer to Moxie and Second Sight. It requires manufacturers to disclose, "at or before the time of sale," the minimum period during which they will provide security and safety updates, and for the higher-risk tiers of robots that period must be at least five years from the date the model is *last sold*, not first sold. (In the draft, the number five sits in square brackets, a drafting convention meaning it is a placeholder for legislators to debate. Reasonable people could argue for longer, especially for medical and assistive robots.) If a manufacturer stops supporting a robot before the end of that period, or goes out of business, it must give owners and the regulator at least 180 days' notice, and, "where doing so does not create an unreasonable safety risk," it must make available to owners "the software, credentials, or documentation necessary to keep the robot's advertised physical functions operational without the manufacturer's servers." For the highest-risk robots, the regulator may require the manufacturer to deposit the software needed for safe operation with an approved escrow agent, so that it can be released if the company disappears.

Imagine Moxie under that rule. Embodied would have had to tell parents, before they bought the robot, how long it promised to support it. It could not have shut down "within days." And when it did shut down, it would have had to hand owners what they needed to keep the robot running locally, rather than leaving that job to former employees' goodwill. Imagine Second Sight under the escrow rule. The information and software needed to keep Argus implants working and to scan their users safely would have been held by a third party, ready to be released, instead of disappearing into a liquidation auction.

Section 906 is the answer to the tractor and the train. It requires a manufacturer to make available to owners and independent repairers, "on fair and reasonable terms, the parts, tools, diagnostic software, and documentation that it makes available to its own authorized repairers." This is the parity principle that the Deere settlement adopted for one company for ten years, applied by law to robots generally. [1] There is one carefully limited exception: for the highest-risk robots, the manufacturer may restrict repair of *safety functions* to people who meet competency requirements set by rule. A robot that can crush a person is not a phone, and the person who repairs its emergency stop should know what they are doing. But the exception is narrow on purpose. It covers safety functions, not the whole machine, and the competency standard is set by the regulator, not by the manufacturer's dealer contracts.

These rules have costs, and the arguments against them deserve a fair hearing. Manufacturers argue that opening up diagnostic software creates cybersecurity risks; that giving untrained people the ability to modify safety systems could get someone hurt; that releasing software and documentation exposes trade secrets; and that long support obligations raise the price of robots and may discourage small companies from entering the market at all. Some of these concerns are real. That is why Section 904 keeps a security exception, why Section 905 releases software only where that does not create an unreasonable safety risk, and why Section 906 allows safety-function repair to be limited to competent people. But the FTC's own review found that manufacturers had offered "scant evidence" for most repair restrictions. [9] The burden should sit with the company that wants to restrict an owner's control, not with the owner who wants to keep using what they bought.

There is also an argument from fairness that I find decisive. A company that sells a robot asks the buyer to trust it: to depend on its products, to let them into the home, sometimes to build them into a child's

routine or a disabled person's body. Trust creates obligations. The tethered robot asks for all of the trust of ownership while giving the owner only some of its powers. The provisions above are an attempt to rebalance that exchange.

I care about this partly because of where I live. In Semarang, as in most Indonesian cities, there is a dense economy of repair. There are motorcycle *bengkel* on nearly every major road, small phone-repair stalls in shopping centers, and technicians who learned their trade by taking things apart. Much of that repair work happens outside formal employment, the kind of informal work Chapter 12 examines closely. The robots that reach Indonesia in the coming years will mostly be designed and sold by foreign companies. If those robots are tethered, every repair will require software and permission controlled from abroad, and the repair economy that has kept Indonesian machines running for generations will be shut out. Some of the rent from our robots will flow overseas for as long as the robots run. If they are repairable, the same machines could create local skilled work. The choice between those futures will be made largely in the design of the machines and the law that governs them, long before most people notice it has been made.

Who holds the off-switch?

Go back to Melitopol. The same digital lock that frustrated farmers in Illinois stopped looters in Chechnya, and most of us cheer the second use while objecting to the first. That is the honest conclusion of this chapter: remote control of robots is neither good nor bad. It is power, and the moral question is always who holds it and in whose interest it is used. Within a single country, law can answer that question, as Sections 904 to 906 try to do: the off-switch belongs to safety and to the owner, the support period must be promised in advance, and repair knowledge must be shared on fair terms.

But the Melitopol story has another layer that a domestic statute cannot reach. The harvesters were made by an American company, sold through a Ukrainian dealer, stolen by an occupying army, and locked from a distance across a war zone. The power to switch off a machine did not stop at a national border. Neither did iRobot's future, which by the end of 2025 was being settled in a Delaware bankruptcy court that would hand the company to its Chinese contract manufacturer. [5] As robots spread around the world, the question "who holds the off-switch?" stops being only a question about companies and customers. It becomes a question about nations: which countries design the robots, write their software, run their cloud services, make their chips, and could, in a crisis, degrade or disable machines that other countries depend on. That is the subject of the next chapter, the geopolitics of embodied AI.

Notes

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Chapter 10: The Geopolitics of Embodied AI

> "Advanced robotic devices collect data that could be leveraged by malign actors to surveil Americans, enhance the capabilities of foreign intelligence services, or to remotely commandeer the robots." — U.S. Executive Branch National Security Determination on foreign-produced advanced robotic devices, 27 July 2026

Beijing, August 2026: a stadium full of robots

On the evening of 22 August 2026, about a thousand humanoid robots walked into the National Speed Skating Oval in Beijing, the building Beijingers call the Ice Ribbon. It was the opening ceremony of the second World Humanoid Robot Games. Over the next five days, 2,056 robots entered by 666 teams from 16 countries competed in 51 events and 1,301 matches.[1] A year earlier, the first Games had drawn 280 teams and a little over 500 robots to 26 events.[1][2] The number of robots had roughly quadrupled in twelve months.

The numbers from the track were stranger still. At the 2025 Games, the best 100-metre time by a humanoid called Tiangong, built by the Beijing Humanoid Robot Innovation Center, was 21.50 seconds. On the final day of the 2026 Games, a larger version, Tiangong Ultra, ran 8.64 seconds. Usain Bolt's men's world record is 9.58. The robot standing high jump went from 0.96 metres to 3.40 metres in a year. The organisers were careful to say that robot and human events are not directly comparable, and they are not. But a referee who had judged the high jump at both Games noticed something more important than the heights: in 2025 many robots were still steered by remote control, and in 2026 more of them were competing on their own.[2]

The Games also had a second programme that got fewer headlines. Twenty-one "application" events tested robots in nine everyday settings, including homes, hotels, factories, shops and emergency response. In the home event, a robot had to tidy, fold clothes and respond to new instructions for twenty minutes or more without a human touching it. A company called Galbot won three of these events fully autonomously. Its co-founder was honest about the limits: robots trained on a particular task can do it, but "their success rate may drop when faced with unfamiliar objects or environments."[2]

The most important number of the week, though, was a ratio. Of the 666 teams, 641 were Chinese, and of the 2,056 robots, 1,975 came from Chinese teams: 157 companies and 200 universities and research institutes.[3] Sixteen countries took part, but this was a Chinese national event with international guests, and it was meant to be. It was hosted by the Beijing municipal government and China Media Group, the state broadcaster.[2] Beijing has also set up a 10-billion-yuan public investment fund for robotics.[4]

I watched clips of the Games on my phone in Semarang, like millions of other people, and I felt two things at once. The first was simple delight. A robot running the 100 metres faster than Bolt is a wonderful piece of engineering. The second feeling arrived more slowly. Nothing about the Games was only about sport. A sprint tests actuators, batteries and balance control. A football match tests perception and real-time decision-making among many robots. A twenty-minute household task tests whether a machine can be trusted in a home. And the whole event, broadcast around the world, was a demonstration to investors, to foreign governments and to Chinese citizens that a particular country intended to lead a particular industry.

A plan written in 2023

That intention had been written down. On 20 October 2023, China's Ministry of Industry and Information Technology (MIIT) issued the *Guiding Opinions on the Innovation and Development of Humanoid Robots* and published it on 2 November.[5] The document set two deadlines. By 2025, China would establish "an initial humanoid robot innovation system," achieve breakthroughs in the robot's "brain, cerebellum and limbs" (meaning its large AI model, its motion control, and its mechanical body), ensure "a safe and efficient supply of core components," and reach mass production at an internationally advanced level. It also aimed to cultivate two or three globally influential companies and two or three industrial clusters. By 2027, China would form "a safe and reliable industry supply chain" with "globally leading comprehensive strength," and humanoids would become "an essential new engine of economic growth."[5]

The phrase that matters most for this chapter is *safe and reliable supply chain*. It appears in the 2025 goals

and again in the 2027 goals. In the language of Chinese industrial policy, "safe" in this context does not only mean that a robot will not fall on someone. It means that the supply of the parts cannot be cut off by another country. The 2023 guideline is, among other things, a plan to make the humanoid robot immune to the kind of pressure the United States had already applied to Chinese semiconductors.

Whether China met the 2025 targets depends on how you count. In industrial robots, the conventional arms that weld and assemble in factories, the scale of China's lead is not in doubt. According to the International Federation of Robotics (IFR), factories around the world installed 603,307 new industrial robots in 2025, a record. China installed 354,226 of them, about 59 percent of the world total. Installations in China grew 20 percent in a single year. And Chinese brands now sell more robots in China than foreign brands do: about 195,000 units in 2025, or 55 percent of the domestic market. Between 2019 and 2021 that share had been around 30 percent.[6][7]

In humanoids, the picture is more uncertain, because humanoids are still mostly an expensive experiment. The IFR's own assessment in September 2026 was blunt: "Most humanoids remain prototypes or pilot-stage technologies rather than industrial-grade products," and for industrial production tasks conventional robots remain "economically superior."[7] Even so, the market that exists is overwhelmingly Chinese. A report by the Taiwanese research institute DSET, citing the market research firm Smart Analytics Global, found that Chinese vendors accounted for 85 percent of global shipments of humanoid and quadruped robots in 2025, while noting that only around 10 percent of humanoids were deployed in real-world applications.[8]

So the honest summary is this. China has not yet proved that humanoid robots are useful at scale. Nobody has. What China has proved is that it can manufacture them, iterate on them quickly and cheaply, put two thousand of them in a stadium, and make the rest of the world nervous.

Washington, July 2026: a list instead of a law

The nervousness had a legal form. On 27 July 2026, an interagency body convened by the White House sent the Federal Communications Commission (FCC) a National Security Determination about "foreign-produced advanced robotic devices." The next day, the FCC added them to its Covered List.[9][10]

The Covered List is an unusual instrument, and it is worth understanding how it works, because it tells you a great deal about how the United States governs technology. It was created by the Secure and Trusted Communications Networks Act of 2019, originally to deal with Chinese telecommunications equipment. Most electronic devices sold in the United States need an FCC "equipment authorisation" because they emit radio waves. Equipment on the Covered List cannot receive one. The FCC cannot add things to the list on its own judgement; it must act when national security agencies tell it to.[9] In practice, that makes the list a ban on importing and selling *new* models of a product category, carried out through radio-frequency paperwork.

The determination's reasoning was the one quoted at the start of this chapter. Networked robots "create extensive vulnerabilities and vectors for attacks that can manipulate the data and physical operation of the advanced robotic system," and they "collect data that could be leveraged by malign actors to surveil Americans … or to remotely commandeer the robots."[9] The listing covers mobile robots "such as humanoids and quadrupeds." It does not affect robots people already own, it does not affect models the FCC had already authorised, and, strikingly, it "does not affect sales to, or use by, the federal government."[9] A producer can escape the list by obtaining a "Conditional Approval" from the Department of War, as the Pentagon now also styles itself.[9][10]

Two details make the July action more sweeping than a simple anti-China measure. First, "foreign-produced" is not limited to adversary countries. The FCC said the risk applies to these products "regardless of the nationality of origin."[9] A robot counts as foreign-produced unless it qualifies as a "domestic end product" under the Buy American Act, meaning, among other things, that more than 65 percent of the cost of its components is domestic.[11] A humanoid assembled in Japan, Germany or Korea is on the list as much as one assembled in Hangzhou, unless its maker wins Conditional Approval. Second, this was the third time in eight months that the FCC had used the Covered List to block a whole category of foreign products by origin, after drones in December 2025 and consumer routers in March 2026.[11]

The Covered List was the fastest instrument, but it was not the only one. On 2 September 2025, the Secretary of Commerce opened an investigation under Section 232 of the Trade Expansion Act of 1962 into whether imports of "robotics and industrial machinery" threaten national security. Its scope is wide: robots and "programmable, computer-controlled mechanical systems," CNC machining centres, presses, tool changers and their parts.[12] Section 232 is the same law that has been used for tariffs on steel, aluminium, cars and, in August 2026, drones. The statute gave Commerce 270 days to report to the President, which meant roughly 30 May 2026.[12][13] As of the end of September 2026, the Congressional Research Service still listed the robotics investigation's outcome as "TBD," and in early October a coalition of technology industry associations wrote to the President asking for "targeted, evidence-based measures" rather than broad tariffs on robots.[13][14] For now, the question hangs over every factory that wants to buy a robot arm from abroad.

Congress has been busy too, although nothing has passed. In November 2025, Senators Bill Cassidy and Chris Coons introduced the Humanoid ROBOT Act (S. 3275), which would bar federal agencies from buying humanoid robots designed, tested or made by entities tied to China, Russia, Iran or North Korea.[15] In March 2026, Senator Tom Cotton introduced the American Security Robotics Act (S. 4235), aimed at federal procurement of adversary-made ground robots.[16] In June 2026, Representatives John Moolenaar, Jay Obernolte and Jennifer McClellan introduced the GUARD Act (H.R. 9129), which would direct national security agencies to decide within a year whether adversary-made humanoid and quadruped robots belong on the FCC Covered List, the very step the executive branch then took on its own a few weeks later.[17] And a pair of bills, H.R. 7334 in the House and S. 4686 in the Senate, would create a national commission to study American competitiveness in robotics.[18]

When I drafted my discussion bill for the United States, the Safe and Accountable Robotics Act, I had to decide what to do with all of this. The honest answer was that I could not write around it. A whole title of my draft ended up being about security: the relationship to the Covered List, to export controls, to Section 232.[19] But the experience left me with an uncomfortable observation, and it is the sociological point of this section. In 2026, the United States has no federal robotics safety statute. It has no federal agency responsible for whether a humanoid robot is safe to stand next to in a kitchen. What it does have, and has used decisively, is national security law. The first comprehensive federal rule about humanoid robots in American homes is not about whether they can hurt you. It is about who made them.

That is not necessarily wrong. A robot with cameras, microphones and a network connection, whose software is updated from a server in another country, really is a different kind of object from a toaster. Chapter 9 argued that ownership without control is hollow; the Covered List is a state's version of the same argument. But it matters which question a society asks first. A country that defines the robot primarily as a security risk will build institutions to watch where robots come from. It will not necessarily build institutions to watch what robots do.

Chips, magnets, and the anatomy of a humanoid

To understand why governments care so much about where robots come from, it helps to take one apart, at least in imagination. A humanoid robot is a body made of motors, gears, sensors and a battery, controlled by a computer running AI models. Two of those ingredients have become instruments of statecraft: the chips that run the brain and the magnets that move the limbs. Each superpower controls one of them.

The brain: chips and export controls

The most capable AI chips are designed mostly by American companies and manufactured mostly in Taiwan. Since 2022, the United States has used export controls to limit China's access to them, on the theory that whoever has more computing power will build better AI. A robot's "brain" depends on the same family of chips, both in the data centres where robot models are trained and in the smaller "edge" processors inside the robot itself. The DSET report noted that China "remains dependent on U.S. frontier edge AI chips and simulation software."[8]

These controls are not fixed walls; they are levers, and in 2026 they were being pulled in both directions. In December 2025, President Trump announced that Nvidia would be allowed to sell its H200 chip, one of its

most powerful, to approved Chinese customers in exchange for 25 percent of the revenue. A Bureau of Industry and Security rule effective 15 January 2026 moved such chips from a "presumption of denial" to case-by-case licensing, with conditions: exports to China capped at half of U.S. sales of the same chip, third-party testing in the United States, and certification that American customers would not be short-changed. A presidential proclamation the next day applied a 25 percent tariff to the chips as they passed through the United States, which was how the government collected its share.[20] Analysts at the Center for a New American Security calculated that the cap alone could let China acquire computing power equivalent to almost 900,000 H200 chips, more than twice what Chinese fabs were expected to produce domestically that year.[20]

Then Beijing hesitated. Chinese customs agents were reportedly told the chips were "not permitted" to enter, and Chinese technology companies were told not to buy them "unless necessary."[20] By mid-May 2026 the U.S. had cleared around ten Chinese firms to buy, but none had received a delivery; in July the head of the Bureau of Industry and Security told Congress that only "very few" had been shipped.[21] Each side was using the chip as a bargaining chip, and each was also trying to protect its own industry. Washington wanted revenue and leverage; Beijing wanted its domestic chipmakers, such as Huawei, to keep growing.

The muscles: rare-earth magnets

If chips are America's chokepoint, magnets are China's. Most powerful electric motors, including the compact, high-torque motors in robot joints, use permanent magnets made from neodymium, iron and boron, often with small amounts of "heavy" rare earths such as dysprosium and terbium added so that the magnet keeps its strength when it gets hot. The International Energy Agency (IEA) lists "magnet rare earths" among the materials most exposed to supply risk, and it explicitly names robotics, alongside semiconductors and AI, as a reason for policy attention.[22] One industry briefing puts China's share of global permanent-magnet production at roughly 90 percent.[23] DSET's researchers were frank about what this means for a non-Chinese robot supply chain: replacing Chinese software, chips and sensors is possible, but reducing reliance on China for motors "would be difficult in the near term because of its dominance in rare-earth magnets."[8]

China used this leverage in 2025. On 4 April, the Ministry of Commerce and the General Administration of Customs issued Announcement No. 18, requiring export licences for seven medium and heavy rare earths (samarium, gadolinium, terbium, dysprosium, lutetium, scandium and yttrium) and for related products, explicitly including samarium-cobalt magnets and neodymium-iron-boron magnet materials containing terbium or dysprosium.[24] The effect was immediate. The IEA reports that some automakers outside China had to reduce output or temporarily halt production.[22] On 9 October 2025, China expanded the controls, adding five more elements, magnet-making equipment and technology, and a provision reaching products made outside China that contain even small amounts of Chinese-origin rare earths. After Presidents Trump and Xi met at the end of October, China suspended the October package for a year by Announcement No. 70, published on 7 November 2025, until 10 November 2026.[22][23] As this chapter was being finished in October 2026, the deadline was close and the outlook unclear. One industry summary reported that a Ministry of Commerce official had said on 28 September that the arrangement would be extended to 10 January 2027, but as of 11 October no formal announcement changing the date had been published, and Western manufacturers were stockpiling magnets in case the pause simply expired.[23] The April 2025 licensing system was never suspended. Every shipment of the listed magnets still needs a Chinese licence.[23]

The IEA's numbers show why magnets are such a powerful weapon. Rare earths make up around 40 percent of the cost of a permanent magnet but less than 1 percent of the value of a car; tripling rare-earth prices would raise the cost of a car by about 0.1 percent.[22] The material is cheap relative to what it enables, which is exactly why cutting it off is so damaging. The IEA estimated that full implementation of China's expanded controls could put about US$6.5 trillion a year of production outside China at risk across the automotive, high-tech, defence and energy sectors. In Europe, heavy rare earths were trading at around five times China's domestic price.[22] And the bottleneck is not only the mineral. The technique used to make heat-resistant magnets with less dysprosium, called grain boundary diffusion, is heavily patented, and the IEA found only one equipment supplier for it outside China, with equipment reportedly costing more than ten times as much.[22]

Mutual dependence, not decoupling

Put the two halves together and you get a strange picture. A humanoid robot built entirely outside China would struggle to get magnets for its joints. A humanoid built entirely inside China would struggle to get the best chips for its brain. Each superpower holds a chokepoint over the other's robots, and each has shown it is willing to squeeze. This is not the "decoupling" of political speeches. It is something closer to a hostage exchange, in which both sides keep their hands on a lever and negotiate summit by summit.

For a sociologist, the lesson is that supply chains are not neutral pipes. They are relationships of power, and they distribute vulnerability. The people who feel that vulnerability first are rarely the presidents at the summits. They are the factory workers on short shifts when the magnets do not arrive, the small robotics start-up that cannot get an export licence processed, and, as the last section of this chapter shows, the communities that live next to the mines and refineries.

Brussels' other weapon: rules as geopolitics

The European Union does not have a humanoid robot champion, and its factories are installing fewer robots, not more. In 2025, robot installations in Europe fell 14 percent to 73,054 units, while Asia's rose 14 percent.[7] What Europe has instead is a large, rich market and the habit of writing detailed rules for it. Those rules are a geopolitical instrument of their own.

The most important for robots is not the famous AI Act but the less famous **Machinery Regulation (EU) 2023/1230**, which replaces the old Machinery Directive and applies from 20 January 2027, with no transition period.[25] It is the first major product law I know of that was written with self-learning machines in mind. Safety components and embedded systems "with fully or partially self-evolving behaviour using machine learning approaches ensuring safety functions" must go through third-party conformity assessment rather than self-certification. A self-evolving control system must not act "beyond its defined task and movement space" and must remain correctable. The regulation adds requirements for autonomous mobile machinery and for protection against the malicious corruption of control systems, which is the European version of the American fear that someone might "remotely commandeer the robots."[25]

The **AI Act** (Regulation (EU) 2024/1689) sits on top of this. It classifies AI by risk: prohibited practices, high-risk systems, systems with transparency duties, and minimal-risk systems that face no new rules.[26] An AI system that is a safety component of a product covered by EU product law, such as a machine, is high-risk. In July 2026, however, the EU postponed the high-risk rules. The "Digital Omnibus on AI," Regulation (EU) 2026/1744, in force since 27 July 2026, moved the rules for high-risk AI embedded in products to 2 August 2028 and the rules for stand-alone high-risk uses, such as AI in hiring, to 2 December 2027. It also clarified how the AI Act fits with the Machinery Regulation, to avoid duplicate requirements.[26][27] Chapter 11 returns to what this delay says about the pace of law.

What makes these rules geopolitical is that they apply to every machine sold in the EU, wherever it was made. The American instrument asks *where* a robot comes from. The European instrument asks *how* it behaves and who is responsible when it does not behave as promised. A Chinese, Japanese or American humanoid can be sold in Europe if it passes the conformity assessment, and cannot if it does not. Scholars call this the "Brussels effect": because the EU market is too large to ignore, manufacturers often design to EU rules and then sell the same product everywhere, so the EU's standards quietly become global ones.

I do not want to romanticise this. European rules can be slow, expensive for small firms, and, as the Omnibus showed, subject to delay when industry complains. But for a country like mine, the European model has one great advantage over the American one. It is a model a middle power can copy. Indonesia cannot put the world's robots on a covered list; it does not have the leverage. It can write behaviour-based rules for robots sold in its market, and it can align them with rules other countries already use.

Japan, Korea and Taiwan: the middle of the supply chain

Between the two superpowers sit three economies that built much of the modern robot industry and now find themselves squeezed by it.

Japan is the country that made the industrial robot ordinary. It remains, by the IFR's account, the world's largest exporter of industrial robots.[28] But in 2025 its home market shrank: installations fell 19 percent to 36,219 units, the lowest in a decade, and Japan slipped from second to third place among the world's robot markets, behind the United States.[6] Its approach to AI law is notably gentle. The AI Promotion Act of 2025 created an AI Strategy Headquarters chaired by the Prime Minister and an AI Basic Plan, but it imposes no penalties; businesses have only a duty to "endeavour" to cooperate.[29] Where Japan does use hard law, it is targeted. Since April 2023, its Road Traffic Act has allowed "Level 4" driverless operation only with a prefectural permit and a designated, trained supervisor who can monitor the vehicle remotely: a named human who answers for the machine.[29]

South Korea has the most robots per worker in the world. In 2024 it had 1,220 industrial robots per 10,000 manufacturing employees, far above the global average of 177.[30] Its installations have been flat at around 30,000 a year since 2019, and the IFR counted 30,000 in 2025.[6] Korea has responded with both industrial policy and law. On 10 April 2025, the Ministry of Trade, Industry and Energy launched the K-Humanoid Alliance, bringing together around forty companies, universities and research institutes with the stated goal of making Korea the world's leading humanoid power by 2030; the ministry expected more than 1 trillion won of investment and set a target of a "robot AI foundation model" by 2028.[31] On the legal side, Korea amended its robot law in 2023 so that certified outdoor delivery robots are legally treated as pedestrians, with mandatory liability insurance for their operators, and its AI Basic Act took effect on 22 January 2026.[29]

Taiwan makes almost no humanoids, but it may be indispensable to everyone who does. Its strengths are semiconductor manufacturing led by TSMC, electronics manufacturers that know how to integrate and scale complex systems, and a dense ecosystem of precision components. DSET identified around 180 Taiwanese entities involved in, or developing capabilities for, the robot supply chain.[8] The government has set aside NT$20 billion for an intelligent robotics programme from 2026 to 2029, and Taiwan's AI Basic Act took effect on 14 January 2026.[32] DSET's analysts argue that Taiwan and other "techno-democracies" have a "golden but narrowing" window of three to five years to build what they call a "non-red" supply chain, one that excludes China.[8] They also point out a fact that is easy to forget in the humanoid excitement: the United States imports more than two-thirds of its industrial robots from Germany and Japan.[8]

That last fact explains why these three economies are in an awkward position. They are America's allies and the most obvious beneficiaries of any effort to replace Chinese robots. But the FCC's July action covers robots "regardless of the nationality of origin," so a Japanese or Korean humanoid needs the same Conditional Approval as a Chinese one.[9] At the same time, Japanese and Korean motor makers depend on the same rare-earth magnets as everyone else. Their companies sell components to Chinese robot makers, whose growing domestic share means fewer orders over time; DSET noted that Taiwanese component sales to China are already declining as Chinese suppliers expand.[8] The middle of a supply chain is a place of opportunity, but also of exposure. You are needed by both sides and protected by neither.

Indonesia: a middle power with nickel in its hands

I live in a country that does not appear on any list of robot superpowers. Indonesia does not publish robot installation numbers that the IFR highlights; it did not send a big team to Beijing; it is not mentioned in the FCC's determination. But Indonesia is inside the geopolitics of embodied AI anyway, through the ground beneath it.

Robots, especially mobile ones, run on batteries, and many lithium-ion batteries use nickel. Indonesia has more nickel than any other country. According to the U.S. Geological Survey, Indonesia mined an estimated 2.6 million tonnes of nickel in 2025 out of a world total of 3.9 million, about two-thirds, and holds 62 million tonnes of the world's more than 140 million tonnes of reserves.[33] The country's share of *refined* nickel has also risen fast: from 23 percent in 2020 to 27 percent in 2023, and is projected to reach 44 percent by 2030.[34]

This did not happen by accident. Indonesia banned exports of raw nickel ore in 2014, relaxed the ban, and banned them again in 2020, forcing companies that wanted Indonesian nickel to process it in Indonesia. The policy, called *hilirisasi* or "downstreaming," is the boldest piece of industrial policy any Southeast Asian country has attempted in my lifetime, and in one sense it worked. Smelters and refineries were built across Sulawesi and North Maluku.

But who built them? The Washington-based research group C4ADS traced the ownership of Indonesia's nickel refining capacity, about 8 million tonnes in 2023, spread nominally across 33 companies. At the second layer of ownership, Chinese companies or shareholders controlled 61 percent of capacity and Indonesian ones just 13 percent. Traced further, Chinese beneficial ownership exceeded 75 percent. Two Chinese groups, Tsingshan and Jiangsu Delong, held stakes in refineries representing more than 70 percent of capacity.[34] Together, Indonesia and China produced 65 percent of the world's refined nickel in 2023.[34] The industry runs largely on laterite ore and coal power. C4ADS also documented more than 90 deaths reported in processing facilities between 2015 and 2023, and the U.S. Department of Labor added Indonesian nickel to its list of goods produced with forced labour in 2024. Meanwhile, the flood of supply helped push nickel prices down 47 percent between January 2023 and July 2024.[34]

Here is the paradox. Indonesia used its sovereignty over a mineral to bring processing onshore, and ended up hosting an industry that is mostly owned by one foreign superpower, sells mostly to that superpower, and has made the mineral cheaper for everyone. I do not think hilirisasi was a mistake. Without it, the processing would simply have happened in China. But it shows what being a "middle power" really means: you can change where things happen, but it is much harder to change who owns them.

Bebas aktif in a divided market

Indonesia's foreign policy has been described since 1948 as *bebas aktif*, "free and active": not aligned with any bloc, but engaged with all. In 2025 and 2026 the country tested that doctrine harder than at any time in my life. In January 2025, Indonesia became the first Southeast Asian full member of BRICS, the group led by Brazil, Russia, India, China and South Africa, a move The Diplomat read as a sign of President Prabowo Subianto's "more proactive approach to foreign policy."[35] A year later, on 19 February 2026, Indonesia signed an Agreement on Reciprocal Trade with the United States. Under it, the U.S. keeps a 19 percent tariff on most Indonesian goods (down from a threatened 32 percent), and Indonesia eliminates tariff barriers on more than 99 percent of American products. Indonesia also agreed to "remove restrictions on exports to the United States for all industrial commodities, including critical minerals," to adopt a ban on imports made with forced labour, and to strengthen workers' freedom of association.[36][37] The White House noted that it might take the agreement into account when deciding on Section 232 actions, the same tool hanging over imported robots.[36]

Jakarta insisted that none of this ends hilirisasi. On the day the agreement was finalised, Energy and Mineral Resources Minister Bahlil Lahadalia said, "There is no change to our policy. We are not opening exports of raw" minerals.[38] Reuters reported that Indonesia also agreed to limit "excess production" by foreign-owned processing plants by holding them to Indonesian mining quotas, a measure that would mostly bite on Chinese-owned refineries.[39] In other words, Indonesia joined a bloc that includes China, signed a deal with the United States that reshapes its Chinese-built nickel industry, and declared that its own policy had not changed. That is *bebas aktif* in 2026: not neutrality, but a constant, careful bargaining with both sides at once.

What a middle power can actually do

When I wrote the *RUU Robotika*, my draft robotics bill for Indonesia, one chapter dealt with robots crossing borders: imports, exports, and the strange new case of a robot in Indonesia being operated by a person abroad.[40] Writing it forced me to think about what Indonesia's real leverage is. I came to three conclusions, and they are the ones I want to leave with the reader.

First, Indonesia's leverage is upstream, not downstream. It will not set the rules for humanoid robots by manufacturing them. It can shape the industry through the minerals that go into the batteries and, increasingly, through the conditions it attaches to processing them: ownership, labour standards, environmental rules and technology transfer. The trade agreement with the United States shows that those conditions are now negotiable, which means they are also a source of power.

Second, Indonesia's protection is in rules, not lists. It cannot ban foreign robots in the American way without banning nearly all robots, since it makes few of its own. What it can do is require any robot sold or operated in Indonesia, whoever made it, to meet behaviour-based safety standards, to be registered, to keep its data

subject to Indonesian law, and to have a legally responsible party in Indonesia. That is closer to the European model, and it works for a country that is a market rather than a maker.

Third, Indonesia's voice is louder in company. Alone, it is a large market with little bargaining power over robot makers. Together with ASEAN neighbours that share standards and certification, it is a market of almost 700 million people that no manufacturer can ignore.

None of these is a dramatic move. None will produce a stadium full of robots. But they are the moves available to a country that holds a mineral everyone needs and a population whose jobs everyone else's robots will affect.

The map and the law

Step back and the map of embodied AI in late 2026 looks like this. China has the factories, the scale, the industrial plan and the magnets. The United States has the chips, the frontier AI models, and a national-security apparatus willing to close its market by list and proclamation. Europe has the rulebook. Japan, Korea and Taiwan have the precision parts and the know-how, and they are trying to become the trusted middle. Indonesia has the nickel and a tradition of refusing to choose.

What strikes me most about this map is how much of it was drawn by instruments that were not designed for robots. A communications law from 2019 became America's robot import ban. A 1962 trade law became the possible basis for robot tariffs. Export control systems built for nuclear materials and military goods now govern magnets and chips. In almost every country, the laws that move fastest on robots are the ones written for something else, by agencies whose job is security or trade, not safety, labour or dignity.

That is the gap between this chapter and the next. Geopolitics decides who gets to build robots and whose robots may enter a market. It says almost nothing about what happens when a robot, wherever it was made, injures a worker, records a family, or replaces a job. Those questions need a different kind of law: one built deliberately, ahead of time, for the machines themselves and for the people who live among them. Chapter 11 asks why that kind of law is so slow to arrive, and how we might build it before we need it rather than after.

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Chapter 11: Law as Social Technology: Responsibility Gaps and Anticipatory Regulation

> "When change is easy, the need for it cannot be foreseen; when the need for change is apparent, change has become expensive, difficult and time-consuming." — David Collingridge, *The Social Control of Technology* (1980)

Four years without a guardian

On 17 October 2022, President Joko Widodo signed Indonesia's Personal Data Protection Law, *Undang-Undang Nomor 27 Tahun 2022 tentang Pelindungan Data Pribadi*, usually called UU PDP. It had sixteen chapters and seventy-six articles, it was modelled in part on the European Union's data protection rules, and it promised Indonesians something they had never had: a single law saying that their personal data belonged to them.[1][2]

The law had taken a long time to arrive. According to the communications ministry's own account, the idea first surfaced in 2012, and the bill entered the five-year national legislation programme for 2014–2019. An academic paper was prepared in 2014. The government's draft was finished around 2016 and sent to the House of Representatives (DPR) with a presidential letter on 24 January 2020. It then went through six extensions of the parliamentary sitting period before the DPR approved it on 20 September 2022.[2][3][4] Measured from the first academic draft, that is about eight years. Measured from the first idea, it is a decade.

What happened next is the story I want to start this chapter with, because it is a story about what comes *after* a law is passed, and it is the part people forget.

UU PDP gave companies two years to adjust, so its obligations became fully binding in October 2024. It also promised an institution. Article 58 said the government would establish a data protection authority, with the details to be set by presidential regulation (*Peraturan Presiden*), and Article 60 gave that authority fifteen powers, from investigating complaints to imposing administrative sanctions.[5] A member of the DPR's Commission I calculated that the authority should have existed by 17 October 2024.[5] It did not. In July 2026, almost four years after the law was signed, a group of advocates and students asked the Constitutional Court to order the government to create the authority within two years. The government told the Court that the draft presidential regulation had been sent to the President on 20 May 2026 and that, in the meantime, a directorate inside the communications ministry was doing the job.[6] The main implementing regulation, *Peraturan Pemerintah* No. 33 of 2026, was finally enacted on 16 July 2026, although copies only began circulating among lawyers in late August.[7] In mid-September 2026, the authority still did not exist. One data-governance expert told CNBC Indonesia that without it, compliance with the new regulation was effectively "voluntary," and predicted that the authority might become effective in 2028 at the earliest.[8][9]

Think about that timeline from the point of view of an ordinary Indonesian. Between 2012 and 2026, the country lived through the arrival of smartphones for almost everyone, the rise of ride-hailing and digital lending apps, the national COVID-19 tracking app, and a series of large data leaks. For most of that period there was no data protection law. For the four years after the law was passed, there was no one to enforce it.

UU PDP is not unusual. The Sexual Violence Crimes Law, UU TPKS, followed a similar path. Komnas Perempuan, the national commission on violence against women, began pushing for it in 2012 and drafted it from 2014. The bill entered the DPR's priority list in 2016, was dropped from that list in July 2020, came back in 2021, was renamed in September 2021, and was finally passed on 12 April 2022. It took about ten years.[10][11] The difference is that once UU TPKS passed, its implementing rules arrived relatively quickly: Komnas Perempuan counted six implementing regulations, three government regulations and three presidential regulations, completed by May 2026.[12]

In the United States the timeline for robots is not even running. As Chapter 10 showed, there is no federal robotics statute. The Occupational Safety and Health Administration says it has no robotics-specific standard and handles robot accidents through the general duty to keep workplaces safe and its rules on machine guarding and lockout.[13] The most consequential federal rule about humanoid robots in 2026 came from the Federal Communications Commission and was about national security, not about whether a robot is safe to stand next to.

I read the UU PDP story in September 2026, when I was revising my own draft robotics bill for Indonesia, the *RUU Robotika*. My draft creates a new agency, the *Badan Robotika Nasional*, and its closing article, Pasal 93, says the agency "must already be established no later than one year" after the law is enacted. Until then, its duties fall to the minister.[14] I had written that article before I knew about the data protection authority. After I read about it, I left the deadline in, but I stopped believing that a deadline written in a law is the same thing as an institution. A legislator can command a future; she cannot make the future obey.

This chapter is about that gap: between the speed of technology and the speed of law, between the law on paper and the institutions that make it real, and between the harm a robot can cause and the person who can be held responsible for it. It argues that the gap cannot be closed by waiting for robots to become common and then legislating. By then it will be too late, in the precise sense the British scholar David Collingridge described in 1980. It argues instead for what I will call *anticipatory law*: building, now, the institutions and rules that will be needed later, while leaving the technical details to be filled in as the technology matures.

The pacing problem and Collingridge's trap

Scholars of technology law have a name for the first gap. They call it the *pacing problem*. Larry Downes put it simply: "technology changes exponentially, but social, economic, and legal systems change incrementally."[15] Law is slow on purpose. It is meant to be deliberated, contested, and stable enough that people can plan their lives around it. A legislature that rewrote its laws every time a company released a new product would not be a legislature; it would be a press office. The legal scholar Ryan Calo has warned that the idea can also be abused: through "libertarian rhetoric," he argues, the pacing problem has become a reason for law not to try at all, which perverts what Collingridge, who was diagnosing a challenge rather than recommending surrender, actually meant.[16]

But the pacing problem is not only about speed. The deeper difficulty is the one Collingridge identified. Early in a technology's life, it is easy to shape. Few people use it, few businesses depend on it, and its design is still changing. But at that stage nobody knows what its harms will be, so there is no clear reason to regulate it. Later, once the harms are visible, the technology has become embedded in factories, supply chains, consumer habits and lobbying budgets, and changing it has become "expensive, difficult and time-consuming."[15] This is the Collingridge dilemma: the moment when we have enough knowledge to act is also the moment when we have the least power to do so.

Robots in 2026 sit right at the edge of this dilemma. Industrial robots are already deep in the second stage. More than five million of them were working in factories around the world in 2025.[17] Their safety is governed mostly by technical standards written by engineers, and those standards work reasonably well for machines bolted to the floor behind fences. Humanoids and other robots that move among people are still in the first stage. Even the International Federation of Robotics, an industry body, describes most humanoids as "prototypes or pilot-stage technologies."[18] We do not yet know how often they will fail, what kinds of harm they will cause, or which jobs they will take. But we can see where the money is going. Chapter 10 described a Chinese state plan to mass-produce them, a stadium with two thousand of them in it, and governments on three continents competing to build supply chains around them. By the time their harms are obvious, they will be very hard to redesign.

The responsibility gap

There is a second gap that robots open up, and it is specific to machines that learn and act on their own. Philosophers call it the *responsibility gap*. Ordinary law assumes that when something goes wrong, there is someone to blame: a driver who was careless, a manufacturer whose product was defective, an employer who

ignored a hazard. A robot that learns its behaviour from data, and acts on the basis of what it has learned, stretches that assumption. Suppose a home-assistance robot drops an elderly man while helping him out of bed. The company that built the robot body says the hardware worked as designed. The company that trained the AI model says the model behaved within its specifications. The company that integrated the two says it followed the manufacturer's instructions. The care agency that deployed the robot says it was certified. The remote operator who was supposed to supervise it, perhaps a person in another country, as Chapter 5 described, says the robot never asked for help. Each claim may be partly true. The man is still on the floor.

The responsibility gap is not mainly a philosophical puzzle. It is a practical one: if everyone can point to someone else, the injured person may end up with no remedy at all. And it is a gap that can be closed by law, if the law is written before the case arrives. The rest of this chapter looks at the tools available for closing it.

Precaution or permission?

The two most famous answers to the pacing problem point in opposite directions.

The first is the *precautionary principle*. Its best-known formulation comes from the 1992 Rio Declaration on Environment and Development, Principle 15: "Where there are threats of serious or irreversible damage, lack of full scientific certainty shall not be used as a reason for postponing cost-effective measures to prevent environmental degradation."[19] The idea is that uncertainty should not be an excuse for inaction. If something might cause serious harm, the burden is on those who want to proceed. Precaution has shaped European regulation of chemicals, food and genetically modified crops, and many critics of AI would like to see it applied to robots.

The second answer is *permissionless innovation*, a phrase associated with the American scholar Adam Thierer. Its claim is the reverse: experimentation with new technologies should generally be allowed by default, and regulators should step in only when concrete harms appear.[15] The argument is that most of the benefits of a technology cannot be predicted in advance, that early rules tend to protect incumbent companies and freeze a technology in its first form, and that harms are better handled by liability, competition and targeted fixes after the fact. Permissionless innovation is roughly the philosophy the U.S. internet grew up under, and it is the explicit philosophy of the current U.S. administration's AI policy. Executive Order 14365, signed on 11 December 2025, declared it the policy of the United States "to sustain and enhance the United States' global AI dominance through a minimally burdensome national policy framework for AI."[20]

Each principle has a characteristic failure. Strong precaution, applied to robots, would mean that no robot could work near people until it was proven safe, which is impossible to prove in advance for a learning machine operating in an open environment. It would push development to countries with looser rules and deprive people of real benefits, such as robots that lift patients so that care workers' backs do not break. Permissionless innovation, applied to robots, means that harms are discovered by the people who suffer them, and that by the time the law reacts, Collingridge's trap has closed. Worse, the costs of that experimentation are not evenly shared. The people who test new technologies first, often without being asked, are frequently those with the least power to refuse: warehouse workers, gig workers, residents of neighbourhoods where delivery robots are piloted, and, as Chapter 12 will argue, people in the Global South.

I do not think either principle, taken alone, is a good guide for robots. Precaution is right that some harms, such as a death or a permanent injury, cannot be undone by a lawsuit. Permission is right that we cannot write good rules for machines that do not yet exist. The more useful question is not "precaution or permission?" but "what can we build *now*, under uncertainty, that will let us act quickly and fairly *later*?" To answer it, we need to look at the shape of rules themselves.

Rules, principles and the sandbox

Rules versus principles

Lawyers distinguish between *rules* and *principles* (or standards). A rule is precise: "the speed limit is 60 kilometres per hour." A principle is general: "drive with reasonable care for the conditions." Rules are predictable, cheap to apply and hard to argue with, but they go out of date and invite people to comply with

the letter while ignoring the purpose. Principles adapt to new situations and capture what we actually care about, but they are uncertain, expensive to enforce and depend heavily on the judgement of whoever applies them.

For a fast-moving technology, the trade-off is sharp. A law that says "a humanoid robot operating within two metres of a person must not exceed 1.5 metres per second" is clear but may be obsolete, or dangerous, within a few years. A law that says "a robot must be designed and operated so as not to create an unacceptable risk to human safety" will never be obsolete, but it does not tell a manufacturer in Semarang what to build.

The practical solution most modern product laws use is a hybrid. The statute states principles, the *essential requirements*, and delegates the precise rules to technical standards that can be updated without amending the law. This is exactly how the EU's new Machinery Regulation works: the regulation sets out what a self-learning safety system must achieve, and harmonised standards written by engineers specify how.[21] It is also how industrial robot safety already works in the United States, where the voluntary consensus standard ANSI/A3 R15.06-2025, the American adoption of the international ISO 10218 standards, does most of the work that a statute would otherwise do.[13]

The EU AI Act's risk tiers

The European Union's AI Act (Regulation (EU) 2024/1689) is the most ambitious attempt so far to combine principles and rules for AI. Its central idea is that obligations should depend on risk. It sorts AI into four levels. At the top are *unacceptable-risk* practices, which are banned outright, such as social scoring, certain kinds of manipulation and, with limited exceptions, real-time remote biometric identification by police in public spaces. Below that are *high-risk* systems, including AI used as a safety component of products, which must meet strict requirements on risk management, data quality, logging, documentation, human oversight and robustness before they can be sold. Then come systems with *transparency* duties, such as chatbots that must tell people they are machines. Finally, *minimal-risk* systems, the vast majority, face no new rules.[22]

The risk-tier approach is a sensible answer to the rules-versus-principles problem. It concentrates detailed rules where harm is most likely and leaves the rest alone. It is also, I should admit, the approach my own drafts copied. The *RUU Robotika* divides robots into four classes, with a fourth, prohibited tier, and the *Safe and Accountable Robotics Act* I drafted for the United States uses three tiers plus a list of prohibited uses, scaled by a robot's mass, speed, closeness to people and how much of its safety-relevant behaviour is learned rather than engineered.[14][23]

But the AI Act also demonstrates the pacing problem in reverse. Its high-risk rules were supposed to apply from August 2026, and for AI embedded in products, from August 2027. In 2026 the EU decided it was not ready. The harmonised standards that companies needed in order to comply had not been finished. The "Digital Omnibus on AI," Regulation (EU) 2026/1744, in force since 27 July 2026, postponed the rules for stand-alone high-risk uses to 2 December 2027 and those for AI embedded in products, including robots, to 2 August 2028.[24] The Commission explained the delay as ensuring that rules "apply when companies have the right support tools to facilitate implementation, such as standards."[22] Here the law was ahead of the engineering, and had to wait for it. That is not a scandal; it is what happens when law tries to anticipate. But it shows that anticipatory law needs a mechanism for adjusting its own timetable.

Sandboxes

One such mechanism is the *regulatory sandbox*. The idea was pioneered by the UK's Financial Conduct Authority, which opened its sandbox to applications in May 2016, calling it "a 'safe space' in which businesses can test innovative products, services, business models and delivery mechanisms while ensuring that consumers are appropriately protected." For its first cohort, the FCA received 69 applications and accepted 24 firms.[25] A sandbox lets a regulator grant limited, temporary relief from certain rules, under supervision, so that both the company and the regulator can learn how a new product actually behaves before permanent rules are written. It is, in effect, a way for the state to do research.

Sandboxes have spread from finance to AI. The AI Act requires every EU member state to set up at least one AI regulatory sandbox; the Digital Omnibus moved that deadline from August 2026 to August 2027 and

added an EU-level sandbox.[22][26] In March 2026, the White House asked Congress to establish "regulatory sandboxes for AI applications" as part of a national framework.[27] My Indonesian draft includes one too, but with a limit I consider essential. Pasal 74 allows participants to be exempted from certain obligations for up to two years, renewable once, but "not from the obligation of liability insurance and responsibility for losses."[14] A sandbox should relax the rules about *how* to build something, not the rules about *who pays* if it hurts someone. Otherwise the sandbox becomes a way to move risk onto the people living next to it.

Liability as infrastructure: the EU's new Product Liability Directive

If rules and sandboxes are ways of preventing harm, liability law is what happens after prevention fails. And it is liability law, more than any other area, that can close the responsibility gap.

The most important development here is the European Union's new **Product Liability Directive**, Directive (EU) 2024/2853, which replaces a directive from 1985. Its changes read almost like a list of answers to the problems robots create.[28]

First, it makes clear that software is a product. The Directive applies to software "irrespective of the mode of its supply or usage," whether it is stored on a device or accessed through the cloud, and it covers AI systems. The 1985 directive had been written for a world of physical goods, and it was unclear whether it covered software at all.

Second, it treats a product as being under the manufacturer's control even after it is sold, when the manufacturer can still change it through software updates. A product can be defective because of a missing update or because of what it learns after it is placed on the market. This matters enormously for robots whose behaviour changes over time.

Third, and most important for the responsibility gap, it eases the burden of proof. Under Article 10, a court must presume that a product is defective if the manufacturer fails to disclose relevant evidence, if the product does not comply with mandatory safety requirements, or if the damage was caused by an obvious malfunction. And where technical or scientific complexity makes it excessively difficult for the injured person to prove defect or causation, a court can presume them if the claimant shows they are likely. All of these presumptions can be rebutted by the manufacturer.[28] In plain terms: when a learning machine causes an injury and its inner workings are too complex for a victim to explain, the law shifts the burden of explanation to the party that built the machine and holds the evidence.

The Directive also has limits. Member states must transpose it by 9 December 2026, and it applies to products placed on the market after that date.[28] It covers injury, damage to property and loss of data, but it does not cover every harm a robot might cause, such as discrimination or loss of a job. And the EU withdrew a separate AI Liability Directive that would have harmonised fault-based claims for AI; its withdrawal was published in October 2025, so fault-based AI liability remains a matter of national law.[29]

Even so, the PLD is the clearest example I know of what anticipatory law looks like in practice. It does not try to predict what robots will do. It establishes, in advance, who will carry the risk when they do something unexpected. I tried to do the same in both of my drafts. The American draft says that a robot is property, never a person, and bars "the robot did it" as a defence. The Indonesian draft adds a robotics compensation fund for victims who cannot identify a responsible party or whose injurer cannot pay, alongside mandatory insurance for operators.[14][23] Liability rules like these are a kind of infrastructure. Like roads, they are most useful if they are built before the traffic arrives.

Two drafts, two constitutions

Over the past months I have drafted two robotics bills: the *RUU Robotika* for Indonesia (version 0.2 has 94 articles in 19 chapters) and the *Safe and Accountable Robotics Act* for the United States (discussion draft 0.1).[14][23] Both are unpublished and unofficial, and both will need many people more qualified than I am to improve them. But writing the same law twice, for two very different constitutional systems, taught me more about law as a social technology than anything I had read. The content of the two drafts is similar. What differs is the machinery each has to work through.

Indonesia: a pyramid of delegation

Indonesia is a civil-law country with a written hierarchy of legislation. Law No. 12 of 2011, as amended, sets the order: the Constitution, then laws (*Undang-Undang*), then government regulations (*Peraturan Pemerintah*), presidential regulations (*Peraturan Presiden*), and below them ministerial regulations (*Peraturan Menteri*) and regional regulations.[30] Criminal penalties can only appear in a law or a regional regulation; they cannot be delegated to a government or ministerial regulation.[30] So every criminal provision in my RUU had to sit in the law itself.

Almost everything else is delegated. My draft contains nine provisions saying that further details "shall be regulated by Government Regulation" and one, on the structure of the new agency and an ethics council, delegated to a Presidential Regulation.[14] This is normal and, in principle, good. It is how a civil-law system separates principles, set by the legislature, from technical rules, set by the executive. It is the Indonesian version of "delegate the technical standards."

But the UU PDP story shows the danger. Delegation only works if the delegate acts. A law that delegates its institutions and details to regulations that are never issued is a law that exists on paper only. That is why my draft includes Pasal 92, which requires all implementing regulations within one year, and Pasal 93, which requires the agency within a year and gives its powers to the minister in the meantime.[14] I know now that these deadlines may be ignored. I have kept them anyway, for two reasons. A deadline gives citizens something to take to the Constitutional Court, as the UU PDP petitioners did. And a fallback rule, under which the minister acts until the agency exists, means the law is never completely without an enforcer.

There is a further obstacle before any of this. A bill has to get onto the national legislation programme (*Prolegnas*) before the DPR will consider it. Reporting on the 2025–2029 Prolegnas and the 2026 priority list found no bill on AI or robotics.[31] The government's planned presidential regulations on AI ethics and a national AI roadmap, which the communications ministry describes as a first step toward a future AI law, had still not been signed as of 12 October 2026.[31] Indonesia's realistic path to a robotics law is long, and it begins with getting the subject onto the agenda.

The United States: federalism and its three tools

The U.S. Constitution distributes power differently. Congress can legislate only on subjects the Constitution assigns to it, and the most important of these for robots is the power to regulate interstate commerce (Article I, Section 8). Robots are built, sold and updated across state lines, so the Commerce Clause gives Congress ample authority, and my draft relies on it expressly.[23] But the states have their own broad powers to protect health and safety, and they have used them. In the absence of federal law, states have written rules for delivery robots on sidewalks, autonomous vehicles on roads and, increasingly, AI in hiring and lending.

Three legal tools govern the relationship between the federal government and the states, and each shaped my draft.

The first is **preemption**: a valid federal law overrides conflicting state law. The design question is how much to preempt. My draft makes federal design and performance standards for robots a *ceiling*: states cannot impose different product requirements, so a manufacturer builds to one national standard. But rules about how robots are *used* (on sidewalks, by local police, in workplaces) and protections for workers and consumers are a *floor* that states may exceed, and state tort claims are expressly preserved.[23] A robot maker should not have to build fifty different robots. A city should still be able to decide whether robots may use its sidewalks, and an injured person should still be able to sue.

The second tool is the **anti-commandeering doctrine**. The Supreme Court has held that Congress cannot order state legislatures to pass laws or order state officers to enforce federal programmes (*New York v. United States*, 1992; *Printz v. United States*, 1997; *Murphy v. NCAA*, 2018).[32] So a federal robot law must be enforced by federal agencies. Section 1303 of my draft states this directly: nothing in the Act requires a state, or any state officer, to enforce it.[23] This is one reason my draft places a National Robotics Safety Office inside the Department of Commerce. Someone federal has to do the work.

The third tool, in 2026, is the fight over preemption itself, and it is happening mostly outside Congress.

Executive Order 14365 directed the Attorney General to create an AI Litigation Task Force "whose sole responsibility shall be to challenge State AI laws" inconsistent with the administration's policy, including on the ground that they "unconstitutionally regulate interstate commerce." It directed the Commerce Department to identify "onerous" state AI laws and made states with such laws ineligible for certain federal broadband funds. It also asked the FCC to consider a federal AI reporting standard that would preempt state rules.[20] The Department of Justice established the task force in January 2026.[23] In March 2026 the White House published legislative recommendations asking Congress to "preempt state AI laws that impose undue burdens," while leaving states their general powers to protect children, prevent fraud and protect consumers.[27]

But an executive order cannot itself preempt state law; only Congress, or a valid federal regulation, can. Congress has so far declined. A proposed moratorium on state AI laws was stripped from the 2025 budget bill by a 99–1 Senate vote, and preemption language was left out of the defence authorisation bill as well.[33] Some states have retreated under pressure; Colorado repealed its pioneering AI Act and replaced it in May 2026 with a narrower law on automated decision-making.[34] As of October 2026, no federal AI preemption statute has been enacted.[23]

What the comparison teaches

Writing both drafts, I came to see that each system has a characteristic weakness that mirrors the other's strength.

Indonesia's centralised hierarchy makes it easy, on paper, to write one national robotics law and fill in the details by regulation. There are no fifty states to negotiate with. But that same centralisation means everything depends on the executive actually issuing the regulations and building the institutions, and when it does not, there is no other level of government to step in. The UU PDP authority is the proof.

America's federalism makes a national law politically hard to pass, and in 2026 the federal government is using that difficulty to *block* regulation rather than create it. But federalism also provides a kind of insurance. While Washington debates, states can experiment. Some of their laws will be bad, and the preemption fight is partly a fight about which ones. But a world in which Congress never legislates and also forbids the states from doing so would be the worst of both systems: a responsibility gap with constitutional protection.

The two systems, in short, fail in different directions. Indonesia's risk is a law without institutions. America's risk is institutions without a law. Both drafts were my attempt to design against their own country's characteristic failure.

The case for anticipatory law

Let me now gather the argument. The pacing problem is real, and Collingridge's dilemma means that waiting until robots' harms are obvious will leave us with little power to change them. Precaution cannot work for learning machines whose risks cannot be proven in advance, and permissionless innovation hands the costs of experimentation to the least powerful. Indonesia's experience shows that a law can take a decade to pass and then wait years more for its institutions. America's experience shows that, without a statute, robots end up governed by laws written for something else.

The alternative I propose is *anticipatory law*. It does not try to predict what robots will do. It builds, ahead of time, the capacities a society will need to respond quickly and fairly when they do something we did not predict. It has five parts.

First, build the institutions now. An institution takes years to staff, fund and earn trust; it cannot be created the week after a disaster. Whether it is a new agency or an office within an existing ministry, someone must be legally responsible for robots before robots are common. The UU PDP authority should be a warning to every legislator who writes "shall be established" and moves on. My own answer is a deadline, a fallback to the minister, and a modest institution rather than a grand one, because a small agency that exists is worth more than a large one that does not.

Second, build registries. You cannot govern what you cannot see. A registry of robots above a certain risk level, recording who made them, who operates them and who answers for them, is cheap to set up while the

numbers are small and very hard to set up once millions of machines are already in use. A registry also makes the rest of the law possible: insurance, recalls, incident investigation and, as Chapter 9 argued, protection for owners against remote disablement.

Third, settle liability in advance. The responsibility gap is a choice, not a natural fact. The law can decide now that a robot is property, never a person; that "the robot did it" is not a defence; that when a learning machine's complexity makes proof impossible, the burden shifts to the party that built and controls it; and that operators of risky robots must carry insurance. The EU's Product Liability Directive shows this can be done without predicting any specific technology.

Fourth, protect workers before displacement, not after. Chapter 3 argued that fear of automation is rational even when unemployment is low, because the costs of transition fall on particular people in particular places. Notice requirements, consultation duties, retraining and a public bridge for displaced workers cannot be improvised once a factory has already automated. They must exist before the decision is made, so that the decision is made in the open.

Fifth, delegate the technical standards, with deadlines and review. The law should state principles and essential requirements, and leave precise technical rules to standards bodies and regulations that can be updated as the technology changes. But delegation must come with deadlines, fallbacks and a duty to report back, and the law itself should be reviewed at fixed intervals. My American draft requires a five-year review of its tiers, insurance amounts and preemption provisions.[23] Both drafts give the institution power to adjust thresholds without a new act of the legislature, within limits the legislature sets.

None of this requires knowing what a 2035 humanoid will be able to do. It only requires deciding, now, who will be watching, who will be responsible and who will be protected. That is what law is good at. A sociologist would say that law is a *social technology*: a tool societies invented for coordinating behaviour among strangers and for distributing risk and responsibility. Like any technology, it has a design, and the design can be good or bad. The argument of this chapter is that the design of robot law should be chosen deliberately and early, rather than inherited by default from communications law, trade law and the accidents that will otherwise do the choosing for us.

There is one more thing the comparison between my two drafts taught me. Both Indonesia and the United States are, in their different ways, deciding about robots that will be built mostly elsewhere. The robots that will work in Indonesian factories and homes will very likely be made in China, Japan, Korea or the West, trained on data gathered elsewhere, and updated from servers abroad. The questions that matter most for Indonesians are not only how to regulate a machine, but how a country that does not make the machines can make sure that their benefits reach its people rather than flowing out with the profits. In a country where most workers are in the informal economy, where labour law was written for factories, and where nickel from Sulawesi goes into batteries for robots assembled somewhere else, anticipatory law has to anticipate something more than accidents. It has to anticipate development itself. That is the subject of Chapter 12.

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Part V: Futures Without Fiction

Chapter 12: Robots in the Global South: Indonesia, Informal Work, and Development

> "Countries are running out of industrialization opportunities sooner and at much lower levels of income compared to the experience of early industrializers." — Dani Rodrik, "Premature Deindustrialization," 2015 [1]

Rp19,400

On October 8, 2026, the Indonesian government published Presidential Regulation (Perpres) No. 27 of 2026 on the Protection of the Digital Platform-Based Transportation Ecosystem, which everyone in Indonesia calls the Perpres Ojol, after the *ojek online*, the app-based motorcycle taxis that carry passengers, food, and parcels through every Indonesian city [2]. Its headline provision caps the fee that an application company may take from a driver at 8 percent, so that drivers keep at least 92 percent of the fare, down from a cut that, according to the government, had been as high as 20 percent. The Minister of Transportation said the 8 percent cap had already been set in a ministerial decree in force since July 1, 2026, and that the Perpres adopted it [2]. The regulation also confirmed that drivers remain *mitra*, partners of the platform, rather than employees, and gave the Ministry of Micro, Small and Medium Enterprises the task of monitoring the partnership agreements between drivers and platforms [3].

Two days later, the chair of a drivers' union, the Serikat Pekerja Angkutan Indonesia (SPAI), Lily Pujiati, told *Liputan6* that the arithmetic on the road did not match the headline. On a passenger trip with a fare of Rp19,400, she said, the platform first deducted Rp5,500, made up of a Rp4,500 application service fee and a Rp1,000 trip insurance fee. Only then did it take its official 8 percent commission from what was left. The driver received Rp12,880, about a third less than the passenger paid [3]. A drivers' association leader said they were still waiting for the implementing ministerial regulations, which were expected before December 31, 2026, to clarify the 8 percent cap, the partnership status, social protection, and sanctions [3].

There is not a robot in this story, and that is why it opens this chapter. The *ojol* driver is the most visible figure in Indonesia's encounter with algorithmic management. Millions of Indonesians already work for a machine, in the sense that an app assigns their tasks, sets their prices, and rates their performance. Their legal status is not employee but partner. Their bargaining happens through street protests and presidential announcements rather than collective agreements. And their income depends on a fee structure that the state can cap but that platforms can rearrange. When robots arrive in Indonesia in large numbers, this is the social arrangement they will enter: a labour market where most people are not employees, where the law's protections are built around a minority who are, and where the most important economic relationships are already mediated by software owned elsewhere.

The previous chapters of this book were written mostly about the United States, Europe, Japan, and China, because that is where most robots are. This chapter asks what changes when the same technology arrives in a country like Indonesia: lower-middle income by most measures, young, mostly informal, rich in the minerals that robots and batteries need, and still hoping to become a high-income country within a generation. The answer is that almost everything changes, because the arrangement a robot creates depends on the society it lands in.

Sixty in every hundred

The single most important fact about work in Indonesia is that most of it is informal. According to Statistics Indonesia (BPS), 59.30 percent of employed Indonesians, about 87.88 million people, worked informally in May 2026, a slight decline in share from 59.42 percent in February 2026, even as the absolute number rose [4]. BPS counts as informal those who work on their own account, those who run a business helped by temporary or unpaid family workers, casual workers, and unpaid family workers. Roughly six of every ten working

Indonesians are therefore outside the formal employment relationship on which most labour protection is built.

The largest single source of work is agriculture. In February 2026, agriculture, forestry, and fisheries employed 42.49 million Indonesians, 28.78 percent of everyone working, more than any other sector [5]. Most of these farms are very small. The 2023 Agricultural Census counted 28.42 million agricultural households, up 8.74 percent from 2013, and 29.34 million individual farm enterprises [6]. The number of *petani gurem*, smallholder households farming less than half a hectare, rose to 16.89 million, an increase of 18.54 percent over the decade [7]. In some provinces, the share is overwhelming: BPS reported that 87.75 percent of agricultural households in Yogyakarta were *gurem*. A BPS official explained the trend plainly: people keep farming, but their land keeps shrinking, sold or divided by inheritance [7]. Farmers are also ageing. Only about 6.18 million farmers, 21.93 percent, were aged 19 to 39, the group the census calls millennial farmers [6].

Outside agriculture, the informal economy is dominated by micro and small enterprises: warung, street vendors, workshops, home producers, and the drivers and merchants who depend on platforms. The Ministry of MSMEs, citing BPS data, said in 2025 that Indonesia had 65.5 million micro, small, and medium enterprises, contributing 61.9 percent of GDP and employing more than 119 million people, about 97 percent of the national workforce [8]. Many of these enterprises are a single person. The Minister of MSMEs said in October 2026 that about 15 million merchants used digital platforms, with delivery services connecting them to customers [3].

This structure changes how automation arrives. In a rich country, the classic automation story begins with a firm, a payroll, and a decision to replace workers on that payroll with machines. The displaced worker has an employer, a contract, and often a union. In Indonesia, most people have none of these. Automation reaches them indirectly: through prices, through platforms, through competition from firms that have automated, and through the slow disappearance of the formal jobs that informal workers hope one day to enter. A smallholder in Central Java is not laid off when a rice-harvesting machine arrives in the next district. Her labour is simply worth less at harvest time. A *warung* owner is not dismissed when an automated warehouse speeds up e-commerce delivery. Her customers simply buy elsewhere. An *ojol* driver is not fired by a robotaxi. If robotaxis ever operate in Indonesian cities, his fares will simply fall, or his account will simply receive fewer orders.

Agriculture shows the double edge most clearly. Indonesian farming is ageing, and the plots are small and shrinking [6][7]. For a 55-year-old smallholder with bad knees, a machine that transplants seedlings or sprays a field is not a threat but a relief, and for a village that cannot find young people willing to work in the mud at harvest, mechanisation may be the only way to keep land in production. At the same time, the landless labourers who earn wages at planting and harvest time, often the poorest people in the village, depend on exactly the tasks that machines take first. The same machine that rescues an ageing owner can erase a labourer's seasonal income. Whether it does one or both depends on who owns it, who operates it, and whether the labourers have anywhere else to go. That is a question of arrangement, not engineering.

This is why the most common policy instincts from rich countries, retraining programmes for displaced employees, severance, notice periods, often miss the people most exposed in Indonesia. They are designed for the employee being let go, and most Indonesians are not employees.

A window that closes

Indonesia's long-term development plan is organised around a date. Law No. 59 of 2024 on the National Long-Term Development Plan 2025–2045 (RPJPN) commits the country to the vision of *Indonesia Emas 2045*, "Golden Indonesia," on the centenary of independence [9]. Its main targets are ambitious: per capita income of about USD 23,000 to 30,300, comparable to developed countries; a poverty rate of 0.5 to 0.8 percent; a Gini ratio of 0.29 to 0.32; and, important for this chapter, a manufacturing share of GDP of 28 percent [9].

The plan rests heavily on a demographic argument. Indonesia's population is still dominated by people of working age, which in principle allows a rising share of earners to support a falling share of dependants, a "demographic dividend." But the window is not open forever, and the planning ministry has been candid about how much of it has already passed. A Bappenas policy book on population quality, prepared for the

long-term plan, states that Indonesia was already in the late phase of the peak of its demographic bonus, that people aged 60 and over exceeded ten percent of the population by 2021, and that under its population projections the demographic bonus ends, at the earliest, in 2039 under the optimistic scenario [10]. The same book contains a harder sentence. At current levels of productivity, it estimates, the demographic bonus contributes on average only about 0.1 percent a year to economic growth and that contribution ends in 2026. Only if productivity rises can the economic benefit be extended [10]. These are Bappenas projections, not certainties, but they come from the institution charged with planning the path to 2045.

Read together, these documents set out the problem that robots enter. Indonesia must raise productivity quickly, before its population ages. The traditional route to rapid productivity growth for a country with many young workers has been labour-intensive manufacturing: move people from low-productivity farms and informal services into factories, where each worker produces more, and use the export earnings to climb toward higher-value industry. That is how Korea, Taiwan, and later China did it. The RPJPN's target of a 28 percent manufacturing share is a commitment to that route [9]. Robots threaten to narrow it, because the more a factory relies on machines, the fewer young workers it needs to absorb.

Nickel, batteries, and the ladder

Indonesia's most visible industrial strategy of the past decade has been *hilirisasi*, downstreaming: forcing the processing of raw materials to happen at home rather than abroad. Nickel is the flagship. Indonesia banned the export of nickel ore from 2020, as the United States Geological Survey's 2026 summary records alongside its later bans on bauxite and copper concentrates [11]. Investment in smelters followed. By 2025, according to the USGS, Indonesia produced about 2.6 million of the world's estimated 3.9 million tons of mined nickel, around two-thirds of global output, and its production had risen an estimated 13 percent in a single year as new operations ramped up [11].

Nickel matters because it is used in many lithium-ion battery chemistries, and the government's ambition is to climb from ore to smelting to battery materials to battery cells to electric vehicles. On July 3, 2024, President Joko Widodo inaugurated the battery cell plant of PT Hyundai LG Indonesia (HLI) Green Power in Karawang, West Java, which he described as the first and largest electric vehicle battery cell factory in Southeast Asia, part of a Korean investment package he put at Rp160 trillion to be completed in stages. "The mines are here, the nickel is here, the bauxite is here, the copper is here," he said, describing a chain that would run from smelters to cathodes and precursors to batteries to car factories, all in Indonesia [12].

This is a serious development strategy, and it has produced real growth in industrial output. But it raises a question that this book's thesis forces us to ask: who works in the new factories, and who owns them? Smelters and battery cell plants are capital-intensive. Modern battery cell production in particular is typically highly automated, because the precision and cleanliness required are hard to achieve by hand. A strategy that captures more of the value chain may therefore capture more output without capturing proportionally more jobs. The value added stays in Indonesia; the question is how much of it reaches Indonesian workers rather than the owners of the capital, much of which is foreign.

The economist Dani Rodrik gave this worry a name. In a 2015 paper, he documented what he called premature deindustrialization: across developing countries, the hump-shaped relationship between industrialization and income had shifted downward and toward lower incomes, so that countries were "running out of industrialization opportunities sooner and at much lower levels of income compared to the experience of early industrializers" [1]. He found Asian countries and manufactures exporters had been largely insulated, while Latin America had been hit hardest, and he identified both globalization and labour-saving technological progress in manufacturing as causes. He warned of potentially significant consequences, including lower growth and "democratic failure" [1].

Whether Indonesia is already deindustrializing prematurely is contested, and the dispute is worth stating honestly. Critics compare today's manufacturing share of GDP, around 18 to 19 percent in recent years, with the higher shares recorded in earlier decades. The Ministry of Industry rejects the diagnosis. Its spokesperson said in May 2026 that manufacturing grew 5.30 percent in 2025, faster than the economy as a whole for the first time since 2011, that its share of GDP rose from 17.92 percent in the second quarter of 2022 to 19.20 percent in the first quarter of 2026, that manufacturing employment rose from 18.7 million to 20.3 million

between 2021 and 2025, and that long-run comparisons are distorted by changes in how BPS classifies and measures the sector [13]. Both sides have a point. What is not in dispute is the distance between roughly 19 percent today and the 28 percent the long-term plan targets for 2045 [9][13].

The nickel story also shows how a development strategy redistributes risk as well as reward. The returns from downstreaming are concentrated where capital is concentrated: in the companies that own the smelters and plants, many with foreign investors, and in the state's revenues. The risks, from land conversion to air and water pollution to occupational hazards in heavy industry, are concentrated where the operations are, in mining and smelting regions far from Jakarta, often among communities that had little say in the decision. Automation inside these plants can reduce some of the most dangerous tasks, which is a genuine gain for the workers who remain. But it also reduces the number of local jobs that were the main argument for accepting the risks in the first place. A downstreaming strategy that becomes heavily automated without any mechanism to share its returns locally risks keeping the costs at home while the jobs, the main local benefit, shrink.

Robots make that distance harder to close in a specific way. The International Federation of Robotics reported in April 2026 that the global average density of industrial robots was 132 per 10,000 manufacturing employees in 2024, with South Korea at 1,220, Singapore at 818, Germany at 449, Japan at 446, and China at 166 after a revision to its labour data [14]. China alone installed 54 percent of the world's new industrial robots in 2024, and its operational stock of around two million units was about four and a half times Japan's [14]. In September 2026, the IFR reported that about five million industrial robots were now operating in factories worldwide [15]. Indonesia does not appear among the leaders in the IFR's published highlights. Its manufacturers face competitors that combine low labour costs with heavy automation, which compresses the advantage that cheap labour used to give a late industrializer.

This is the core of the development dilemma. If Indonesian factories do not automate, they may lose competitiveness to those that do. If they automate as fast as China, they may not create the millions of formal jobs that a young, mostly informal workforce needs. Neither path, on its own, delivers the 2045 vision. What matters, as throughout this book, is the arrangement: who owns the machines, who is trained to work with them, and where the gains go.

Laws written for factories

Indonesia's legal response to automation so far has three layers, and each was designed for a world that is not quite the one Indonesia lives in.

The first layer is **labour law**. Indonesia's regulation on fixed-term contracts, outsourcing, working time, and dismissal, Government Regulation No. 35 of 2021, implements the Job Creation Law. It defines an employment relationship as one based on a work agreement containing work, wages, and command [16]. If a dismissal cannot be avoided, the employer must notify the worker, and the union where the worker is a member, in writing at least fourteen working days in advance, stating the reason and the compensation. A worker who rejects the dismissal must respond in writing within seven working days, and a dispute must first be settled through bipartite negotiation between employer and worker or union before it moves to the industrial relations dispute mechanism [16]. Severance and long-service pay are set on a schedule, and for micro and small enterprises they are left to agreement between the parties [16].

These are real protections, and for a formal employee replaced by an automated line they matter. But look at what they assume: a single employer, a written contract, a dismissal event, and a union at the workplace. For the 87.88 million informal workers, none of these exist [4]. There is no notice letter when a rice harvester reduces the demand for harvest labour. There is no bipartite negotiation between a *warung* owner and the e-commerce warehouse that took her customers. For *ojol* drivers, the Perpres has explicitly confirmed partner status rather than employment [3]. The notice-and-negotiation model of labour law is built around a moment, the layoff, that most Indonesians exposed to automation will never experience. It catches the minority and misses the majority.

The second layer is **AI and digital policy**, led by the Ministry of Communication and Digital Affairs (Komdigi). Indonesia launched a National Strategy for Artificial Intelligence in 2020, setting directions

through 2045 and naming five priority areas: health services, bureaucratic reform, education and research, food security, and mobility and smart cities [17]. In December 2023, the then Ministry of Communication and Informatics issued Circular Letter No. 9 of 2023 on AI ethics, a non-binding guideline without sanctions [18]. Since 2025, Komdigi has been preparing two presidential regulations, one on a National AI Roadmap for 2026–2029 and one on AI ethics; according to GovInsider, the roadmap regulation would also create a National AI Coordination Task Force [19]. In July 2026, the deputy minister, Nezar Patria, said the government had finished drafting the regulations and described the AI Perpres as a "test case" and a first step toward a comprehensive AI law [20]. At the end of September 2026, GovInsider still described both as drafts, and no signed text had been published [19].

This is reasonable sequencing for software. But it has two limits for the subject of this chapter. It is about artificial intelligence in general, mostly software, and has little to say about physical robots: their safety, their liability, their registration, or their effect on work. And it is organised around adoption and ethics rather than distribution. A roadmap that tells ministries how to use AI responsibly does not say who should own the machines or how the gains should be shared.

The third layer is **platform regulation**, of which the Perpres Ojol is the newest example. It is the closest Indonesia has come to regulating algorithmic management, and its approach is instructive: rather than reclassifying drivers as employees, the state caps the platform's share, mandates social protection, and assigns a ministry to supervise partnership agreements [2][3]. It accepts informality and tries to make it fairer. As the SPAI's arithmetic shows, enforcement will depend on details that remain to be written [3].

Put these layers together and a gap appears exactly where the next decade's automation will land. Labour law protects employees from dismissal. AI policy guides software adoption. Platform regulation sets fees for one sector. Nothing yet addresses the slow, diffuse displacement of informal livelihoods by machines owned by others, or the question of whether Indonesians will own any of those machines themselves. The RUU Robotika I drafted tries to fill part of this gap, with a duty on the central government to run retraining programmes for workers affected by automation, fiscal incentives for employers who retrain and redeploy them, a right for workers to stop work when a nearby robot endangers them, and a ban on using robots for degrading continuous surveillance or for dismissal decisions without a human decision [21]. But those provisions, too, mostly protect people who have an employer. Writing them made me realise how much of Indonesia they do not reach.

Who should own the robots

If robots are social arrangements, then the most important design choice is ownership. A robot owned by a distant firm and deployed in Indonesia sends most of its returns abroad and displaces local work. The same robot owned by a cooperative of the farmers whose harvest it gathers keeps the returns in the village and turns displacement into a reduction of drudgery. The technology is identical. The arrangement is not.

What follows are proposals. None of them is current Indonesian law, and none has been tested at scale. They are offered in the spirit of this book's argument that law should build institutions before a technology becomes entrenched, not after.

An automation transition fund. My draft RUU Robotika already creates a Robotics Compensation Fund, a *Dana Kompensasi Robotika*, to pay victims of robot incidents when no other source of compensation is available [21]. I now think it needs a sibling: a fund dedicated to the transition of workers and communities affected by automation, including those who are not employees. The draft's own drafter's notes reject a per-unit "robot tax," on the grounds that no country applies one and that the European Parliament rejected the idea in 2017, and instead propose adjusting existing investment incentives such as super-deduction and tax holiday facilities [21]. The same logic can finance a transition fund: a share of the fiscal incentives granted to highly automated investments, including in the downstream nickel and battery industries, could be earmarked for it. Unlike severance, which is paid by one employer to one employee, the fund could support people whose losses have no employer attached: smallholders whose harvest labour is displaced, drivers whose orders fall, small producers who lose a market. Its eligibility rules would need to be based on sector and region rather than on a dismissal letter, and it should be governed with representation from informal workers' organisations, not only employers and formal unions.

Robot-owning cooperatives. Indonesia already has an unusually large cooperative infrastructure, and it has just been expanded. Presidential Instruction No. 9 of 2025, signed on March 27, 2025, ordered ministries and regional governments to accelerate the formation of 80,000 village and urban ward cooperatives, the *Koperasi Desa/Kelurahan Merah Putih*, with activities including staple-goods supply, savings and loans, clinics, pharmacies, cold storage and warehousing, and logistics [22]. By July 2026, the cooperative information system recorded 83,382 cooperatives formed, with transactions worth Rp56.69 billion in 2026 so far, the largest by value being subsidised NPK and urea fertiliser [23]. Most of these cooperatives are young, and their business models are still forming. That is an opportunity. A village cooperative that already handles fertiliser, storage, and logistics is a natural owner of shared agricultural machines: spraying drones, transplanters, harvesters, grain dryers, and, in time, more autonomous field robots. Smallholders with half a hectare cannot buy such machines individually; a cooperative of several hundred can. If the cooperative owns the machine, members share both the labour savings and the rental income from neighbouring villages, and the data the machine collects about local fields stays under local control. A national programme could offer concessional finance for cooperative purchases of approved robots, with maintenance training tied to the purchase so that repair skills, and the jobs that come with them, stay in the region. Chapter 9 explained why ownership without control is hollow; a cooperative that owns a machine it cannot repair, or whose software can be switched off from abroad, owns very little. Any such programme should therefore require repairability and local data access as conditions of financing.

Access for UMKM and the 3T regions. The risk of automation in a country as unequal in geography as Indonesia is that it deepens the divide between Java's industrial corridors and the rest. The 3T regions, *terdepan, terluar, tertinggal*, the frontier, outermost, and least developed areas, are least likely to attract private investment in robots and most likely to lose markets to more automated competitors elsewhere. The RPJPN explicitly targets reduced regional inequality, including a larger share of GDP from eastern Indonesia [9]. Three tools could help. First, shared robotics service centres, attached to vocational schools, polytechnics, or cooperatives, where micro and small enterprises can rent automated equipment by the hour rather than buy it, as a food processor might rent time on a packaging line. Second, public procurement that reserves a share of contracts for robotics solutions developed or assembled by Indonesian small firms, so that domestic capability grows with demand. Third, a deliberate priority in the AI roadmap and in any future robotics law for applications that serve the 3T regions directly: agricultural machines suited to small and sloping plots, inspection drones for remote infrastructure, and remote-assistance services that can bring scarce expertise to places that lack it. Chapter 5 showed that remote operation can also be an export: Indonesians can operate and train machines located abroad. A policy that builds those skills in the 3T regions, with the protections that ghost work has lacked, could turn a channel of exploitation into a channel of development.

A share in the data. Chapter 5 described Indonesians already being paid by the hour to record their own household chores as training data for foreign humanoid robots. That is the first, quiet form in which robotics is reaching Indonesian homes: not as machines, but as a demand for human demonstrations. The value of those demonstrations lies in the models they help build, which will be owned and monetised elsewhere. A development-minded policy would not try to stop this work, which provides real income, but it could insist on transparency about who buys the data and for what purpose, on clear consent for anything recorded inside a home, and on encouraging Indonesian cooperatives and firms to aggregate and license such data collectively rather than selling it hour by hour. Indonesia's personal data protection law gives a starting point for consent; collective bargaining over data is the harder and more important step.

There are honest objections to each proposal. Funds can be captured by the politically connected. Cooperatives in Indonesia have a long history of mismanagement, and the speed of the Merah Putih programme raises real questions about governance. Shared service centres can become white elephants. None of these objections is a reason to leave ownership to chance. They are reasons to design institutions carefully: with transparency about who receives money, with members who actually control their cooperatives, and with evaluation built in from the start.

I live in Semarang, in a province where rice fields, garment factories, and *ojol* drivers sit within a few kilometres of each other. The choices described in this chapter are not abstract to me. Robots could arrive in Central Java as machines owned by distant firms, compressing the value of the work my neighbours do while the gains flow elsewhere. Or they could arrive as tools owned by the cooperatives, small firms, and young

engineers of the region, taking over the most exhausting tasks while widening what ordinary people can do. Which of these happens will not be decided by the robots. It will be decided, mostly in the next ten years, by fiscal rules, cooperative law, procurement policy, and whether lawmakers notice that most Indonesian workers are not employees.

This chapter has been about institutions: funds, cooperatives, regulations, and plans. Underneath all of them is a question that institutions alone cannot answer. If the market will increasingly price human labour against machines, and if most people in a country like Indonesia already work without the protections that a formal job provides, what is a human being worth, and what are we owed simply for being human? The final chapter turns to that question, and to the outline of a social contract for an age of machines.

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Chapter 13: The Value of a Human — Toward a Robotic Social Contract

> "The question is not whether machines will replace us, but whether we will remember what we were for." — the author

The question beneath all the others

Each chapter of this book began with a concrete scene: a fence around a robot arm, a warehouse quota, a teleoperator's screen, a grandmother and a seal-shaped robot, a robot vacuum's map of a living room, a wheelchair stuck behind a delivery robot, a farmer locked out of his own tractor, an FCC list, a draft bill, an *ojol* driver in Semarang. Underneath each scene ran the same question, usually unspoken: what is a human being for, and what is a human being owed, when machines can do more and more of what humans do?

Economics tends to answer by pricing labor. A worker is worth the value of what they produce. By that measure, the robot is a threat in principle. Whenever a machine can produce the same output more cheaply, the market value of the human doing that task falls toward the cost of the machine. Much of the anxiety about automation is the fear that this logic, followed to its end, leaves people worth nothing.

But that measure has never been the whole story, and it was never the story that societies actually lived by. We do not value children by their output, or the elderly, or the sick. We do not think a person loses dignity on the day they retire. Nearly every moral and religious tradition holds that human worth is not earned and cannot be lost by being outperformed. Indonesia's state philosophy puts *kemanusiaan yang adil dan beradab*, just and civilized humanity, as its second principle, before any principle about the economy.

The real danger, then, is not that robots will make humans worthless. It is that we will let the market's narrow measure of worth become the only one our institutions recognize: that we will keep tying income,

status, security, and belonging to paid labor even as paid labor is automated, and then act surprised when people feel discarded.

What the evidence actually says

It is worth being careful here, because both optimists and pessimists overstate their case.

The evidence reviewed in Chapter 3 does not show that robots have produced mass unemployment. In most rich countries, employment rates in 2026 are not lower than they were before industrial robots spread. It does show that robots have displaced specific workers in specific places, often permanently, and that the people displaced rarely captured the gains. Studies of US local labor markets found measurable negative effects of robot exposure on employment and wages. Studies of Germany found that displacement was absorbed more through reduced hiring of young workers and shifts within firms than through layoffs. The effects depend on institutions.

The evidence reviewed in Chapter 4 shows that the share of income going to labor has fallen in many economies, and that ownership of productive capital, including robots, is highly concentrated. That is the core distributive fact. When a robot replaces a worker, the wage that used to go to a household now flows to whoever owns the robot and the software that runs it.

The evidence reviewed in Chapters 5 and 12 shows that "automation" often relocates human labor rather than removing it. It moves labor into teleoperation centers, data-labeling platforms, and maintenance roles, frequently across borders and into informal arrangements beyond the reach of labor law.

So the honest summary is neither "robots will take all the jobs" nor "technology always creates more jobs than it destroys." It is that robots redistribute: work, income, risk, and control. Whether that redistribution is fair is not a technical question. It is decided by ownership structures, laws, bargaining power, and public investment.

Five things worth keeping human

If machines can do much of what we do, we should decide deliberately what we want humans to keep doing, not because machines cannot, but because the doing itself matters. I propose five.

1. Judgment over life, liberty, and force. Decisions to injure, detain, or kill a person should remain with an accountable human. That is why both of my draft bills ban autonomous force and require a human decision before a robot applies force likely to cause injury. A machine can be accurate. It cannot be answerable.

2. The relational core of care. Chapter 6 argued that robots can lift a patient but should not replace the person who sits with her. Care consists of more than tasks. It is recognition: someone choosing to attend to you. A simulation of that choice, however convincing, is not the choice.

3. Authorship of the rules. The design of the arrangements that govern robots — the standards, the bills, the terms of service — must remain a human, democratic activity. Delegating rule-making to the systems being ruled, or to the handful of firms that build them, would be the deepest kind of automation and the most dangerous.

4. Meaningful work, chosen. Not all work is meaningful, and much of it should be automated gladly. But the ability to do work that expresses skill and earns recognition is part of a good life. A society should make sure automation removes drudgery rather than removing people's access to meaningful work.

5. Presence. Being there for a friend, a neighbor, a citizen, a stranger in trouble. No robot can be present on your behalf. Those of us who build technology have a duty to design it to send people back into each other's company rather than replace it.

A robotic social contract

A social contract is the set of mutual promises that make a society livable: what citizens owe each other and what institutions owe citizens. Industrial societies built one in the twentieth century around wage labor.

You worked, and in return you got wages, protections, pensions, and a voice through unions and votes. Automation strains that contract because it weakens the link between work and income for some people while strengthening the link between capital and income for others.

Drawing on the whole book, I propose the outline of a new contract in ten clauses. These are not legislative text. They are commitments that legislation, firms, and communities can translate into practice.

1. **Robots are property, and someone is always responsible.** No robot is a legal person. Every robot operating among people has an identifiable human or institution answerable for it, and "the machine did it" is never a defense.
2. **Those who carry the risk get a say.** Workers, neighbors, patients, and the public, who share space with robots, have a right to information, to stop dangerous machines, and to participate in deciding where robots are deployed.
3. **Harm is compensated quickly and simply.** Mandatory insurance, access to the robot's event data, and fair presumptions mean an injured person does not need a team of engineers and lawyers to be made whole.
4. **The gains of automation are shared.** Through some combination of public robotics funds, automation dividends, profit-sharing, worker ownership, and taxation of the rents that concentrated robot ownership produces, a fair portion of productivity gains returns to the people and communities whose labor and data made the machines possible.
5. **Transitions are supported, not left to chance.** Advance notice of automation, real retraining, income support, and wage insurance are guaranteed. In countries with large informal workforces, they are delivered through the state, cooperatives, and platforms rather than through employers who do not exist.
6. **Ownership means control.** If you buy a robot, its maker cannot switch off what you paid for, abandon it without warning, or stop you from repairing it.
7. **The home is not a mine.** Data from inside homes and bodies is collected only with real consent and is never the price of a machine's basic function.
8. **No hidden humans, no fake humans.** People have a right to know when a robot is controlled by a remote human and when a "human" is a machine. The workers inside the machine have rights wherever they sit.
9. **Access is broad.** Small businesses, farmers, cooperatives, public services, and remote regions get access to robots so that automation does not widen the gap between those who own machines and those who are displaced by them.
10. **The rules stay democratic and anticipatory.** Societies build the institutions, registries, liability rules, and worker protections now, while technical details evolve through delegated standards, revisited regularly, with the affected in the room.

From Semarang, looking forward

I live in a city of about 1.7 million people on the north coast of Java, in a country where most workers have no formal contract and where the average age is around thirty. For a place like this, robots could go two ways. They could arrive as imported machines owned by distant firms, displacing factory and service jobs before the country has finished industrializing, with the gains flowing abroad. Or they could arrive as tools owned and shaped locally, by cooperatives, small firms, public institutions, and a generation trained to build and govern them, taking on dangerous and exhausting work while expanding what ordinary people can do.

Neither path is fate. Both are being decided now, in policy and investment choices that rarely make the news. The main reason I wrote two bills and this book is that I came to believe the choice belongs to us, and that the worst outcome is to wait until the machines arrive and then discover that the arrangement was settled without us.

The value of a human was never what we produced. It was that we could choose what to make of the world and of each other. Robots do not take that away. Only we can give it up. The task of this generation is to build machines that serve that choice, and institutions that make sure everyone gets to make it.

Notes

This chapter synthesizes the arguments and evidence of Chapters 1 through 12. Sources for the empirical claims summarized here appear in the notes to those chapters. The Pancasila text is from the Preamble to the 1945 Constitution of the Republic of Indonesia. The McLuhan/Culkin attribution in the Introduction follows John M. Culkin, "A Schoolman's Guide to Marshall McLuhan," *Saturday Review*, March 18, 1967.

Glossary

Actor-network theory (ANT). An approach from science and technology studies, associated with Bruno Latour, that treats humans and non-human things as participants in networks that produce social outcomes.

Anticipatory regulation. Building legal institutions, registries, liability rules, and protections before a technology is widespread, while delegating technical detail to standards that can be updated.

Automation dividend. A proposal to return part of the gains from automation to the public, for example as payments from a fund that owns a share of robotic capital.

Collingridge dilemma. A technology is easy to control early, when its effects are unknown, and hard to control later, when its effects are clear but it is entrenched.

Covered List. The US Federal Communications Commission's list of equipment deemed a national security risk, which cannot receive new equipment authorizations. In July 2026, foreign-produced advanced robotics devices were added.

Embodied AI. Artificial intelligence that perceives and acts in the physical world through a body, such as a robot.

Ghost work. Hidden human labor, such as data labeling, content moderation, and remote assistance, that makes automated systems appear autonomous. The term comes from Mary L. Gray and Siddharth Suri.

Humanoid. A robot with a roughly human body plan: a torso, two arms, and usually two legs.

Informal work. Work without a formal employment contract or social protection. In Indonesia, it covers most of the workforce.

Labor share. The fraction of national income paid to workers as wages and benefits, as opposed to returns to capital.

Preemption. In US law, the displacement of state law by federal law on the same subject.

Remote ID. A requirement that a machine broadcast its identity and its operator's information while operating in public.

Respondeat superior. The legal doctrine holding an employer responsible for the acts of an employee within the scope of employment. It is often used as an analogy for responsibility for robots.

Responsibility gap. The worry, set out by Andreas Matthias, that learning machines act in ways no human could predict or control, leaving no one clearly responsible for the harm.

Right to repair. The principle that owners and independent repairers should have access to the parts, tools, software, and information needed to fix what they own.

Robot density. The number of industrial robots per 10,000 manufacturing employees, as measured by the International Federation of Robotics.

Sandbox (regulatory). A supervised program that lets firms test new products under relaxed rules with safeguards.

So-so automation. Automation that displaces workers while producing only small productivity gains (Acemoglu and Restrepo).

Teleoperation. Controlling a robot remotely, often from another city or country.

VLA model. A vision-language-action model: an AI model that takes in images and instructions and outputs robot actions.